
 $^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10**

$J^\pi=0^+$ for ^{34}S ground state.

- 1976Me12:** 0.7-2.1-MeV proton beams were produced from the 1.1-MV Cockcroft-Walton generator and a 3-MV Van de Graaff accelerator. Targets were ZnS (46% ^{34}S) and CdS (37% ^{34}S) on tantalum backings. γ rays were detected using a 10-cm by 10-cm NaI detector and a 40-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced level energies for 33 bound states and 47 resonances, J, π , resonance strengths, γ branching ratios, mixing ratios, and lifetimes for 25 bound states using DSAM.
- 1976Sp08:** 1.21, 1.95-2.91-MeV protons were produced from the Utrecht University 3-MV Van de Graaff accelerator. Targets were Ag₂S (95% ^{34}S) evaporated onto a tantalum backing. γ rays were detected using a 10 cm by 10 cm diameter NaI scintillator and a 120- or 60-cm³ Ge(Li) detector. Measured reaction yields, E_γ . Deduced Q value, 84 resonance energies, resonance strengths, widths, J, π , and γ -branching ratios.
- 1976Sp09:** similar setup as **1976Sp08** with 2.33-2.79-MeV protons. Measured reaction yields, E_γ , $\gamma(\theta)$. Deduced levels, J, π , and γ -branching ratios.
- 1972Hu10:** 0.7-2.1-MeV protons were produced from the 4-MV Van de Graaff at the Centre d'Etudes Nucleaires de Bordeaux-Gradignan. Targets were 100- $\mu\text{g}/\text{cm}^2$ Ag₂S (90% ^{34}S). γ rays were detected using a 12.7-cm by 12.7-cm NaI(Tl) detector and a 60-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced 59 resonances, 32 bound states, resonance strengths, and γ branching ratios.
- 1972Hu11:** similar setup as **1972Hu10**. Deduced lifetimes using DSAM.
- 1973Fa07:** 1-2-MeV protons were produced from the Helsinki 2-MV Van de Graaff accelerator. Targets were ^{34}S on tantalum backings. γ rays were detected using a 55-cm³ Ge(Li) detector. Measured reaction yields, E_γ , I_γ . Deduced 28 resonances, resonance strengths, γ -branching ratios, and 9 lifetimes using DSAM.
- 2001Vo24:** 1.183-2.820-MeV protons were produced from the 4-MeV electrostatic Van de Graaff accelerator (NSC KhPEI). Enriched ^{34}S target (100%). A 63 cm³ Ge(Li) detector and a 15-cm-diam by 10-cm-long NaI(Tl) detector. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced 12 resonances and resonance strengths.
- 2002Vo17:** 2.187-MeV protons were produced from the ESU-4 accelerator (NSC KhPEI). Enriched ^{34}S target. A 63 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, resonance strengths, widths.
- 1981Bi05:** 1.4-2.8-MeV protons were produced from the Groningen 5-MV Van de Graaff accelerator. Targets were 4 and 1- $\mu\text{g}/\text{cm}^2$ Sb₂S₃ (90% in ^{34}S) evaporated onto 10- $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected using three silicon surface barrier detectors. γ rays were detected using a 10-cm by 10-cm NaI(Tl) detector and a Ge(Li) detector. Measured $\sigma(E_p, \theta)$, E_γ , $\gamma(\theta)$ for two γ rays. Deduced levels, J, π , widths, and resonance strengths.
- 1971Wi13:** protons were produced from the Ontario Cancer Institute. Targets were CdS. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -linear polarization. Deduced 1.020-, 1.214-, 1.267-, and 1.510-MeV resonances, J^π , branchings, mixing ratios, and half-lives using DSAM.
- 1971Pr11:** 1.211-MeV protons were produced from the 2-MeV ARL Van de Graaff at the Aerospace Research Laboratories (ARL). Targets were Ag₂S (85.6% ^{34}S). γ rays were detected using an 80-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, $p\gamma$ -delay. Deduced resonances, levels, J, π , branchings.
- 1963Ha32:** 0.7-1.2-MeV protons were produced from the Laboratorium Technische Fysika. Target was ^{34}S . γ rays were detected using NaI(Tl) scintillation spectrometers. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced levels, branchings.
- 1967Da12:** 0.8-2.5-MeV protons were produced from the 2.5 MeV Van de Graaff generator of the U.A.R. Atomic Energy Establishment. Targets were ^{34}S on tantalum backings. γ rays were detected using two 15.2-cm-diam by 12.7-cm-thick NaI(Tl) crystals. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced resonances, levels, branchings.
- 1960An06:** 0.7-1.9-MeV protons were produced from the electrostatic generator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were isotopic ^{34}S (4.2%). γ rays were detected using a 2-cm-thick by 3-cm-diam CsI(Tl) crystal. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced levels, J, branchings.
- 1966Wa09:** 1.214 MeV proton beam. Measured E_γ , γ -polarization. Deduced level energy and J, π for the level of 3163 and 7540 level.
- 1966En04:** 0.3-2.1-MeV protons were produced from the Utrecht 850-keV Cockcroft-Walton generator and the 3-MV Van de Graaff generator. Targets were natural sulphur compound on tantalum backings. γ rays were detected using a cylindrical 10-cm-by-10-cm NaI crystal. Measured $\sigma(E_p)$, E_γ . Deduced the resonance strength for $E_p=1214$ keV.
- 1966Az01:** 1.214-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS with enriched ^{34}S (46%). γ rays were detected using a Compton polarimeter. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J, π , branchings.
- 1967Wa22:** 1.214- and 1.512-MeV protons were produced from the Aerospace Research Laboratories. Targets were enriched CdS (40% ^{34}S) and Ag₂S (86% ^{34}S). γ rays were detected using two large volume NaI(Tl) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -polarization. Deduced levels, J, π , branchings.
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$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

- 1967Ta08:** 1.020-, 1.213- and 1.513-MeV protons were produced from the 3-MV Van de Graaff accelerator of the Ontario Cancer Institute. Targets were CdS (46% ^{34}S) on thick metal foils. γ rays were detected using a 7.6-cm-long by 7.6-cm-diam NaI(Tl) detector. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -polarization. Deduced levels, J , π , branchings, mixing ratios.
- 1977Ko35:** 1.055, 1.183, 1.214, and 1.416 MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Target of ^{34}S . A 40 cm³ coaxial Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced levels, branchings, mixing ratios.
- 1977Ko34:** 1.0-1.4 MeV protons. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π , branchings, mixing ratios.
- 1970Ko33:** 2.317-, 2.223-, and 2.336-MeV protons were produced from the 4-MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were ^{34}S on tantalum backings. γ rays were detected using a 12-cm³ coaxial Ge(Li) detector. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced 3 levels, J , and branching ratios.
- 1967Ko19,1967Ko20,1967Ko23:** 2.079-MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were made by implanting ^{34}S ions in a tantalum substrate. γ rays were detected using a 70-mm by 50-mm and a 70-mm by 60-mm NaI(Tl) detectors. Measured E_γ , I_γ , $p\gamma\gamma(\theta)$. Deduced levels, J , π , branchings, mixing ratios.
- 1968Az02:** 1.214- and 1.513-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS (37% ^{34}S) evaporated onto gold backings. γ rays were detected using a 5-cm-diam by 5-cm-long NaI(Tl) crystal. Measured E_γ , $\gamma\gamma$ -delay. Deduced lifetimes of the level of 3163 keV using the delayed coincidence method and DSAM.
- 1969Mi23:** 2.079-MeV protons. ^{34}S target. Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, branchings.
- 1971Ko32,1971Ko46:** 0.8-2.6-MeV protons. Targets were made by evaporating ^{34}S ions onto tantalum backings. γ rays were detected using Ge(Li) and NaI(Tl) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , and $p\gamma\gamma(\theta)$. Deduced levels, J , π , level widths, and branchings.
- 1971Ba23:** 1.214-MeV protons were produced from the 3 MeV Dynamitron of Carleton and Ottawa Universities. Target of CdS (93% ^{34}S) on a gold backing. A 28 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced lifetime of the 3163 keV level using the delayed coincidence method.
- 1974Ai04:** 1.214-MeV protons. Targets were Ag₂S, CdS, and ZnS of 300- $\mu\text{g}/\text{cm}^2$. Ge(Li) detector. Measured $\sigma(E_\gamma)$. Deduced resonance strengths.
- 1975Ke11:** 1.214-MeV protons were produced from the Helsinki University 2.5-MV van de Graaff. Targets of CdS and ZnS on tantalum backings. γ rays were detected using a 120 cm³ Ge(Li) detector with FWHM=2.9 keV at 2.6 MeV. Measured $\sigma(E_\gamma)$. Deduced resonance strengths.
- 1979Pa16:** 1.211-MeV protons were produced from the University of Melbourne 5U Pelletron accelerator. Natural sulphur target. NaI(Tl) and Ge(Li) detector. Measured $\sigma(E_p)$. Deduced resonance strength.
- 1988Va06:** 0.4-0.7 and 1.47 MeV protons were produced from the Utrecht 3 MV Van de Graaff accelerator. Targets of Ag₂S (90% ^{34}S) of 10 and 34 $\mu\text{g}/\text{cm}^2$ on Ta backings. A 80-cm³ coaxial Philips Ge(Li) or an γ -X intrinsic Ge detector, FWHM=1.85 and 1.75 keV at 1.33 MeV. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π , branchings, mixing ratios.
- 1988Co03:** 1.211-MeV protons were produced from the Geel 7-MV CN-type Van de Graaff. Targets of ^{34}S . Ge(Li) detector. Measured $\gamma\gamma$ -coin. Deduced weighting function.
- 1996Vo20:** 1.891, 1.900 and 2.070-MeV protons were produced by an electrostatic accelerator ESU-4 at the Physical and Technical Institute. Target made by implanting ^{34}S ions in a tantalum backing. A 63 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, gamma widths.
- 1996Ka21:** 1-3 MeV protons were produced from the electrostatic accelerator of the Institute of Physica and Technology (Kharkov). Targets were 20- $\mu\text{g}/\text{cm}^2$ Ag₂S. γ rays were detected using a 60 cm³ Ge(Li) detector and a 150×100-mm² NaI(Tl) as a monitor. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced J^π for 3 resonances, resonance strengths.
- 2000Vo21:** 1.794 and 2.275 MeV protons. Measured E_γ , I_γ . Deduced gamma widths, transition strengths.
- 2001Ka69:** 0.8-3 MeV protons produced from the ESU-4 accelerator (NSC KhPEI). Target of ^{34}S . A 15-cm-diam by 19-cm-long NaI(Tl) detector. Measured E_γ , I_γ . Deduced levels, transition strengths.
- Others: 1962Du05, 1963Du08, 1964Hy01, 1966Ko22, 1966Ko15, 1967Ma48, 1969Ko26, 1969Ah02, 1969KrZW, 1972LeYW, 1974Ii01, 1974Bi16, 1981Va02.
- Theoretical studies: 2020Ri06 (shell model).

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) ^{35}Cl Levels

Resonance strength $\omega\gamma=(2J+1)\Gamma_\gamma\Gamma_p/\Gamma$.

Resonance strengths are mainly obtained from resonance γ -ray yields. $\omega\gamma$ in [1976Sp08](#), [1996Ka21](#), and [2001Vo24](#) is all normalized to $\omega\gamma=9.7$ eV 7 at $E_p=1212$ keV. $\omega\gamma$ in [1972Hu10](#), [1973Fa07](#), and [1976Me12](#) is normalized to $\omega\gamma=21$ eV 3 at $E_p=1212$ keV from [1966En04](#).

E(level) [†]	J^π [‡]	$T_{1/2}$	Comments
0	$3/2^+$		
1219.39 7	$1/2^+$	105 fs 14	E(level): weighted average of 1219.1 5 (1972Hu10) and 1219.3 1 (1976Me12), and 1219.44 7 (1981Bi05). $T_{1/2}$: lifetime $\tau=152$ fs 20: weighted average of 80 fs 40 (1971Wi13 , DSAM), 175 fs 20 (1972Hu11 , DSAM), 150 fs 20 (1973Fa07 , DSAM), and 145 fs 30 (1976Me12 , DSAM).
1763.14 4	$5/2^+$	0.347 ps 63	E(level): weighted average of 1762.8 5 (1972Hu10), 1763.4 7 (1976Me12), and 1763.14 4 (1981Bi05). J^π : from $\gamma(\theta)$ in 1971Wi13 . $T_{1/2}$: lifetime $\tau=0.50$ ps 9: weighted average of 0.50 ps 10 (1972Hu11 , DSAM), 0.40 ps 14 (1973Fa07 , DSAM), and 0.53 ps 9 (1976Me12 , DSAM).
2645.70 8	$(5/2,7/2)$	0.146 ps 21	E(level): weighted average of 2645.6 5 (1972Hu10), 2644.7 13 (1976Me12), and 2645.71 8 (1981Bi05). J^π : from $\gamma(\theta)$ in 1971Wi13 . Other: $3/2$ from $\gamma(\theta)$ in 1967Ko20 . $T_{1/2}$: lifetime $\tau=0.210$ ps 30: weighted average of 0.255 ps 65 (1972Hu11 , DSAM) and 0.200 ps 30 (1976Me12 , DSAM).
2693.9 3	$(3/2,5/2)$	13.9 fs 21	E(level): weighted average of 2694.0 3 (1972Hu10), 2694.7 12 (1976Me12), and 2693.8 3 (1981Bi05). J^π : from $\gamma(\theta)$ in 1971Wi13 . $T_{1/2}$: lifetime $\tau=20$ fs 3: weighted average of 21 fs 3 (1972Hu11 , DSAM), 19 fs 4 (1973Fa07 , DSAM), and 20 fs 4 (1976Me12 , DSAM).
3002.75 8	$5/2$	11.1 fs 21	E(level): weighted average of 3002.7 7 (1972Hu10), 3003.7 8 (1976Me12), and 3002.74 8 (1981Bi05). J^π : $5/2$ from $\gamma(\theta)$ in 1967Ko20 and 1971Wi13 . $T_{1/2}$: lifetime $\tau=16$ fs 3: weighted average of 22 fs 3 (1972Hu11 , DSAM), 14 fs 2 (1973Fa07 , DSAM), and 16 fs 4 (1976Me12 , DSAM).
3162.99 7	$7/2^-$	25.7 ps 28	$T=1/2$ E(level): weighted average of 3162.5 5 (1972Hu10), 3163.9 7 (1976Me12), and 3162.99 6 (1981Bi05). J^π : spin from $\gamma(\theta)$ in 1966Wa09 , 1967Wa22 , 1967Ta08 , and 1971Wi13 ; parity from 3163γ , $M2+E3$ to $3/2^+$ ground state in 1966Wa09 and 1967Wa22 . $T_{1/2}$: lifetime $\tau=37$ ps 4 (1971Ba23 , delayed coincidence). Others: 140 ps 40 (1968Az02 , delayed coincidence), >0.6 ps (1968Az02 , DSAM), >0.2 ps (1971Wi13 , DSAM), >2 ps (1972Hu11 , DSAM), >0.9 ps (1973Fa07 , DSAM), and >1 ps (1976Me12). 1971Ba23 states that the earlier 1968Az02 value was obtained using NaI(Tl) detectors with broad time resolution and the discrepancy between 1968Az02 and 1971Ba23 is not unreasonable.
3918.48 10	$3/2^+$	4.9 fs 14	E(level): weighted average of 3917.9 6 (1972Hu10), 3920.7 13 (1976Me12), and 3918.48 7 (1981Bi05). J^π : spin= $3/2$ determined by the feeding $5162\gamma(\theta)$ from $5/2^+$, 9081 level in 1976Sp09 ; parity= $+$ determined by 3918γ , $M1+E2$ to $3/2^+$ ground state.
3942.9 2		0.159 ps 42	$T_{1/2}$: lifetime $\tau=7$ fs 2 (1972Hu11 , DSAM). Other: <6 fs (1976Me12 , DSAM). E(level): weighted average of 3942.6 9 (1972Hu10), 3944.1 11 (1976Me12), and 3942.9 2 (1981Bi05). $T_{1/2}$: lifetime $\tau=0.23$ ps 6: weighted average of 0.39 ps 11 (1972Hu11 , DSAM), 0.145 ps 50 (1973Fa07 , DSAM), and 0.29 ps 5 (1976Me12 , DSAM).
3967.3 6		11.8 fs 28	E(level): from 1972Hu10 . Other: 3979.0 15 (1976Me12). 1976Me12 matched their 3979 level with the 3967 level observed in other work. No other work has observed

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
			a level at 3979 keV.
4059.2 4		13.9 fs 21	T _{1/2} : lifetime $\tau=17$ fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM). E(level): weighted average of 4058.4 8 (1972Hu10), 4059.4 4 (1976Me12), and 4059.2 3 (1981Bi05).
4113.7 10	7/2		T _{1/2} : lifetime $\tau=20$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM). E(level): weighted average of 4111 3 (1972Hu10) and 4114.0 10 (1976Me12).
4173.45 19		34.7 fs 63	J ^π : spin=7/2 from feeding 4516 γ (θ) from 7/2, 8630 level in 1976Sp09. T=1/2 E(level): weighted average of 4173.0 6 (1972Hu10), 4174.7 10 (1976Me12), and 4173.45 19 (1981Bi05).
4177.9 2	3/2	27.0 fs 28	T _{1/2} : lifetime $\tau=50$ fs 9: weighted average of 105 fs +35–30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM). E(level): weighted average of 4177.2 9 (1972Hu10), 4180.1 15 (1976Me12), and 4177.9 2 (1981Bi05).
4347.76 6	5/2,9/2	0.229 ps 42	J ^π : from γ (θ) in 1967Wa22. T _{1/2} : lifetime $\tau=39$ fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM). E(level): weighted average of 4347.5 8 (1972Hu10), 4347.2 12 (1976Me12), and 4347.76 6 (1981Bi05).
4624.4 3			J ^π : spin=5/2,9/2 based on feeding 4282 γ (θ) from 7/2, 8630 level in 1976Sp09. T _{1/2} : lifetime $\tau=0.33$ ps 6 (1976Me12, DSAM). Other: >0.6 ps (1972Hu11, DSAM). E(level): weighted average of 4624.2 20 (1976Me12) and 4624.4 3 (1981Bi05). Other: 4625 5 (1972Hu10).
4769.86 23		76 fs 20	15% of I γ from this level decays to unknown final levels (1976Me12). E(level): weighted average of 4768.9 12 (1972Hu10), 4766.9 15 (1976Me12), and 4769.90 15 (1981Bi05).
4839.10 11		9.7 fs 35	T _{1/2} : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 (1972Hu11, DSAM) and 105 fs 17 (1976Me12, DSAM). E(level): weighted average of 4838 3 (1972Hu10), 4841.7 19 (1976Me12), and 4839.09 11 (1981Bi05).
4854.4 19		4.9 fs 14	T _{1/2} : lifetime $\tau=14$ fs 5 (1976Me12, DSAM). E(level): weighted average of 4853 2 (1972Hu10) and 4855.7 19 (1976Me12).
4880.96 15		5.6 fs 14	T _{1/2} : lifetime $\tau=7$ fs 2 (1976Me12, DSAM). Other: <6 fs (1972Hu11, DSAM). E(level): weighted average of 4880.9 1 (1972Hu10), 4885 2 (1976Me12), and 4881.14 20 (1981Bi05).
5010.1 18		7.6 fs 21	T _{1/2} : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 (1972Hu11, DSAM), 7 fs 3 (1973Fa07, DSAM), and 10 fs 3 (1976Me12, DSAM). E(level): weighted average of 5008 5 (1972Hu10) and 5010.4 18 (1976Me12).
5163.34 22	7/2		T _{1/2} : lifetime $\tau=11$ fs 3 (1976Me12, DSAM). E(level): weighted average of 5163 5 (1972Hu10), 5166.7 15 (1976Me12), and 5163.31 14 (1981Bi05).
5215.8 15		<4.85 fs	J ^π : spin=3/2,7/2 from feeding 3466 γ (θ) from 7/2, 8630 level in 1976Sp09. Spin=3/2 and $\delta(\text{O}/\text{Q})=0.12$ 1 would require a large E3 or M3 component, which is ruled out by RUL. 10% of I γ transitions to unknown levels (1976Me12).
5403.5 10		11.8 fs 28	E(level): weighted average of 5214 3 (1972Hu10) and 5216.2 15 (1976Me12). T _{1/2} : lifetime $\tau<7$ fs (1976Me12, DSAM). E(level): weighted average of 5400 5 (1972Hu10) and 5403.6 10 (1976Me12).
5520			T _{1/2} : lifetime $\tau=17$ fs 4 (1976Me12, DSAM). 20% of I γ from this level decays (1976Me12). 50% of I γ from this level decays to unknown levels (1972Hu10).
5586 2			E(level): from 1976Sp08.
5599.7 15		2.08 fs 69	E(level): weighted average of 5587 3 (1972Hu10) and 5586 2 (1976Me12). E(level): weighted average of 5598 3 (1972Hu10) and 5600.1 15 (1976Me12).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	E _p (lab) (keV)	Comments
5646 2		2.77 fs 69		T _{1/2} : lifetime $\tau=3$ fs 1 (1976Me12, DSAM). E(level): weighted average of 5645 3 (1972Hu10) and 5646 2 (1976Me12).
5655 2		13.9 fs 35		T _{1/2} : lifetime $\tau=4$ fs 1 (1976Me12, DSAM). E(level): weighted average of 5651 3 (1972Hu10) and 5656 2 (1976Me12). J ^π : Mirror level of ^{35}S ground state and thus 3/2 ⁺ (2001Vo24). Other: spin=5/2 based on the ϕ eeding 2051g(θ) from the 5/2, 7706 level in 1977Ko34.
5683 2				T _{1/2} : lifetime $\tau=20$ fs 5 (1976Me12, DSAM). $\Gamma_{\gamma}=0.033$ eV 10 (2001Vo24). 14% of I γ from this level decays to unknown final levels (1976Me12). E(level): weighted average of 5682 5 (1972Hu10) and 5683 2 (1976Me12).
5723.52 18				40% of I γ from this level decays to unknown final levels (1976Me12). E(level): from 1981Bi05. 38% of I γ from this level decays to unknown final levels (1976Sp09).
5758 3				E(level): weighted average of 5754 5 (1972Hu10) and 5759 3 (1976Me12). 25% of γ from this level decays to unknown final levels and the other 75% is attributed to two γ transitions (1976Me12). Other: 100% of γ from this level decays to unknown final levels (1972Hu10).
5806 2		3.47 fs 69		E(level): weighted average of 5802 5 (1972Hu10) and 5806 2 (1976Me12).
6106.2 21		8.3 fs 21		T _{1/2} : lifetime $\tau=5$ fs 1 (1976Me12, DSAM). E(level): weighted average of 6102 3 (1972Hu10) and 6107.2 15 (1976Me12).
6181 3				T _{1/2} : lifetime $\tau=12$ fs 3 (1976Me12, DSAM). E(level): from 1976Me12.
6492 3				55% of I γ from this level decays to unknown final levels (1976Me12). E(level): weighted average of 6489 5 (1972Hu10) and 6493 3 (1976Me12).
6866.7 6			510.6 6	E _p (lab) (keV): from 1988Va06.
7066.2 7			716.0 7	E _p (lab) (keV): weighted average of 716.0 7 (1972Hu10) and 716.0 10 (1976Me12). $\omega\gamma=0.2$ eV 1 (1972Hu10). $\omega\gamma=0.3$ eV 1 (1976Me12).
7103.4 7	3/2		754.3 7	J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 754.1 7 (1972Hu10) and 754.6 10 (1976Me12). $\omega\gamma=0.04$ eV (1963Ha32). $\omega\gamma=0.5$ eV 3 (1972Hu10). $\omega\gamma=0.5$ eV 2 (1976Me12).
7178.6 8	1/2		831.8 8	J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 831.5 8 (1972Hu10), 832.2 15 (1976Me12), and 832 1 (2001Vo24). $\omega\gamma=0.14$ eV 2 (2001Vo24). $\Gamma_{\gamma} \geq 0.07$ eV (2001Vo24). $\omega\gamma=0.4$ eV 2 (1972Hu10). $\omega\gamma=0.3$ eV 1 (1976Me12).
7185			838	E _p (lab) (keV): from 1963Ha32.
7194.6 7	1/2,3/2		848.2 7	$\omega\gamma=0.056$ eV (1963Ha32). J ^π : from $\gamma(\theta)$ in 1960An06. E _p (lab) (keV): weighted average of 848.2 7 (1972Hu10) and 848.1 10 (1976Me12). $\omega\gamma=0.17$ eV (1963Ha32).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
7225.5 8	5/2	880.0 8	$\omega\gamma=1.1$ eV 3 (1972Hu10). $\omega\gamma=1.2$ eV 4 (1976Me12). J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 880.5 8 (1972Hu10) and 879.3 10 (1976Me12). $\omega\gamma=0.073$ eV (1963Ha32). $\omega\gamma=0.4$ eV 2 (1972Hu10). $\omega\gamma=0.3$ eV 1 (1976Me12).
7234.0 8	5/2 ⁺	888.8 8	J ^π : spin=5/2 from $\gamma(\theta)$ in 1960An06 and 6014 γ to 1/2 ⁺ , 1219 level in 1972Hu10 and 1976Me12. E _p (lab) (keV): weighted average of 889.0 8 (1972Hu10) and 888.5 10 (1976Me12). $\omega\gamma=0.23$ eV (1963Ha32). $\omega\gamma=1.3$ eV 4 (1972Hu10). $\omega\gamma=1.9$ eV 6 (1976Me12).
7269.2 1		925.1 1	E _p (lab) (keV): from 1976Sp08.
7272.7 9	(1/2,3/2)	928.6 9	J ^π : from $\gamma(\theta)$ in 1960An06. E _p (lab) (keV): weighted average of 928.2 9 (1972Hu10) and 929.0 10 (1976Me12). $\omega\gamma=0.31$ eV (1963Ha32). $\omega\gamma=1.4$ eV 4 (1972Hu10). $\omega\gamma=2.2$ eV 7 (1976Me12).
7362.0 8	3/2	1020.6 8	J ^π : from $\gamma(\theta)$ in 1971Wi13 and 1977Ko35. E _p (lab) (keV): weighted average of 1020.4 8 (1972Hu10), 1020.5 11 (1973Fa07), and 1021.0 10 (1976Me12). $\omega\gamma=0.48$ eV (1963Ha32). $\omega\gamma=3.2$ eV 10 (1972Hu10). $\omega\gamma=2.5$ eV (1973Fa07). $\omega\gamma=3.1$ eV 9 (1976Me12).
7396.0 10	7/2	1055.6 10	J ^π : from $\gamma(\theta)$ in 1977Ko35. E _p (lab) (keV): weighted average of 1054.9 10 (1972Hu10), 1055.8 12 (1973Fa07), and 1056.9 16 (1976Me12). $\omega\gamma=0.67$ eV (1963Ha32). $\omega\gamma=0.4$ eV 2 (1972Hu10). $\omega\gamma=0.4$ eV (1973Fa07). $\omega\gamma=0.5$ eV 2 (1976Me12).
7451.1 6	3/2	1112.3 6	J ^π : from $\gamma(\theta)$ in 1972LeYW. E _p (lab) (keV): weighted average of 1112.2 10 (1972Hu10), 1112.8 13 (1973Fa07), and 1112.2 6 (1976Me12). $\omega\gamma=0.3$ eV 2 (1972Hu10). $\omega\gamma=0.3$ eV (1973Fa07). $\omega\gamma=0.4$ eV 1 (1976Me12).
7496?		1158	E(level): from 1960An06.
7501.1 8		1163.8 8	E _p (lab) (keV): from 1973Fa07. Other: 1165 (1972Hu10). $\omega\gamma=0.1$ eV (1973Fa07). $\omega\gamma=0.6$ eV 3 for E _p =1165+1166 doublet (1972Hu10).
7502.9 7		1165.7 7	E _p (lab) (keV): from 1973Fa07. Other: 1165.4 7 (1976Me12), 1166 (1972Hu10). $\omega\gamma=0.6$ eV 3 for E _p =1165+1166 doublet (1972Hu10). $\omega\gamma=0.5$ eV (1973Fa07). $\omega\gamma=0.8$ eV 2 (1976Me12), possible doublet.
7518.8 7	7/2	1182.0 7	J ^π : from $\gamma(\theta)$ in 1977Ko35. E _p (lab) (keV): weighted average of 1183.2 11 (1972Hu10), 1182.2 12 (1973Fa07), and 1181.4 7 (1976Me12). $\omega\gamma=0.2$ eV 1 (1972Hu10). $\omega\gamma=0.3$ eV (1973Fa07). $\omega\gamma=0.4$ eV 1 (1976Me12).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^{\dagger}$	$J^{\pi\dagger}$	$T_{1/2}$	$E_p(\text{lab})$ (keV)	Comments
7548.9 6	$7/2^{-}$	<0.69 fs	1213.0 6	$T=3/2$ J^{π}: spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, and 1977Ko35; parity from 4386γ, M1+E2 to $7/2^{-}$, 3163 level in 1966Wa09 and 1967Wa22. $T_{1/2}$: lifetime $\tau < 1$ fs (1973Fa07, DSAM). Others: $\tau < 13$ fs (1971Wi13, DSAM); $\Gamma=5.8$ eV 13, $\Gamma_p=3.8$ eV 13, $\Gamma_{\gamma}=2.0$ eV 13 (1975Ke11). $E_p(\text{lab})$ (keV): weighted average of 1213.7 10 (1972Hu10), 1213.6 6 (1973Fa07), 1212.4 7 (1976Me12), and 1212 1 (2001Vo24). $\omega\gamma=2.0$ eV (1963Ha32). $\omega\gamma=21$ eV 3 (1966En04) and used as the reference by 1972Hu10, 1973Fa07, and 1976Me12. This value is not supported by the resonant absorption experiments in 1976Sp09 and 1981Bi05. $\omega\gamma=2.6$ eV 4 (1967Wa22). $\omega\gamma=9.7$ eV 9 (1974Al04). $\omega\gamma=9.8$ eV 10 (1975Ke11). $\omega\gamma=9$ eV 1 (1979Pa16). $\omega\gamma=9.7$ eV 7 (1996Ka21). $\omega\gamma=9.9$ eV 7 (2001Vo24). $\omega\gamma=9.5$ eV 12 (1981Bi05), deduced from $\omega\gamma$ at $E_p=1891$ from 1981Bi05 and a relative strength of 1212/1891 from 1978En02. $\Gamma_{\gamma}=1.8$ eV 7 (2001Vo24). 1971Pr11 reports a 3166γ with $I_{\gamma}=0.37$ 13 to final level at 4382.
7561.3 6	$1/2, 3/2$		1225.8 6	J^{π}: from $\gamma(\theta)$ in 1977Ko35. $E_p(\text{lab})$ (keV): weighted average of 1225.6 10 (1972Hu10), 1226.3 6 (1973Fa07), and 1225.3 7 (1976Me12). $\omega\gamma=1.5$ eV 5 (1972Hu10). $\omega\gamma=1.1$ eV (1973Fa07). $\omega\gamma=1.5$ eV 5 (1976Me12). $\omega\gamma=0.7$ eV 2 (2001Vo24). $A_2=2$ (2001Vo24).
7600.8 8	$5/2$	<13.9 fs	1266.5 8	J^{π}: spin=$5/2$ from $\gamma(\theta)$ in 1971Wi13 and 1976Me12. Other: spin=$3/2$ from 1977Ko34. $T_{1/2}$: lifetime $\tau < 20$ fs (1971Wi13, DSAM). $E_p(\text{lab})$ (keV): weighted average of 1266.5 10 (1972Hu10), 1267.9 11 (1973Fa07), and 1265.7 8 (1976Me12). $\omega\gamma=3.4$ eV, $\Gamma_{\gamma}\approx 0.57$ eV (1971Wi13). $\omega\gamma=3.2$ eV 10 (1972Hu10). $\omega\gamma=2.9$ eV (1973Fa07). $\omega\gamma=2.8$ eV 8 (1976Me12).
7618.7 5	$5/2$		1284.9 5	J^{π}: from $\gamma(\theta)$ in 1977Ko34. Spin=$3/2, 5/2$ from $\gamma(\theta)$ in 1976Me12. $E_p(\text{lab})$ (keV): weighted average of 1285.5 11 (1972Hu10), 1286.1 11 (1973Fa07), and 1284.5 5 (1976Me12). $\omega\gamma=1.2$ eV 4 (1972Hu10). $\omega\gamma=1.1$ eV (1973Fa07). $\omega\gamma=1.7$ eV 5 (1976Me12).
7656.6 8			1323.9 8	$E_p(\text{lab})$ (keV): weighted average of 1323.7 12 (1972Hu10), 1323.5 12 (1973Fa07), and 1324.2 8 (1976Me12). $\omega\gamma=0.5$ eV 3 (1972Hu10). $\omega\gamma=0.5$ eV (1973Fa07). $\omega\gamma=0.8$ eV 2 (1976Me12).
7671.9 8	$5/2$		1339.7 8	J^{π}: spin=$5/2$ from $\gamma(\theta)$ in 1977Ko34. Spin=$5/2, 7/2$ from $\gamma(\theta)$ in 1976Me12. $E_p(\text{lab})$ (keV): weighted average of 1339.8 11 (1972Hu10), 1339.4 11 (1973Fa07), and 1339.7 8 (1976Me12). $\omega\gamma=1.5$ eV 5 (1972Hu10). $\omega\gamma=1.4$ eV (1973Fa07). $\omega\gamma=1.3$ eV 4 (1976Me12).
7685.5 8	$3/2^{-}$		1353.7 8	J^{π}: $3/2^{-}$ from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=$3/2$ from $\gamma(\theta)$ in 1976Me12.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	$E_p(\text{lab})$ (keV)	Comments
				Spin=3/2,5/2 from $\gamma(\theta)$ in 1977Ko34 . $E_p(\text{lab})$ (keV): weighted average of 1353.7 <i>11</i> (1972Hu10), 1353.4 <i>11</i> (1973Fa07), and 1353.8 <i>8</i> (1976Me12). $\Gamma_p=445$ eV <i>11</i> (1981Bi05). $\omega\gamma=2.5$ eV <i>8</i> (1972Hu10). $\omega\gamma=2.4$ eV (1973Fa07). $\omega\gamma=2.8$ eV <i>8</i> (1976Me12). $E_p(\text{lab})$ (keV): weighted average of 1362.4 <i>12</i> (1972Hu10) and 1362.3 <i>8</i> (1976Me12). $\omega\gamma=0.4$ eV <i>2</i> (1972Hu10). $\omega\gamma=0.6$ eV <i>2</i> (1976Me12). J^π : 5/2 ⁺ from $\sigma(E_p,\theta)$ in 1981Bi05 . Spin=5/2 from $\gamma(\theta)$ in 1977Ko34 . Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12 and 5/2 is favored. $E_p(\text{lab})$ (keV): weighted average of 1375.7 <i>11</i> (1972Hu10), 1375.0 <i>11</i> (1973Fa07), and 1375.1 <i>8</i> (1976Me12). $\omega\gamma=3.7$ eV <i>11</i> (1972Hu10). $\omega\gamma=3.1$ eV (1973Fa07). $\omega\gamma=4.4$ eV <i>13</i> (1976Me12). $\Gamma_p=4$ eV <i>1</i> (1981Bi05). J^π : from $\gamma(\theta)$ for $J_f(2645.7)=7/2$, $J_f(3163)=7/2$, $J_f(3942.9)=9/2$ (1977Ko35). $E_p(\text{lab})$ (keV): weighted average of 1415.6 <i>12</i> (1972Hu10), 1414.2 <i>12</i> (1973Fa07), and 1414.7 <i>9</i> (1976Me12). $\omega\gamma=1.3$ eV <i>4</i> (1972Hu10). $\omega\gamma=1.2$ eV (1973Fa07). $\omega\gamma=1.7$ eV <i>5</i> (1976Me12). $E(\text{level})$: from 1960An06 . $E_p(\text{lab})$ (keV): weighted average of 1449.0 <i>11</i> (1972Hu10), 1447.5 <i>12</i> (1973Fa07), and 1447.5 <i>9</i> (1976Me12). $\omega\gamma=1.5$ eV <i>5</i> (1972Hu10). $\omega\gamma=(1.4)$ eV (1973Fa07). $\omega\gamma=1.5$ eV <i>5</i> (1976Me12). J^π : proposed by 1976Me12 based on strengths of primary transitions. $E_p(\text{lab})$ (keV): weighted average of 1453.6 <i>11</i> (1972Hu10), 1451.8 <i>15</i> (1973Fa07), and 1452.3 <i>13</i> (1976Me12). $\omega\gamma=0.8$ eV <i>4</i> (1972Hu10). $\omega\gamma=(0.9)$ eV (1973Fa07). $\omega\gamma=0.9$ eV <i>3</i> (1976Me12). J^π : from $\gamma(\theta)$ in 1988Va06 , which rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12 . $E_p(\text{lab})$ (keV): weighted average of 1469.6 <i>12</i> (1972Hu10), 1468.2 <i>12</i> (1973Fa07), and 1468.2 <i>9</i> (1976Me12). $\omega\gamma=1.5$ eV <i>5</i> (1972Hu10). $\omega\gamma=1.6$ eV (1973Fa07). $\omega\gamma=1.4$ eV <i>4</i> (1976Me12). $T=3/2$ J^π : spin from $\gamma(\theta)$ in 1967Wa22 . $T_{1/2}$: lifetime $\tau<5$ fs (1971Wi13 , DSAM). $E_p(\text{lab})$ (keV): weighted average of 1509.1 <i>15</i> (1973Fa07), 1510.2 <i>10</i> (1976Me12), and 1510 <i>1</i> (2001Vo24). $\omega\gamma=3.4$ eV <i>6</i> (1967Wa22). $\omega\gamma\approx 7.5$ eV, $\Gamma_\gamma=1.9$ eV (1971Wi13). $\omega\gamma=11$ eV <i>3</i> for $E_p=1510$ doublet (1972Hu10). $\omega\gamma=1.3$ eV (1973Fa07). $\omega\gamma=4.8$ eV (1976Me12). $\omega\gamma=2.2$ eV <i>3</i> (2001Vo24). $\Gamma_\gamma=0.55$ eV <i>6</i> (2001Vo24). $E_p(\text{lab})$ (keV): weighted average of 1512.3 <i>15</i> (1973Fa07) and 1511.6 <i>9</i>
7693.9 <i>8</i>			1362.3 <i>8</i>	
7706.4 <i>8</i>	5/2 ⁺		1375.2 <i>8</i>	
7744.8 <i>9</i>	7/2		1414.8 <i>9</i>	
7748			1418	
7777.1 <i>9</i>			1448.0 <i>9</i>	
7781.7 <i>11</i>	5/2 ⁻		1452.8 <i>11</i>	
7797.1 <i>9</i>	1/2		1468.6 <i>9</i>	
7837.2 <i>10</i>	3/2	<3.47 fs	1509.9 <i>10</i>	
7839.0 <i>9</i>			1511.8 <i>9</i>	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
			(1976Me12).
			$\omega\gamma=11$ eV 3 for E _p =1510 doublet (1972Hu10).
			$\omega\gamma=8.7$ eV (1973Fa07).
			$\omega\gamma=6.1$ eV (1976Me12).
7868.7 10		1542.3 10	E _p (lab) (keV): weighted average of 1542.3 13 (1972Hu10), 1542.3 12 (1973Fa07), and 1542.2 10 (1976Me12).
			$\omega\gamma=0.5$ eV 3 (1972Hu10).
			$\omega\gamma=(0.5)$ eV (1973Fa07).
			$\omega\gamma=1.0$ eV 3 (1976Me12).
7873?		1547	E(level): from 1960An06.
7880.8 9	3/2,5/2	1554.8 9	J ^π : from $\gamma(\theta)$ in 1976Me12.
			E _p (lab) (keV): weighted average of 1554.9 12 (1972Hu10), 1554.8 11 (1973Fa07), and 1554.7 9 (1976Me12).
			$\omega\gamma=1.6$ eV 5 (1972Hu10).
			$\omega\gamma=1.3$ eV (1973Fa07).
			$\omega\gamma=2.2$ eV 7 (1976Me12).
7885?		1559	E(level): from 1960An06.
7889.1 15		1563.3 15	E _p (lab) (keV): from 1973Fa07.
			$\omega\gamma=(0.1)$ eV (1973Fa07).
7899.2 7	(3/2 ⁻ ,5/2)	1573.7 7	E _p (lab) (keV): weighted average of 1574.4 12 (1972Hu10), 1573.2 11 (1973Fa07), and 1573.7 7 (1976Me12).
			$\omega\gamma=1.4$ eV 4 (1972Hu10).
			$\omega\gamma=1.0$ eV (1973Fa07).
			$\omega\gamma=2.2$ eV 7 (1976Me12).
7903?		1578	E(level): from 1960An06.
7923.3 9		1598.6 9	E _p (lab) (keV): weighted average of 1598.9 13 (1972Hu10), 1598.7 12 (1973Fa07), and 1598.4 9 (1976Me12).
			$\omega\gamma=0.8$ eV 4 (1972Hu10).
			$\omega\gamma=(0.7)$ eV (1973Fa07).
			$\omega\gamma=1.0$ eV 3 (1976Me12).
7930?		1605	E(level): from 1960An06.
7949?		1625	E(level): from 1960An06.
7970.3 10		1646.9 10	E _p (lab) (keV): weighted average of 1647.3 12 (1972Hu10), 1645.9 13 (1973Fa07), and 1647.1 10 (1976Me12).
			$\omega\gamma=1.2$ eV 4 (1972Hu10).
			$\omega\gamma=(1.0)$ eV (1973Fa07).
			$\omega\gamma=1.9$ eV 6 (1976Me12).
7987.8 10	3/2	1665.0 10	J ^π : from $\gamma(\theta)$ in 1976Me12.
			E _p (lab) (keV): weighted average of 1665.2 12 (1972Hu10), 1663.3 12 (1973Fa07), and 1666.1 10 (1976Me12).
			$\omega\gamma=1.5$ eV 5 (1972Hu10).
			$\omega\gamma=(1.3)$ eV (1973Fa07).
			$\omega\gamma=1.2$ eV 4 (1976Me12).
7995.6 10	5/2	1673.0 10	E _p (lab) (keV): weighted average of 1673.7 11 (1972Hu10), 1672.4 13 (1973Fa07), and 1672.7 10 (1976Me12).
			$\omega\gamma=2.7$ eV 8 (1972Hu10).
			$\omega\gamma=(2.3)$ eV (1973Fa07).
			$\omega\gamma=2.1$ eV 6 (1976Me12).
8001.0 9	(7/2,9/2 ⁺)	1678.6 9	E _p (lab) (keV): weighted average of 1679.7 11 (1972Hu10), 1678.2 13 (1973Fa07), and 1678.1 9 (1976Me12).
			$\omega\gamma=4.1$ eV 12 (1972Hu10).
			$\omega\gamma=(2.6)$ eV (1973Fa07).
			$\omega\gamma=2.9$ eV 9 (1976Me12).
8005.0 10	5/2 ⁺	1682.7 10	J ^π : from $\sigma(\text{E}_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1981Bi05. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12.
			E _p (lab) (keV): weighted average of 1684.3 13 (1972Hu10), 1681.9 11 (1973Fa07), and

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
			1682.5 10 (1976Me12). $\omega\gamma=(3.4)$ eV (1973Fa07). $\omega\gamma=4.8$ eV 15 (1976Me12). $\omega\gamma=4.5$ eV 14 (1972Hu10). $\Gamma_p=21$ eV 3 (1981Bi05). E _p (lab) (keV): weighted average of 1714.0 11 (1972Hu10), 1713.5 14 (1973Fa07), and 1714.9 10 (1976Me12). $\omega\gamma=0.9$ eV 5 (1972Hu10). $\omega\gamma(0.9)$ eV (1973Fa07). $\omega\gamma=0.8$ eV 2 (1976Me12).
8035.7 10		1714.3 10	J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1717.9 11 (1972Hu10), 1716.3 11 (1973Fa07), and 1717.3 11 (1976Me12). $\omega\gamma=3.5$ eV 11 (1972Hu10). $\omega\gamma=3.0$ eV (1973Fa07). $\omega\gamma=2.9$ eV 9 (1976Me12).
8038.5 11	3/2	1717.2 11	E _p (lab) (keV): weighted average of 1756.0 10 (1972Hu10), 1755.4 11 (1973Fa07), and 1755.7 11 (1976Me12). $\omega\gamma=1.6$ eV 5 (1972Hu10). $\omega\gamma(1.3)$ eV (1973Fa07). $\omega\gamma=1.6$ eV 5 (1976Me12).
8075.9 10		1755.7 10	E _p (lab) (keV): from 1960An06. E _p (lab) (keV): weighted average of 1776.7 11 (1972Hu10), 1776.9 11 (1973Fa07), and 1776.0 11 (1976Me12). $\omega\gamma=2.5$ eV 8 (1972Hu10). $\omega\gamma=2.5$ eV (1973Fa07). $\omega\gamma=2.4$ eV 7 (1976Me12).
8080?		1760	J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1787.7 11 (1972Hu10), 1786.6 11 (1973Fa07), and 1786.9 11 (1976Me12). $\omega\gamma=4.2$ eV 13 (1972Hu10). $\omega\gamma=3.6$ eV (1973Fa07). $\omega\gamma=3.7$ eV 11 (1976Me12).
8096.1 11	(5/2,7/2 ⁺)	1776.5 11	E _p (lab) (keV): weighted average of 1794.8 13 (1972Hu10), 1793.3 14 (1973Fa07), and 1794.2 11 (1976Me12). $\omega\gamma=1.2$ eV 4 (1972Hu10). $\omega\gamma(1.1)$ eV (1973Fa07). $\omega\gamma=1.6$ eV 5 (1976Me12).
8106.4 11	3/2	1787.1 11	E _p (lab) (keV): weighted average of 1829.1 13 (1972Hu10), 1828.6 12 (1973Fa07), and 1829.5 11 (1976Me12). $\omega\gamma=2.4$ eV 7 (1972Hu10). $\omega\gamma=1.5$ eV (1973Fa07). $\omega\gamma=1.7$ eV 5 (1976Me12).
8113.3 11	(1/2 ⁺ ,3/2,5/2 ⁺)	1794.2 11	E _p (lab) (keV): weighted average of 1839.0 12 (1972Hu10), 1838.8 12 (1973Fa07), and 1839.1 11 (1976Me12). $\omega\gamma=2.1$ eV 6 (1972Hu10). $\omega\gamma=2.1$ eV (1973Fa07). $\omega\gamma=1.9$ eV 6 (1976Me12).
8147.2 11	3/2 ⁻	1829.1 11	E _p (lab) (keV): weighted average of 1862.7 13 (1972Hu10), 1862.1 13 (1973Fa07), and 1862.1 12 (1976Me12). $\omega\gamma=0.8$ eV 4 (1972Hu10). $\omega\gamma(0.7)$ eV (1973Fa07). $\omega\gamma=0.9$ eV 3 (1976Me12).
8156.8 11	(5/2 ⁺ ,7/2 ⁻)	1839.0 11	J ^π : from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1996Ka21 and 2001Vo24. E _p (lab) (keV): weighted average of 1893.2 11 (1972Hu10), 1891.3 12
8179.4 12		1862.3 12	
8208.2 10	5/2 ⁺	1891.9 10	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π _z	E _p (lab) (keV)	Comments
			(1976Me12), 1892.0 11 (1973Fa07), and 1891 1 (2001Vo24). $\omega\gamma=8.0$ eV 24 (1972Hu10). $\omega\gamma=6.0$ eV (1973Fa07). $\omega\gamma=9.0$ eV 27 (1976Me12). $\omega\gamma=4.1$ eV 5, $\Gamma_p=39$ eV 5 (1981Bi05). $\omega\gamma=2.6$ eV 7 (1996Ka21). $\omega\gamma=4.2$ eV 8 (2001Vo24), sum of $\omega\gamma$ for the resonances at E _p =1891 and 1893, $\Gamma_\gamma=0.70$ eV 14 (2001Vo24). J ^π : from $\sigma(\text{E}_p, \theta)$ in 1981Bi05. $\Gamma_p=120$ eV 11 (1981Bi05). $\omega\gamma=6.0$ eV (1973Fa07).
8209	1/2 ⁺	1893	J ^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21.
8216.3 10	5/2	1900.3 10	E _p (lab) (keV): weighted average of 1901.8 11 (1972Hu10), 1899.7 12 (1973Fa07), 1899.6 11 (1976Me12), and 1900 1 (2001Vo24). $\omega\gamma=6.2$ eV 19 (1972Hu10). $\omega\gamma=4.7$ eV (1973Fa07). $\omega\gamma=7.1$ eV 21 (1976Me12). $\omega\gamma=1.7$ eV 5 (1996Ka21). $\omega\gamma=3.3$ eV 5 (2001Vo24). $\Gamma_\gamma=0.55$ eV 8 (2001Vo24).
8242.4 12		1927.1 12	E _p (lab) (keV): weighted average of 1928.2 13 (1972Hu10), 1926.8 13 (1973Fa07), and 1926.5 12 (1976Me12). $\omega\gamma=1.9$ eV 6 (1972Hu10). $\omega\gamma=1.9$ eV (1973Fa07). $\omega\gamma=2.2$ eV 7 (1976Me12). $\omega\gamma=1.0$ eV 3 (2001Vo24). $\Gamma_\gamma=0.25$ eV 8 (2001Vo24). 20% of I γ transitions to unknown levels (1973Fa07).
8251? 5		1936 5	E _p (lab) (keV): from 1967Da12.
8268.9 5		1954.4 5	E _p (lab) (keV): weighted average of 1955.9 13 (1972Hu10), 1952 2 (1973Fa07), 1954.3 12 (1976Me12), and 1954.4 5 (1976Sp08). $\omega\gamma=1.8$ eV 5 (1972Hu10). $\omega\gamma=(1.9)$ eV (1973Fa07). $\omega\gamma=2.0$ eV 6 (1976Me12). $\omega\gamma=0.6$ eV 3 (1976Sp08).
8277.2 5		1963.0 5	E _p (lab) (keV): weighted average of 1964.1 14 (1972Hu10), 1962 2 (1973Fa07), 1962.8 12 (1976Me12), and 1962.9 5 (1976Sp08). $\omega\gamma=1.6$ eV 5 (1972Hu10). $\omega\gamma=(1.7)$ eV (1973Fa07). $\omega\gamma=1.5$ eV 5 (1976Me12). $\omega\gamma=0.7$ eV 4 (1976Sp08).
8282.4 5	(3/2 ⁻ , 5/2)	1968.3 5	E _p (lab) (keV): weighted average of 1970.0 13 (1972Hu10), 1971 2 (1973Fa07), 1967.8 12 (1976Me12), and 1967.9 5 (1976Sp08). $\omega\gamma=2.4$ eV 7 (1972Hu10). $\omega\gamma=(2.4)$ eV (1973Fa07). $\omega\gamma=1.3$ eV 4 (1976Me12). $\omega\gamma=0.4$ eV 2 (1976Sp08).
8287.82 30	1/2 ⁻	1973.89 30	J ^π : from $\sigma(\text{E}_p, \theta)$ in 1981Bi05, which also rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1975.6 13 (1972Hu10), 1973.8 12 (1976Me12), and 1973.80 30 (1976Sp08). $\omega\gamma=2.9$ eV 9 (1972Hu10). $\omega\gamma=1.6$ eV 5 (1976Me12). $\omega\gamma=0.9$ eV 5 (1976Sp08). $\Gamma_p=48$ eV 6 (1981Bi05).
8297.2 9		1983.5 9	E _p (lab) (keV): weighted average of 1983.7 5 (1976Sp08) and 1980 2 (1973Fa07). $\omega\gamma=(3.0)$ eV (1973Fa07).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
8298.2 5	3/2	1984.6 5	$\omega\gamma=0.6$ eV 3 (1976Sp08). J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1986.0 13 (1972Hu10), 1983.0 20 (1973Fa07), 1984.5 12 (1976Me12), and 1984.5 5 (1976Sp08). $\omega\gamma=4.0$ eV 12 (1972Hu10). $\omega\gamma=(1.7)$ eV (1973Fa07). $\omega\gamma=3.5$ eV 11 (1976Me12). $\omega\gamma=0.9$ eV 5 (1976Sp08).
8318.1 3		2005.1 3	E _p (lab) (keV): weighted average of 2006.7 12 (1972Hu10), 2004.6 13 (1973Fa07), 2006.0 13 (1976Me12), and 2005.0 3 (1976Sp08). $\omega\gamma=4.2$ eV 13 (1972Hu10). $\omega\gamma=4.6$ eV (1973Fa07). $\omega\gamma=4.2$ eV 13 (1976Me12). $\omega\gamma=2.7$ eV 5 (1976Sp08).
8323.1 13		2010.2 13	E _p (lab) (keV): weighted average of 2010.5 13 (1972Hu10) and 2009.9 13 (1976Me12). $\omega\gamma=1.0$ eV 3 (1972Hu10). $\omega\gamma=1.6$ eV 5 (1976Me12).
8346? 5		2034 5	E _p (lab) (keV): from 1967Da12.
8381.8 8	5/2 ⁺	2070.7 8	J ^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1967Ko23. E _p (lab) (keV): weighted average of 2073.5 12 (1972Hu10), 2070.4 5 (1976Sp08), and 2070 1 (2001Vo24). $\omega\gamma\approx 4$ eV (1967Ko23). $\omega\gamma=14.1$ eV 42 (1972Hu10). $\omega\gamma=6.1$ eV 12 (1976Sp08). $\omega\gamma=6.1$ eV 6 (2001Vo24). $\Gamma_\gamma=1.02$ eV 10 (2001Vo24).
8389.4 16		2078.5 16	E _p (lab) (keV): unweighted average of 2080.0 14 (1972Hu10) and 2076.9 5 (1976Sp08). $\omega\gamma=0.6$ eV 3 (1972Hu10). $\omega\gamma=0.4$ eV 2 (1976Sp08).
8402.9 5		2092.4 5	E _p (lab) (keV): from 1976Sp08. $\omega\gamma=0.7$ eV 4 (1976Sp08).
8405.7 14		2095.3 14	E _p (lab) (keV): unweighted average of 2096.6 12 (1972Hu10) and 2093.9 5 (1976Sp08). $\omega\gamma=6.4$ eV 19 (1972Hu10). $\omega\gamma=2.3$ eV 5 (1976Sp08).
8409.6 18		2099.3 18	E _p (lab) (keV): unweighted average of 2101.0 14 (1972Hu10) and 2097.5 5 (1976Sp08). $\omega\gamma=3.6$ eV 11 (1972Hu10). $\omega\gamma=1.4$ eV 3 (1976Sp08).
8416.7 5		2106.6 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8430.3 5		2120.6 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8434.8 5		2125.2 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8464.3 5		2155.6 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
8484.4 5	3/2 ⁺	2176.3 5	J ^π : from 2001Vo24 based on proposed isobaric analog to 3/2 ⁺ in ^{35}S . E _p (lab) (keV): weighted average of 2176.4 5 (1976Sp08) and 2176 1 (2001Vo24). $\omega\gamma=2.1$ eV 4 (1976Sp08). $\omega\gamma=2.1$ eV 4 (2001Vo24). $\Gamma_\gamma=0.36$ eV 6 (2001Vo24).
8486.1 5		2178.0 5	$\omega\gamma=1.9$ eV 4 (1976Sp08).
8506.5 5		2199.1 5	$\omega\gamma=0.3$ eV 2 (1976Sp08).
8514.3 5		2207.1 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8533.9 5		2227.3 5	$\omega\gamma=0.10$ eV 5 (1976Sp08).
8571.9 5		2266.4 5	$\omega\gamma=2.9$ eV 6 (1976Sp08).
8580.5 5		2275.3 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8585.8 5		2280.7 5	$\omega\gamma=0.10$ eV 5 (1976Sp08).
8590.0 5		2285.0 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8612.0 5		2307.7 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8613.7 5		2309.4 5	$\omega\gamma=2.4$ eV 5 (1976Sp08).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
8618.1 5	(3/2 ⁺ , 5/2 ⁺)	2314.0 5	J ^π : most probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33 . $\omega\gamma=3.8$ eV 8 (1976Sp08).
8630.1 3	7/2	2326.3 3	J ^π : spin=3/2, 7/2 from $\gamma(\theta)$ in 1976Sp09 ; 1976Sp09 ruled out 3/2 based on 4687 γ to 9/2 ⁺ , 3943 level. Other: most probable spin=3/2 from $\gamma(\theta)$ in 1970Ko33 . $\omega\gamma=2.7$ eV 5 (1976Sp08).
8640	5/2	2336	J ^π : probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33 . E _p (lab) (keV): from 1970Ko33 .
8641.4 5	3/2, 5/2	2338.0 5	J ^π : spin=3/2 from $\gamma(\theta)$ in 1976Sp09 ; spin=5/2 is also possible. $\omega\gamma=0.9$ eV 5 (1976Sp08).
8686.2 5		2384.1 5	$\omega\gamma=2.2$ eV 4 (1976Sp08).
8688.3 5		2386.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8696.9 5		2395.1 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8706.3 5		2404.8 5	$\omega\gamma=0.2$ eV 1 (1976Sp08).
8717.7 5		2416.5 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8750.9 5		2450.7 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8767.0 5		2467.3 5	$\omega\gamma=0.3$ eV 2 (1976Sp08).
8772.9 5		2473.4 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8779.8 5		2480.5 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8787.2 5		2488.1 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
8798.4 5		2499.6 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8820.9 5		2522.8 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8824.2 5		2526.2 5	$\omega\gamma=3.7$ eV 7 (1976Sp08).
8829.3 5		2531.4 5	$\omega\gamma=2.1$ eV 4 (1976Sp08).
8833.1 5		2535.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8838.1 5	7/2 ⁻	2540.5 5	J ^π : 5/2 ⁻ , 7/2 ⁻ from $\sigma(\text{E}_p, \theta)$ in 1981Bi05 . Spin=7/2 from $\gamma(\theta)$ in 1981Bi05 . $\omega\gamma=1.5$ eV 3 (1976Sp08).
			$\Gamma_p=4$ eV 1 (1981Bi05).
8856.1 5		2559.0 5	$\omega\gamma=4.6$ eV 9 (1976Sp08).
8868.6 5		2571.9 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8884.1 5		2587.9 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
8886.1 5		2589.9 5	$\omega\gamma=1.4$ eV 3 (1976Sp08).
8893.3 5	(5/2, 7/2) ⁺	2597.3 5	J ^π : from 2001Vo24 with no supporting arguments. E _p (lab) (keV): weighted average of 2597.4 5 (1976Sp08) and 2597 1 (2001Vo24). $\omega\gamma=1.2$ eV 2 (1976Sp08).
			$\omega\gamma=1.2$ eV 4 (2001Vo24).
			$\Gamma_\gamma=0.20$ eV 6 (2001Vo24).
8904.8 5		2609.2 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8906.8 5	5/2 ⁺	2611.2 5	J ^π : from 2001Vo24 with no supporting arguments. E _p (lab) (keV): weighted average of 2611.3 5 (1976Sp08) and 2611 1 (2001Vo24). $\omega\gamma=1.5$ eV 3 (1976Sp08).
			$\omega\gamma=1.5$ eV 3 (2001Vo24).
			$\Gamma_\gamma=0.25$ eV 5 (2001Vo24).
8919.8 5		2624.6 5	$\omega\gamma=1.1$ eV 2 (1976Sp08).
8933.3 5		2638.5 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8953.0 5	3/2	2658.8 5	J ^π : from $\gamma(\theta)$ in 1976Sp09 . $\omega\gamma=2.4$ eV 5 (1976Sp08).
8957.9 5		2663.8 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
8981.9 5		2688.6 5	$\omega\gamma=2.2$ eV 4 (1976Sp08).
8984.2 5		2690.9 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8988.4 5		2695.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8992.4 5		2699.4 5	$\omega\gamma=2.3$ eV 5 (1976Sp08).
8996.7 5		2703.8 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
9001.1 5		2708.3 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	$E_p(\text{lab})$ (keV)	Comments
9019.3 5		3.1 keV 3	2727.1 5	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma=0.8$ eV 4 (1976Sp08).
9024.5 5			2732.4 5	$\omega\gamma=1.1$ eV 2 (1976Sp08).
9029.9 5			2738.0 5	$\omega\gamma=2.6$ eV 5 (1976Sp08).
9033.1 5			2741.3 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
9038.3 5			2746.6 5	$\omega\gamma=0.8$ eV 4 (1976Sp08).
9048.4 5			2757.0 5	$\omega\gamma=3.2$ eV 6 (1976Sp08).
9081.4 4	5/2 ⁺	65 eV 20	2791.0 4	J^π : from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1976Sp09. $T_{1/2}$: width $\Gamma_p=62$ eV 20, $\Gamma_\gamma=2.3$ eV 4 (1976Sp09, resonant absorption). $E_p(\text{lab})$ (keV): weighted average of 2791.0 4 (1976Sp08) and 2791 1 (2001Vo24). $\omega\gamma=16.0$ eV 32 (1976Sp08). $\omega\gamma=16.3$ eV 4 (1996Ka21). $\omega\gamma=16.0$ eV 3 (2001Vo24). $\omega\gamma=13.4$ eV 22 (1976Sp09). $\Gamma_p=59$ eV 7 (1981Bi05). $\Gamma_\gamma=2.7$ eV 5 (2001Vo24).
9088.5 5			2798.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
9099.3 5			2809.4 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
9100.4 5			2810.6 5	$\omega\gamma=2.1$ eV 4 (1976Sp08).
9107.4 5			2817.8 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
9110.0 5			2820.5 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
9124.0 5			2834.9 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
9135.1 5		2.0 keV 4	2846.3 5	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma=0.2$ eV 1 (1976Sp08).
9138.3 5			2849.6 5	$\omega\gamma=1.9$ eV 4 (1976Sp08).
9155.7 5			2867.5 5	$\omega\gamma=2.0$ eV 4 (1976Sp08).
9157.1 5	5/2		2869.0 5	J^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21. $\omega\gamma=6.7$ eV 13 (1976Sp08). $\omega\gamma=6.4$ eV 1 (1996Ka21).
9163.3 5			2875.3 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
9184.2 5			2896.9 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
9188.8 5			2901.6 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
9194.1 5			2907.1 5	$\omega\gamma=4.0$ eV 8 (1976Sp08).

[†] For resonances above $S(\text{p})(^{35}\text{Cl})=6370.81$ 4 (2021Wa16), $E(\text{level})=E_{\text{c.m.}}+S(\text{p})(^{35}\text{Cl})$, where $E_{\text{c.m.}}=E_p(\text{lab})\times m(^{34}\text{S})/[m_p+m(^{34}\text{S})]$. Weighted average is taken when multiple $E_p(\text{lab})$ measurements are available. For levels below 8.27 MeV ($E_p<1.95$ MeV), $E_p(\text{lab})$ is mainly from 1972Hu10, 1973Fa07, and 1976Me12; for levels above 8.27 MeV ($E_p>1.95$ MeV), $E_p(\text{lab})$ is mainly from 1976Sp08, unless otherwise noted.

[‡] From the comparison of experimental angular distributions of γ -rays with theoretical predictions or deduced from γ -feedings, unless otherwise noted.

 $\gamma(^{35}\text{Cl})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
1219.39	1/2 ⁺	1219.37	100	0	3/2 ⁺			E_γ : other: 1220 40 (1963Ha32). $A_2=-0.03$ 3, $A_4=-0.002$ 50 (1971Wi13). $A_2=-0.01$ 2, $A_4=+0.005$ 24 (1967Ta08).
1763.14	5/2 ⁺	543.75 1763.09	<0.2 100	1219.39 0	1/2 ⁺ 3/2 ⁺	D+Q	-0.42 17	I_γ : from 1972Hu10 and 1976Me12. I_γ : from 1972Hu10 and 1976Me12. Mult., δ : from $\gamma\gamma(\theta)$ in 1967Ko20 and 1967Ko23.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								Other: $+2.8 < \delta < +4.0$ (1971Wi13). $A_2 = -0.24$ 6, $A_4 = -0.08$ 7 (1971Wi13). $A_2 = +0.25$ 3, $A_4 = -0.49$ 8 (1967Ko20,1967Ko23,1970Ko33). I_γ : weighted average of 8 1 (1972Hu10) and 9.4 10 (1976Me12).
2645.70	(5/2,7/2)	882.55	8.7 10	1763.14	5/2 ⁺			I_γ : weighted average of 92 1 (1972Hu10) and 90.6 10 (1976Me12). Mult., δ : D+Q with $\delta = -0.62 + 38-36$ from $\gamma(\theta)$ for $J_1=3/2$ (1967Ko20). $A_2 = +0.124$ 17, $A_4 = -0.230$ 24 (1967Ko20,1967Ko23).
		1426.28	<2	1219.39	1/2 ⁺			E_γ : other: 933 7 (1970Ko33). I_γ : weighted average of 13 2 (1972Hu10) and 13.0 10 (1976Me12). $A_2 = -0.05$ 38 (1971Wi13). E_γ : other: 1475 7 (1970Ko33). I_γ : weighted average of 6 2 (1972Hu10) and 8.0 10 (1976Me12). Other: 66 14 from 1963Ha32.
		2645.59	91.3 10	0	3/2 ⁺			I_γ : weighted average of 81 3 (1972Hu10) and 79 2 (1976Me12). Other: 34 9 from 1963Ha32.
2693.9	(3/2,5/2)	930.8	13.0 10	1763.14	5/2 ⁺			δ : $+0.1 < \delta < +0.26$ or $-35 < \delta < -7$ for $J=3/2$, $\delta < -5$ for $J=5/2$ (1971Wi13). $A_2 = +0.20$ 10, $A_4 = -0.16$ 11 (1971Wi13). I_γ : from 1972Hu10. Other: $I_\gamma < 2$ (1976Me12).
		1474.5	7.6 10	1219.39	1/2 ⁺			I_γ : other: $I_\gamma \leq 5$ (1972Hu10). E_γ : other: 3006 7 (1970Ko33). I_γ : from 1972Hu10 and 1976Me12.
		2693.8	80 2	0	3/2 ⁺	D+Q		Mult., δ : from $\gamma\gamma(\theta)$ in 1967Ko20. $A_2 = +0.45$ 18, $A_4 = +0.05$ 19 (1971Wi13). $A_2 = -0.025$ 11, $A_4 = -0.084$ 16 (1967Ko20).
3002.75	5/2	1239.59	≤ 1	1763.14	5/2 ⁺			I_γ : from 1971Pr11. Other: <3 (1976Me12).
		1783.31	<3	1219.39	1/2 ⁺			I_γ : from 1971Pr11. Others: 11 2 (1972Hu10), 8.9 20 (1976Me12). $A_2 = +0.31$ 8, $A_4 = -0.06$ 8 (1971Wi13).
		3002.61	100	0	3/2 ⁺	D+Q	-0.22 1	I_γ : from 1971Pr11. Others: ≤ 0.2 (1972Hu10), <0.2 (1976Me12). Mult., δ : from $\gamma(\theta)$ in 1971Pr11, which states that the anomalous $\gamma(\theta)$ can be understood if it occurs by a mixture of M2 and E1 radiation. In view of the weakness of this transition, such a mixture is not unreasonable.
3162.99	7/2 ⁻	160.24	1.7 2	3002.75	5/2			E_γ : other: 3163 7 (1970Ko33). I_γ : from 1971Pr11. Others: 89 2 (1972Hu10), 91 2 (1976Me12). Mult.: Q+O from $\gamma(\theta)$ in 1966Az01, 1966Wa09, 1967Wa22, and 1967Ta08;
		517.29	8 1	2645.70	(5/2,7/2)			
		1399.82	0.30 4	1763.14	5/2 ⁺	E1+M2	+0.44 12	
		1943.54	<0.2	1219.39	1/2 ⁺			
		3162.84	90 1	0	3/2 ⁺	M2+E3	+0.25 4	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								magnetic character from $\gamma(\text{pol})$ in 1966Wa09 and 1967Wa22 . δ : from 1966Az01 . Others: +0.25 5 (1967Ta08); +0.16 1 (1966Wa09,1967Wa22). Asymmetry ratio $R=1.426$ 7, $N_{90}/N_0=1.25$ 11 (1971Wi13). $\text{POL}=-0.75$ 20, $\Delta\pi=\text{yes}$ (1967Wa22). $\text{POL}=-1.27$ 10, $\Delta\pi=\text{yes}$ (1966Wa09). $0.00<\text{POL}<0.384$ (1966Az01,1967Ta08). $A_2=+0.40$ 8, $A_4=-0.23$ 8 (1971Wi13). $A_2=+0.25$ 3, $A_4=+0.02$ 5 (1967Ta08). $A_2=+0.55$ 3, $A_4=-0.008$ 38 (1966Az01).
3918.48	3/2 ⁺	1272.76 2155.27	<3 19.5 23	2645.70 (5/2,7/2) 1763.14 5/2 ⁺				I_γ : weighted average of 17 3 (1972Hu10), 25 3 (1976Me12), and 18.0 21 (1976Sp09). I_γ : weighted average of 83 3 (1972Hu10), 75 3 (1976Me12), and 82.0 10 (1976Sp09). Mult.: D+Q from $\gamma(\theta)$ in 1976Sp09 ; $\delta=-0.21$ 2 or +20 +10-5 (1976Sp09). E1+M2 ruled by RUL.
		3918.25	81.5 15	0 3/2 ⁺	3/2 ⁺	M1+E2		
3942.9		1297.2	8 2	2645.70 (5/2,7/2)				$A_2=+0.07$ 2, $A_4=-0.01$ 3 (1976Sp09). I_γ : weighted average of 7 3 (1972Hu10) and 8 2 (1976Me12). I_γ : weighted average of 93 3 (1972Hu10) and 92 2 (1976Me12).
		2179.7	92 2	1763.14 5/2 ⁺				
		2723.4 ^d	<6	1219.39 1/2 ⁺				
		3942.7 ^d	<10	0 3/2 ⁺				
3967.3		1273.4	2.1 5	2693.9 (3/2,5/2)				I_γ : weighted average of 5 3 (1972Hu10) and 2.0 5 (1976Me12).
		1321.6 ^d	<2	2645.70 (5/2,7/2)				
		2204.1	≤ 1	1763.14 5/2 ⁺				I_γ : from 1972Hu10 . Other: <2 (1976Me12).
		2747.8	79 5	1219.39 1/2 ⁺				I_γ : weighted average of 74 5 (1972Hu10) and 83 5 (1976Me12).
		3967.1	19 3	0 3/2 ⁺				I_γ : weighted average of 21 3 (1972Hu10) and 15 5 (1976Me12).
4059.2		1365.3	1.2 4	2693.9 (3/2,5/2)				E_γ : other: 2296 7 (1970Ko33). I_γ : other: 3 2 (1972Hu10). E_γ : other: 2838 7 (1970Ko33). I_γ : other: 97 2 (1972Hu10). 45 5 (1971Wi13).
		1413.5 ^d	<1	2645.70 (5/2,7/2)				E_γ : other: 4058 7 (1970Ko33). I_γ : other: 20 2 (1971Wi13).
		2296.0	4.8 15	1763.14 5/2 ⁺				I_γ : weighted average of 50 20 (1972Hu10), 44 7 (1976Me12), and 48 3 (1976Sp09). Mult., δ : from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.56$ 3, $A_4=+0.05$ 4 (1976Sp09).
		2839.7	94 2	1219.39 1/2 ⁺				
		4059.0	<2	0 3/2 ⁺				
4113.7	7/2	2350.5	47 3	1763.14 5/2 ⁺		D+Q	-0.16 2	I_γ : weighted average of 50 20 (1972Hu10), 44 7 (1976Me12), and 48 3 (1976Sp09). Mult., δ : from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.56$ 3, $A_4=+0.05$ 4 (1976Sp09).
		2894.2 ^d	<10	1219.39 1/2 ⁺				
		4113.4	53 3	0 3/2 ⁺		Q(+O)	0.00 10	I_γ : weighted average of 50 20 (1972Hu10) and 56 7 (1976Me12), and 52 3 (1976Sp09). Mult., δ : Q(+O) from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.46$ 25, $A_4=-0.23$ 25 (1976Sp09).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)**

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
4173.45		1479.5	26 8	2693.9	(3/2,5/2)			I_γ : weighted average of 27 8 (1972Hu10) and 25 8 (1976Me12).
		1527.71	<10	2645.70	(5/2,7/2)			I_γ : weighted average of 18 6 (1972Hu10) and 15 5 (1976Me12).
		2410.22	16 5	1763.14	5/2 ⁺			I_γ : from 1972Hu10. Other: <4 (1976Me12); 32 3 (1971Wi13).
		2953.93	≤3	1219.39	1/2 ⁺			I_γ : weighted average of 55 10 (1972Hu10) and 60 10 (1976Me12).
		4173.18	58 10	0	3/2 ⁺			$A_2=-0.01$ 3, $A_4=-0.006$ 40 (1967Ta08).
4177.9	3/2	1175.1	<0.5	3002.75	5/2			I_γ : weighted average of 7 2 (1972Hu10) and 10 3 (1976Me12).
		1484.0	8 2	2693.9	(3/2,5/2)			I_γ : other ≤1 (1972Hu10).
		1532.2	<0.5	2645.70	(5/2,7/2)			I_γ : weighted average of 31 5 (1972Hu10) and 30 5 (1976Me12).
		2414.7	<1	1763.14	5/2 ⁺	D+Q	+0.11 4	Mult., δ : from $\gamma(\theta)$ in 1967Wa22.
		2958.4	31 5	1219.39	1/2 ⁺			I_γ : weighted average of 62 6 (1972Hu10) and 60 5 (1976Me12).
		4177.6	61 5	0	3/2 ⁺	D+Q	+0.06 3	Mult., δ : from $\gamma(\theta)$ in 1967Wa22.
								POL=+0.12 31 (1967Wa22).
4347.76	5/2,9/2	1184.75	70 2	3162.99	7/2 ⁻			I_γ : weighted average of 70 5 (1972Hu10), 72 5 (1976Me12), and 69 2 (1976Sp09).
		1344.98 ^d	<4	3002.75	5/2			I_γ : weighted average of 30 5 (1972Hu10), 28 5 (1976Me12), and 31 2 (1976Sp09).
		1653.8 ^d	<4	2693.9	(3/2,5/2)			Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09.
		1702.02	31 2	2645.70	(5/2,7/2)	D+Q	-0.018 12	$A_2=-0.34$ 3, $A_4=-0.00$ 4 (1976Sp09).
		2584.52	≤3	1763.14	5/2 ⁺			I_γ : from 1972Hu10. Other: <5 (1976Me12).
		3128.22 ^d	<4	1219.39	1/2 ⁺			I_γ : from 1972Hu10. Other: <10 (1976Me12).
		4347.47	≤5	0	3/2 ⁺			I_γ : other: 100 (1972Hu10).
4624.4		4624.1	85 10	0	3/2 ⁺			I_γ : other: 53 10 (1972Hu10) and 47% of I_γ to unknown levels (1972Hu10).
4769.86		1606.83	65 10	3162.99	7/2 ⁻			
		1767.06	35 10	3002.75	5/2			
		2075.9	<10	2693.9	(3/2,5/2)			
		3550.28	<10	1219.39	1/2 ⁺			
		4769.51	<10	0	3/2 ⁺			
4839.10		1836.30	<7	3002.75	5/2			I_γ : weighted average of 50 20 (1972Hu10) and 60 5 (1976Me12).
		2145.1	<4	2693.9	(3/2,5/2)			Other: 48 4 (1976Sp09).
		2193.33	<3	2645.70	(5/2,7/2)			
		3075.82	59 5	1763.14	5/2 ⁺			I_γ : weighted average of 50 20 (1972Hu10) and 40 5 (1976Me12).
		3619.51	<7	1219.39	1/2 ⁺			Other: 18 3 (1976Sp09).
		4838.74	41 5	0	3/2 ⁺			
4854.4		3634.8	75 5	1219.39	1/2 ⁺			
		4854.0	25 5	0	3/2 ⁺			
4880.96		1878.16	9 3	3002.75	5/2			

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
4880.96		2187.0	<10	2693.9	(3/2,5/2)			
		2235.18	29 5	2645.70	(5/2,7/2)			I_γ : other: 30 10 (1972Hu10).
		3117.67	62 5	1763.14	5/2 ⁺			I_γ : other: 70 10 (1972Hu10).
		3661.36 ^d	<4	1219.39	1/2 ⁺			
		4880.60	<5	0	3/2 ⁺			
5010.1		836.6	<25 ^a	4173.45				
		950.9	<10 ^a	4059.2				
		1847.1	<50 ^a	3162.99	7/2 ⁻			
		2007.3	<20 ^a	3002.75	5/2			
		2316.1	<25 ^a	2693.9	(3/2,5/2)			
		2364.3	<5	2645.70	(5/2,7/2)			
		3246.8	<10	1763.14	5/2 ⁺			
		3790.5	<7	1219.39	1/2 ⁺			
		5009.7	100	0	3/2 ⁺			I_γ : from 1976Me12 and 1972Hu10.
5163.34	7/2	2000.29	40 5	3162.99	7/2 ⁻	D+Q	+0.44 20	I_γ : weighted average of 40 20 (1972Hu10), 36 5 (1976Me12), and 54 10 (1976Sp09). Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09. $A_2=+0.52$ 5, $A_4=-0.02$ 6 (1976Sp09).
		2160.52	10 4	3002.75	5/2			I_γ : weighted average of 10 5 (1972Hu10), 10 4 (1976Me12), and 10 5 (1976Sp09).
		3400.02	43 5	1763.14	5/2 ⁺	D(+Q)	0.00 3	I_γ : weighted average of 50 20 (1972Hu10), 44 5 (1976Me12), and 36 10 (1976Sp09). Mult., δ : D(+Q) from $\gamma(\theta)$ in 1976Sp09. $A_2=-0.23$ 6, $A_4=-0.08$ 7 (1976Sp09).
		3943.71	<10	1219.39	1/2 ⁺			
		5162.93	<10	0	3/2 ⁺			
5215.8		205.7	<23 ^a	5010.1				
		1042.3	<60 ^a	4173.45				
		1156.6	<10 ^a	4059.2				
		2052.8	<15 ^a	3162.99	7/2 ⁻			
		2213.0	<25 ^a	3002.75	5/2			
		2521.8	<15 ^a	2693.9	(3/2,5/2)			
		2570.0	<15 ^a	2645.70	(5/2,7/2)			
		3452.5	<6	1763.14	5/2 ⁺			
		3996.2	<4	1219.39	1/2 ⁺			
		5215.4	100	0	3/2 ⁺			
5403.5		2400.7	25 5	3002.75	5/2			I_γ : other: 25 10 (1972Hu10).
		3640.2	<10	1763.14	5/2 ⁺			I_γ : other: 25 10 from 1972Hu10.
		4183.8	55 10	1219.39	1/2 ⁺			
		5403.1	<7	0	3/2 ⁺			
5586		2940	40 10	2645.70	(5/2,7/2)			I_γ : weighted average of 40 20 (1972Hu10) and 40 10 (1976Me12).
		3823	60 10	1763.14	5/2 ⁺			I_γ : weighted average of 60 20 (1972Hu10) and 60 10 (1976Me12).
		4366	<10	1219.39	1/2 ⁺			
		5586	<10	0	3/2 ⁺			
5599.7		3836.3	27 10	1763.14	5/2 ⁺			I_γ : weighted average of 20 10 (1972Hu10) and 33 10 (1976Me12).
		5599.2	74 10	0	3/2 ⁺			I_γ : weighted average of 80 10 (1972Hu10) and 67 10 (1976Me12).
5646		2483	100	3162.99	7/2 ⁻			I_γ : from 1972Hu10, 1976Me12, 1971Pr11.
		2643		3002.75	5/2			I_γ : 80 from 2001Vo24 only; not observed by others.
		3883		1763.14	5/2 ⁺			I_γ : 6 from 2001Vo24 only; not observed by others.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
5646		5646	<8	0	3/2 ⁺			
5655		2652	80 5	3002.75	5/2			
		3892	6 2	1763.14	5/2 ⁺			
		4435	<6	1219.39	1/2 ⁺			
5683		1505	60 10	4177.9	3/2			
5723.52		3029.5	24 3	2693.9	(3/2,5/2)			I_γ : from 1976Sp09.
		3960.14	38 4	1763.14	5/2 ⁺			I_γ : from 1976Sp09.
5758		1580	30 10	4177.9	3/2			
		4538	<6	1219.39	1/2 ⁺			
		5758	45 10	0	3/2 ⁺			
5806		5806	100	0	3/2 ⁺			I_γ : from 1972Hu10 and 1976Me12.
6106.2		4342.8	6 2	1763.14	5/2 ⁺			
		4886.4	36 10	1219.39	1/2 ⁺			I_γ : other: 40 10 (1972Hu10).
		6105.6	58 10	0	3/2 ⁺			I_γ : other: 60 10 (1972Hu10).
6181		6180	45 10	0	3/2 ⁺			
6492		6491	100	0	3/2 ⁺			
7066.2		1902.8	0.9	5163.34	7/2			
		2892.6	0.5	4173.45				
		2952.4	0.6	4113.7	7/2			
		3006.9	1.4	4059.2				I_γ : other: 1.0 5 from 1972Hu10.
		3903.0	6	3162.99	7/2 ⁻			I_γ : other: 6.0 6 from 1972Hu10.
		4063.2	3	3002.75	5/2			I_γ : other: 2 1 from 1972Hu10.
		4372.0	5	2693.9	(3/2,5/2)			I_γ : other: 6.0 6 from 1972Hu10.
		4420.2	1.6	2645.70	(5/2,7/2)			I_γ : other: 3.0 15 from 1972Hu10.
		5302.6	14	1763.14	5/2 ⁺			I_γ : other: 16.0 16 from 1972Hu10.
		5846.3	18	1219.39	1/2 ⁺			I_γ : other: 18.0 18 from 1972Hu10.
		7065.4	49	0	3/2 ⁺			I_γ : other: 48.0 48 from 1972Hu10.
7103.4	3/2	2925.4	1.5	4177.9	3/2			
		3184.8	1.5	3918.48	3/2 ⁺			I_γ : other: 2 1 (1972Hu10).
		4100.4	3	3002.75	5/2			I_γ : other: 3.0 15 from 1972Hu10.
		4409.2	11	2693.9	(3/2,5/2)	D(+Q)	0.00 3	I_γ , Mult., δ : from 1976Me12. Other: 13.0 13 (1972Hu10).
								$A_2=+0.34$ 7, $A_4=+0.03$ 11 (1976Me12).
		5339.8	4	1763.14	5/2 ⁺			I_γ : other: 4 2 from 1972Hu10.
		5883.5	64	1219.39	1/2 ⁺	D(+Q)	0.00 3	I_γ , Mult., δ : from 1976Me12. Other: 67.0 67 (1972Hu10).
								$A_2=-0.53$ 8, $A_4=+0.01$ 11 (1976Me12).
		7102.6	15	0	3/2 ⁺			I_γ : other: 11.0 11 (1972Hu10).
								$A_2=+0.54$ 5, $A_4=-0.01$ 8 (1976Me12).
7178.6	1/2	1072.4	3	6106.2				I_γ : other: 4 2 (1972Hu10).
		1775.1	1.7	5403.5				
		2339.4	1.5	4839.10				I_γ : other: 2 (2001Vo24).
		3005.0	0.8	4173.45				I_γ : other: 1 (2001Vo24).
		3119.3	22	4059.2				I_γ : others: 22 (2001Vo24), 24.0 24 (1972Hu10).
								$A_2=-0.02$ 4, $A_4=+0.10$ 66 (1976Me12).
								$A_2=-0.01$ 4, $A_4=+0.09$ 24 (2001Vo24).
		3211.1	9	3967.3				I_γ : others: 9 (2001Vo24), 10.0 10 (1972Hu10).
		3260.0	3	3918.48	3/2 ⁺			I_γ : others: 3 (2001Vo24), 2 1 (1972Hu10).
		4484.4	4	2693.9	(3/2,5/2)			I_γ : others: 4 (2001Vo24), 4 2 (1972Hu10).
		5958.7	17	1219.39	1/2 ⁺			I_γ : others: 17 (2001Vo24), 16.0 16 (1972Hu10).
								$A_2=-0.01$ 3, $A_4=+0.06$ 7 (2001Vo24).
								$A_2=-0.01$ 5, $A_4=+0.07$ 8 (1976Me12).
		7177.8	38	0	3/2 ⁺			I_γ : others: 38 (2001Vo24), 40.0 40 (1972Hu10).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								$A_2=-0.01$ 6, $A_4=-0.02$ 7 (1976Me12). $A_2=-0.02$ 4, $A_4=-0.01$ 5 (2001Vo24).
7185		1969	<0.5 ^a	5215.8				
		2175	<0.5 ^a	5010.1				
		3011	<5 ^a	4173.45				
		3126	26 ^a 6	4059.2				
		4022	<3 ^a	3162.99	7/2 ⁻			
		4182	<3 ^a	3002.75	5/2			
		4491	2 ^a 1	2693.9	(3/2,5/2)			
		4539	6 ^a 2	2645.70	(5/2,7/2)			
		5421	4 ^a 3	1763.14	5/2 ⁺			
		5965	16 ^a 7	1219.39	1/2 ⁺			
7194.6	1/2,3/2	7184	46 ^a 10	0	3/2 ⁺			
		2184.4	0.8	5010.1				I_γ : other: 1.0 5 (1972Hu10).
		2340.1	2	4854.4				I_γ : other: 2 1 (1972Hu10).
		2355.4	1.8	4839.10				
		3016.6	16	4177.9	3/2			I_γ : other: 15.0 15 (1972Hu10).
		3135.3	3	4059.2				I_γ : other: 3.0 15 (1972Hu10).
		3227.1	1.8	3967.3				I_γ : other: 2 1 (1972Hu10).
		3276.0	4	3918.48	3/2 ⁺			I_γ : other: 5.0 25 (1972Hu10).
		4500.4	1.1	2693.9	(3/2,5/2)			
		5431.0	0.5	1763.14	5/2 ⁺			
7225.5	5/2	5974.7	67	1219.39	1/2 ⁺			I_γ : other: 72.0 72 (1972Hu10).
		7193.8	2	0	3/2 ⁺			
		1119.3	1.8	6106.2				I_γ : other: 3.0 15 (1972Hu10).
		1420		5806				I_γ : 1.0 5 (1972Hu10).
		1543	2.0	5683				I_γ : other: 2 1 (1972Hu10).
		1571	0.8	5655				
		1822.0	0.9	5403.5				I_γ : other: 1.0 5 (1972Hu10).
		2009.6	7	5215.8				I_γ : other: 6.0 30 (1972Hu10).
		3047.5	8	4177.9	3/2			I_γ : other: 7.0 35 (1972Hu10).
		3166.2	2	4059.2				
7234.0	5/2 ⁺	3306.9	1.5	3918.48	3/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		4222.5		3002.75	5/2			I_γ : 3.0 15 (1972Hu10).
		4531.3	10	2693.9	(3/2,5/2)	D+Q	+0.36 4	I_γ , Mult., δ : from 1976Me12. Other: 10.0 10 (1972Hu10).
								$A_2=+0.43$ 8, $A_4=-0.21$ 9 (1976Me12).
		5461.9	7	1763.14	5/2 ⁺	D+Q	-0.36 7	I_γ , Mult., δ : from 1976Me12. Other: 8.0 8 (1972Hu10).
								$A_2=+0.02$ 10, $A_4=-0.09$ 17 (1976Me12).
		7224.7	59	0	3/2 ⁺	D+Q	+0.36 4	I_γ , Mult., δ : from 1976Me12. Other: 58.0 58 (1972Hu10).
								$A_2=+0.40$ 7, $A_4=+0.04$ 12 (1976Me12).
		1579	0.2	5655				
		3056.0		4177.9	3/2			I_γ : 1.0 5 (1972Hu10).
7272.7	(1/2,3/2)	4539.8	1.8	2693.9	(3/2,5/2)			I_γ : other: 2 1 (1972Hu10).
		4588.0	3	2645.70	(5/2,7/2)			I_γ : other: 2 1 (1972Hu10).
		5470.4	0.2	1763.14	5/2 ⁺			
		6014.1	1.8	1219.39	1/2 ⁺			I_γ : other: 2 1 (1972Hu10).
		7233.2	93	0	3/2 ⁺	D+Q	-0.10	I_γ : other: 93.0 93 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1960An06.
		1515	1.2	5758				I_γ : other: 1.5 8 (1972Hu10).
		1869.2	0.2	5403.5				I_γ : other: 0.5 3 (1972Hu10).
		2262.5	1.4	5010.1				I_γ : other: 1.5 8 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7272.7	(1/2,3/2)	2433.5	1.0	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3099.1	0.4	4173.45				I_γ : other: 0.5 3 (1972Hu10).
		3213.3	0.9	4059.2				I_γ : other: 1.0 5 (1972Hu10).
		3305.2	0.8	3967.3				I_γ : other: 1.0 5 from 1972Hu10.
		4578.5	1.1	2693.9	(3/2,5/2)			I_γ : other: 1.0 5 (1972Hu10).
		6052.8	24	1219.39	1/2 ⁺			I_γ : other: 23.0 23 (1972Hu10).
		7271.9	69	0	3/2 ⁺			I_γ : other: 69.0 69 (1972Hu10).
7362.0	3/2	1707	0.3	5655				I_γ : other: 0.5 3 (1972Hu10).
		2351.8	0.6	5010.1				I_γ : other: 0.5 3 (1972Hu10).
		2522.8	1.0	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3183.9	0.5	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		3302.6	2	4059.2				I_γ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).
		3394.5	3	3967.3				I_γ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).
		3443.3	0.8	3918.48	3/2 ⁺			I_γ : other: 0.5 3 (1972Hu10).
		4359.0	0.3	3002.75	5/2			I_γ : other: 0.5 3 (1972Hu10).
		4667.8	0.5	2693.9	(3/2,5/2)			
		5598.4	8	1763.14	5/2 ⁺	D(+Q)		I_γ : others: 10.0 10 (1972Hu10); 9 (1971Wi13).
								Mult., δ : $\delta=-0.01$ 10 or $\delta>+2.5$ (1977Ko34).
		6142.0	73	1219.39	1/2 ⁺	D+Q		$A_2=-0.20$ 24, $A_4=+0.92$ 25 (1977Ko34).
								I_γ : others: 70.0 70 (1972Hu10); 72 (1971Wi13).
								Mult., δ : $\delta=-0.05$ 3 or $+2.5$ 3 (1977Ko34); $\delta=-20$ or $+0.22<\delta<+0.26$ (1971Wi13).
								$A_2=-0.05$ 3, $A_4=-0.04$ 5 (1971Wi13).
								$A_2=-0.25$ 11, $A_4=+0.31$ 11 (1977Ko34).
		7361.2	10	0	3/2 ⁺	D+Q		I_γ : others: 10.0 10 (1972Hu10); 11 (1971Wi13).
								Mult., δ : $-0.16<\delta<-0.12$ or $+7.5<\delta<+10$ (1971Wi13).
								$A_2=+0.19$ 3, $A_4=-0.08$ 5 (1971Wi13).
7396.0	7/2	3048.1	11	4347.76	5/2,9/2			I_γ : other: 10.0 10 (1972Hu10).
		3282.1	8	4113.7	7/2			I_γ : other: 9.0 9 (1972Hu10).
		3452.9	5	3942.9				I_γ : other: 8.0 8 (1972Hu10).
		4232.7	47	3162.99	7/2 ⁻	D+Q	+0.28 25	I_γ : other: 49.0 49 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		4393.0	14	3002.75	5/2	D(+Q)	+0.011 18	I_γ : other: 14.0 14 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
								I_γ : other: 10.0 10 (1972Hu10).
		4750.0	13	2645.70	(5/2,7/2)			
		5632.4	2	1763.14	5/2 ⁺			
7451.1	3/2	3532.4	5.0 25	3918.48	3/2 ⁺			I_γ : from 1972Hu10.
		4448.1	10 1	3002.75	5/2			I_γ : from 1972Hu10.
		4756.9	8.0 8	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.
		5687.5	4 2	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
		6231.1	73 7	1219.39	1/2 ⁺			I_γ : from 1972Hu10.
7501.1		3558.0	25 3	3942.9				I_γ : from 1972Hu10.
		4498.0	13.0 13	3002.75	5/2			I_γ : from 1972Hu10.
		4855.0	55 6	2645.70	(5/2,7/2)			I_γ : from 1972Hu10.
		5737.5	7.0 7	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
7502.9		1396.7 ^c	7 ^c	6106.2				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
Other: 10.0 10 for 7503 level only (1972Hu10).								
7502.9		1903.1 ^c	2 ^c	5599.7				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		2492.7 ^c	2 ^c	5010.1				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3324.8 ^c	3 ^c	4177.9	3/2			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3443.5 ^c	50 ^c	4059.2				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 70.0 70 for 7503 level only (1972Hu10).
		3559.8 ^c	2 ^c	3942.9				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4339.6 ^c	4 ^c	3162.99	7/2 ⁻			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4499.8 ^c	4 ^c	3002.75	5/2			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4856.8 ^c	7 ^c	2645.70	(5/2,7/2)			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		5739.3 ^c	0.7 ^c	1763.14	5/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		6282.9 ^c	18 ^c	1219.39	1/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 20.0 20 for 7503 level only (1972Hu10).
		7502.0 ^c	0.3 ^c	0	3/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
7518.8	7/2	1919.0	2 1	5599.7				I_γ : from 1972Hu10.
		1933	3.0 15	5586				I_γ : from 1972Hu10.
		3170.9	2 1	4347.76	5/2,9/2			I_γ : from 1972Hu10.
		3575.7	2 1	3942.9				I_γ : from 1972Hu10.
		4355.5	68 7	3162.99	7/2 ⁻	D(+Q)	+0.1 2	I_γ : from 1972Hu10. Mult., δ : from 1977Ko35.
		4872.7	3.0 15	2645.70	(5/2,7/2)			I_γ : from 1972Hu10.
		5755.2	20 2	1763.14	5/2 ⁺	D+Q	+0.098 22	I_γ : from 1972Hu10. Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
7548.9	7/2 ⁻	1903	0.55 6	5646				I_γ : from 1971Pr11. Others: 0.6 (1976Me12); 0.5 3 (1972Hu10); 1 (2001Vo24,2002Vo17).
		1963	0.1	5586				I_γ : other: 0.2 1 (1972Hu10).
		2778.9	1.4 1	4769.86				I_γ : from 1971Pr11. Others: 1.3 (1976Me12); 1.5 8 (1972Hu10).
		4385.6	94 1	3162.99	7/2 ⁻	M1+E2	-0.07 2	I_γ : from 1971Pr11. Others: 95 (1976Me12); 95.0 95 (1972Hu10); ≥ 93 (1971Wi13). Mult.: D+Q from $\gamma(\theta)$ in 1967Wa22 and 1966Wa09; magnetic character from $\gamma(\text{pol})$ in 1967Wa22.
δ : from 1966Wa09 and 1967Wa22. Others: $\delta=+0.08$ 10 (1977Ko35), $\delta=-0.03$ 3 from 2001Vo24 and 2002Vo17.								
$N_{90}/N_0=0.72$ 6 (1971Wi13).								
POL=+0.76 6, $\Delta\pi=\text{no}$ (1966Wa09).								
POL=+0.93 10, $\Delta\pi=\text{no}$ (1967Wa22).								
0.021<POL<0.813 (1966Az01).								

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								$A_2=+0.42\ 3$, $A_4=0.00\ 3$ (2001Vo24,2002Vo17).
								$A_2=+0.42\ 3$, $A_4=-0.001\ 27$ (1966Az01).
								$A_2=+0.42\ 3$, $A_4=+0.00\ 3$ (1967Ta08).
7548.9	$7/2^-$	4545.8	2.0 1	3002.75	5/2	D(+Q)	+0.6 4	I_γ : from 1971Pr11. Others: 2 (1976Me12); 1.8 9 (1972Hu10); 1.3 (1971Wi13).
		4902.8	0.68 7	2645.70	(5/2,7/2)			Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		5785.3	0.33 4	1763.14	5/2 ⁺			I_γ : from 1971Pr11. Others: 0.5 (1976Me12); 0.5 3 (1972Hu10); ≤ 5 (1971Wi13).
		7548.0	0.27 3	0	3/2 ⁺	Q(+O)	+0.1 2	I_γ : from 1971Pr11. Others: 0.3 (1976Me12); 0.3 2 (1972Hu10); ≈ 0.4 (1971Wi13).
								I_γ : from 1971Pr11. Others: 0.2 (1976Me12); 0.2 1 (1972Hu10); ≈ 0.3 (1971Wi13).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
7561.3	$1/2,3/2$	3383.2	0.9	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		3501.9	6	4059.2				I_γ : other: 5.0 25 (1972Hu10).
		3593.8	0.5	3967.3				
		3642.6	1.6	3918.48	3/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		4867.0	22	2693.9	(3/2,5/2)			I_γ : other: 21.0 21 (1972Hu10).
		6341.3	35	1219.39	1/2 ⁺			$A_2=+0.04\ 3$, $A_4=-0.05\ 3$ (1976Me12).
		7560.4	34	0	3/2 ⁺			I_γ : other: 36.0 36 (1972Hu10).
								$A_2=-0.02\ 2$, $A_4=-0.05\ 3$ (1976Me12).
								I_γ : other: 36.0 36 (1972Hu10).
								$A_2=+0.01\ 2$, $A_4=-0.03\ 3$ (1976Me12).
7600.8	$5/2$	2437.4	1.9	5163.34	7/2			
		2719.7	2	4880.96				
		2761.6	2	4839.10				
		2830.8	1.3	4769.86				
		2976.3	1.8	4624.4				
		3422.7	3	4177.9	3/2			I_γ : other: 2 1 (1972Hu10).
		3427.2		4173.45				I_γ : 2 1 from 1972Hu10.
		3541.4	2	4059.2				I_γ : other: 1.5 8 (1972Hu10).
		4437.5	5	3162.99	7/2 ⁻			I_γ : others: 4 2 (1972Hu10); 8 (1971Wi13).
								Mult., δ : Q+O with $\delta=-0.25\ 15$ for $J_i=3/2$ (1977Ko34).
								$A_2=+0.19\ 2$, $A_4=+0.08\ 2$ (1977Ko34).
								$A_2=+0.35\ 32$, $A_4=-0.44\ 32$ (1971Wi13).
		4597.7	8	3002.75	5/2	D+Q	+0.50 22	I_γ : others: 9.0 9 (1972Hu10); 12 (1971Wi13).
								Mult., δ : from 1971Wi13; $\delta=+0.85\ 42$ for $J_i=3/2$ (1977Ko34).
								$A_2=+0.75\ 7$, $A_4=-0.06\ 7$ (1971Wi13).
								$A_2=+0.53\ 21$, $A_4=-0.04\ 25$ (1976Me12).
								$A_2=+0.42\ 3$, $A_4=-0.10\ 3$ (1977Ko34).
		4906.5	18	2693.9	(3/2,5/2)	D+Q	+0.32 7	I_γ : others: 19.0 19 (1972Hu10); 24 (1971Wi13).
								δ : from 1971Wi13; other: $\delta=-5.3\ 9$ or $+0.01\ 5$ for $J_i=3/2$ (1977Ko34).
								$A_2=+0.33\ 1$, $A_4=-0.05\ 1$ (1977Ko34).
								$A_2=+0.18\ 4$, $A_4=-0.02\ 5$ (1976Me12).
								$A_2=+0.26\ 10$, $A_4=-0.05\ 10$ (1971Wi13).
		4954.7	1.9	2645.70	(5/2,7/2)			I_γ : other: 1.5 8 (1972Hu10).
		5837.1	19	1763.14	5/2 ⁺	D+Q	+0.31 7	I_γ : others: 20 2 (1971Wi13), 2 1 (1972Hu10); 25 (1971Wi13).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7685.5	$3/2^-$	1927		5758				I_γ : 0.5 3 (1972Hu10).
		2085.7	2	5599.7				I_γ : other: 2 1 (1972Hu10).
		2281.9	2	5403.5				I_γ : other: 2 1 (1972Hu10).
		2469.6	0.7	5215.8				I_γ : other: 0.5 3 (1972Hu10).
		2675.3	0.6	5010.1				
		3507.4	3	4177.9	$3/2$			I_γ : other: 2 1 (1972Hu10).
		3511.9	4	4173.45				I_γ : other: 5.0 25 (1972Hu10).
								$A_2=+0.01$ 14, $A_4=+0.23$ 16 (1977Ko34).
		3626.1	6	4059.2				I_γ : other: 6.0 6 (1972Hu10).
		4682.4	10	3002.75	$5/2$	D(+Q)	-0.01 5	I_γ : other: 9.0 9 (1972Hu10).
								Mult., δ : other: $\delta=-2.4$ +20-4 (1976Me12);
								$\delta=+0.01$ 7 (1977Ko34).
								$A_2=-0.04$ 6, $A_4=-0.09$ 7 (1976Me12).
								$A_2=-0.06$ 12, $A_4=+0.08$ 13 (1977Ko34).
7693.9		5921.8	0.7	1763.14	$5/2^+$			I_γ : other: 1.0 5 (1972Hu10).
		6465.5	1	1219.39	$1/2^+$			I_γ , Mult., δ : from 1976Me12. Other: $I_\gamma=72.0$ 72
		7684.6	70	0	$3/2^+$	D+Q	+6.0 7	(1972Hu10); $\delta<-3.5$ for $J_i=3/2$ (1977Ko34).
								$A_2=+0.25$ 1, $A_4=-0.03$ 1 (1976Me12).
								$A_2=+0.18$ 11, $A_4=-0.04$ 11 (1977Ko34).
7706.4	$5/2^+$	2094.1	1.0 5	5599.7				I_γ : from 1972Hu10.
		3634.5	1.0 5	4059.2				I_γ : from 1972Hu10.
		4690.8	2.0 10	3002.75	$5/2$			I_γ : from 1972Hu10.
		7693.0	96 10	0	$3/2^+$			I_γ : from 1972Hu10.
		2051	4	5655				I_γ : other: 4 2 (1972Hu10).
								Mult., δ : D+Q with $\delta\leq-2$ or ≥ 13 for
								$J_f(5655)=5/2$ (1977Ko34), but $J_f(5655)=(3/2)^+$
								from the Adopted Levels.
								$A_2=+0.14$ 15, $A_4=-0.39$ 17 (1977Ko34).
		2825.3	0.4	4880.96				
		2867.2	1.3	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3081.9	0.8	4624.4				
		3528.3	2	4177.9	$3/2$			
		3592.5	2	4113.7	$7/2$			I_γ : other: 1.0 5 (1972Hu10).
		3787.7	0.6	3918.48	$3/2^+$			
		4703.3	1	3002.75	$5/2$			I_γ : other: 1.0 5 (1972Hu10).
		5060.3	4	2645.70	$(5/2, 7/2)$	D+Q	-1.01 10	I_γ : other: 4 2 (1972Hu10).
								Mult., δ : or ≥ 13 (1977Ko34).
7744.8	$7/2$							$A_2=-0.03$ 6, $A_4=+0.08$ 8 (1976Me12).
								$A_2=-0.18$ 17, $A_4=+0.41$ 18 (1977Ko34).
		5942.7	0.4	1763.14	$5/2^+$			I_γ : other: 1.0 5 (1972Hu10).
		6486.4	0.5	1219.39	$1/2^+$			I_γ : other: 88.0 88 (1972Hu10).
		7705.5	83	0	$3/2^+$	D(+Q)	+0.1 1	Mult., δ : from $\gamma(\theta)$ in 1977Ko34.
								$A_2=-0.54$ 3, $A_4=-0.07$ 3 (1976Me12).
								$A_2=-0.63$ 4, $A_4=+0.15$ 3 (1977Ko34).
		2863.7	20	4880.96				I_γ : other: 19.0 19 (1972Hu10).
		2974.8	2	4769.86				I_γ : other: 1.0 5 (1972Hu10).
		3630.9	6	4113.7	$7/2$			I_γ : other: 5.0 25 (1972Hu10).
		3801.7	10	3942.9		D(+Q)	-0.16 17	I_γ : other: 12.0 12 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		4581.5	16	3162.99	$7/2^-$	D+Q	+0.18 8	Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		5098.7	43	2645.70	$(5/2, 7/2)$	D(+Q)	+0.1 30	Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		5981.1	2.4	1763.14	$5/2^+$			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
7744.8	7/2	6524.8	0.6	1219.39	1/2 ⁺	
7777.1		2122	0.5	5655		
		2613.7	1.1	5163.34	7/2	
		2937.9	2	4839.10		I_γ : other: 2 1 (1972Hu10).
		3007.1	3	4769.86		I_γ : other: 2 1 (1972Hu10).
		3599.0	7	4177.9	3/2	I_γ : other: 7.0 7 (1972Hu10).
		3717.7	1.1	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		4613.8	2	3162.99	7/2 ⁻	
		4774.0	4	3002.75	5/2	I_γ : other: 4 2 (1972Hu10).
		5082.8	9	2693.9	(3/2,5/2)	I_γ : other: 7.0 7 (1972Hu10).
		5131.0	1.3	2645.70	(5/2,7/2)	I_γ : other: 2 1 (1972Hu10).
		6013.4	40	1763.14	5/2 ⁺	I_γ : other: 46.0 46 (1972Hu10).
		6557.1	11	1219.39	1/2 ⁺	I_γ : other: 13.0 13 (1972Hu10).
		7776.2	18	0	3/2 ⁺	I_γ : other: 16.0 16 (1972Hu10).
7781.7	5/2 ⁻	1675.5	1.4	6106.2		
		2181.9		5599.7		I_γ : 1.0 5 from 1972Hu10.
		2378.1		5403.5		I_γ : 1.0 5 from 1972Hu10.
		2565.8	1.5	5215.8		I_γ : other: 1.0 5 (1972Hu10).
		3011.7	7	4769.86		I_γ : other: 7.0 7 (1972Hu10).
		3433.8	2	4347.76	5/2,9/2	I_γ : other: 2 1 (1972Hu10).
		3608.1	36	4173.45		I_γ : other: 40.0 40 (1972Hu10).
		3667.8	6	4113.7	7/2	I_γ : other: 3.0 15 (1972Hu10).
		3722.3	1.2	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		3863.0	11	3918.48	3/2 ⁺	I_γ : other: 10.0 10 (1972Hu10).
		4618.4	5	3162.99	7/2 ⁻	I_γ : other: 5.0 25 (1972Hu10).
		5135.6	8	2645.70	(5/2,7/2)	I_γ : other: 9.0 9 (1972Hu10).
		6018.0	18	1763.14	5/2 ⁺	I_γ : other: 20.0 20 (1972Hu10).
		6561.7	1.2	1219.39	1/2 ⁺	
		7780.8	1.7	0	3/2 ⁺	
7797.1	1/2	2039		5758		I_γ : 0.5 3 from 1972Hu10.
		2142	1.3	5655		I_γ : other: 1.0 5 (1972Hu10).
						$A_2=+0.03$ 8 (1988Va06).
		2393.5	2	5403.5		I_γ : other: 2 1 (1972Hu10).
						$A_2=+0.04$ 5 (1988Va06).
		2786.9	2	5010.1		I_γ : other: 2 1 (1972Hu10).
						$A_2=-0.07$ 5 (1988Va06).
		2942.6	1.3	4854.4		I_γ : other: 0.5 3 (1972Hu10).
		2957.9	1.1	4839.10		
		3737.7	0.3	4059.2		
		6577.1	9	1219.39	1/2 ⁺	I_γ : other: 9.0 9 (1972Hu10).
						δ : $\delta(Q/D)=+0.21$ 4 or -1.9 2 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.
						$A_2=-0.02$ 6, $A_4=-0.14$ 7 (1976Me12).
						$A_2=-0.03$ 3 (1988Va06).
		7796.2	83	0	3/2 ⁺	I_γ : other: 85.0 85 (1972Hu10).
						δ : $\delta(Q/D)=-0.33$ 2 or -19 $+5-10$ for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.
						$A_2=-0.06$ 2, $A_4=-0.03$ 2 (1976Me12).
						$A_2=+0.006$ 14 (1988Va06).
7837.2	3/2	1656 ^c	0.5 ^c	6181		I_γ : other: 1 from 2001Vo24. Initial level is 7837 and 7839 doublet (1976Me12).
		2237.4 ^c	0.5 ^c	5599.7		I_γ : other: 1 (2001Vo24). Initial level is 7837 and 7839 doublet (1976Me12).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	E_γ [†]	I_γ [#]	E_f	J_f^π	Mult. ^b	δ ^b	Comments
7837.2	2621.3 ^c	1.4 ^c	5215.8				I_γ : other: 1 (2001Vo24); 1.0 5 (1972Hu10). Initial level is 7837 and 7839 doublet (1976Me12).
	3659.1 ^c	28 ^c	4177.9	3/2	M1+E2	-0.05 3	I_γ : other: 28 (2001Vo24); 35.0 35 (1972Hu10). Initial level is 7837 and 7839 doublet (1976Me12). Mult., δ : D+Q from $\gamma(\theta)$ in 1967Wa22; magnetic character from $\gamma(\text{pol})$ in 1967Wa22. Other: $\delta=-0.04$ 3 or +0.27 5 from 2001Vo24. $A_2=+0.17$ 1, $A_4=-0.01$ 2 (2001Vo24). POL=+0.66 18, $\Delta\pi=\text{no}$ (1967Wa22).
	3663.5 ^c	3 ^c	4173.45				I_γ : others: 3 (2001Vo24); 3.0 15 (1972Hu10); 34 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12).
	3777.8 ^c	2 ^c	4059.2				$A_2=+0.17$ 1, $A_4=-0.005$ 13 (1967Ta08). I_γ : others: 2 (2001Vo24); 2 1 (1972Hu10); 4 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12).
	3894.1 ^c	0.5 ^c	3942.9				I_γ : other: 1 (2001Vo24). Initial level is 7837 and 7839 doublet (1976Me12).
	3918.5 ^c	0.2 ^c	3918.48	3/2 ⁺			I_γ : others: 4 (2001Vo24); 4 2 (1972Hu10); 4 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12).
	4834.1 ^c	4 ^c	3002.75	5/2			I_γ : others: 2 (2001Vo24); ≤ 1 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12).
	6073.5 ^c	1.9 ^c	1763.14	5/2 ⁺			I_γ : others: 37 (2001Vo24); 27.0 27 (1972Hu10); 33 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12).
	6617.1 ^c	37 ^c	1219.39	1/2 ⁺	D+Q	+0.06 3	Mult., δ : from $\gamma(\theta)$ in 1967Wa22. POL=+0.17 45 (1967Wa22). $A_2=-0.46$ 4, $A_4=+0.04$ 6 (1967Ta08).
	7836.3 ^c	21 ^c	0	3/2 ⁺	D(+Q)	+0.02 3	I_γ : others: 21 (2001Vo24); 27.0 27 (1972Hu10); 25 (1971Wi13). Initial level is 7837 and 7839 doublet (1976Me12). Mult., δ : from $\gamma(\theta)$ in 1967Wa22. Other: 0.00 3 from 2001Vo24. POL=-1.1 8 (1967Wa22). $A_2=+0.41$ 3, $A_4=+0.012$ 40 (1967Ta08). $A_2=+0.40$ 2, $A_4=+0.01$ 4 (2001Vo24).
7839.0	1658 ^c	0.5 ^c	6181				
	2239.2 ^c	0.5 ^c	5599.7				
	2623.1 ^c	1.4 ^c	5215.8				
	3660.9 ^c	28 ^c	4177.9	3/2			
	3665.3 ^c	3 ^c	4173.45				
	3779.6 ^c	2 ^c	4059.2				
	3895.9 ^c	0.5 ^c	3942.9				
	3920.3 ^c	0.2 ^c	3918.48	3/2 ⁺			
	4835.9 ^c	4 ^c	3002.75	5/2			
	6075.3 ^c	1.9 ^c	1763.14	5/2 ⁺			
	6618.9 ^c	37 ^c	1219.39	1/2 ⁺			
	7838.1 ^c	21 ^c	0	3/2 ⁺			
7868.7	2652.8	1.0	5215.8				
	2858.5	3	5010.1				I_γ : other: 4 2 (1972Hu10).
	3695.0	5	4173.45				I_γ : other: 4 2 (1972Hu10).
	6105.0	17	1763.14	5/2 ⁺			I_γ : other: 18.0 18 (1972Hu10).
	6648.6	13	1219.39	1/2 ⁺			I_γ : other: 12.0 12 (1972Hu10).
	7867.8	61	0	3/2 ⁺			I_γ : other: 62.0 62 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
7880.8	3/2,5/2	2226	12	5655		I_γ : other: 9.0 9 (1972Hu10).
		3041.6	1.7	4839.10		I_γ : other: 3.0 15 (1972Hu10).
		3256.2	2	4624.4		I_γ : other: 1.0 5 (1972Hu10).
		3702.7	3	4177.9	3/2	
		3766.9	1.3	4113.7	7/2	I_γ : other: 4 2 (1972Hu10).
		3821.4	2	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		3962.1	2	3918.48	3/2 ⁺	I_γ : other: 2 1 (1972Hu10).
		4877.7	11	3002.75	5/2	I_γ : other: 10.0 10 (1972Hu10).
		5186.5	25	2693.9	(3/2,5/2)	I_γ : other: 30.0 30 (1972Hu10).
						$A_2=+0.50$ 3, $A_4=+0.09$ 4 (1976Me12).
		6117.1	30	1763.14	5/2 ⁺	I_γ : other: 31.0 31 (1972Hu10).
						$A_2=+0.08$ 3, $A_4=-0.03$ 4 (1976Me12).
		7879.9	10	0	3/2 ⁺	I_γ : other: 9.0 9 (1972Hu10).
						$A_2=-0.06$ 4, $A_4=+0.04$ 5 (1976Me12).
7899.2	(3/2 ⁻ ,5/2)	2253	1.5	5646		
		2735.8	0.9	5163.34	7/2	
		3018.1	0.8	4880.96		
		3274.6	0.5	4624.4		
		3721.1	8	4177.9	3/2	I_γ : other: 8.0 8 (1972Hu10).
		3725.5	8	4173.45		I_γ : other: 6.0 6 (1972Hu10).
		3839.8	3	4059.2		I_γ : other: 2 1 (1972Hu10).
		4735.9	4	3162.99	7/2 ⁻	I_γ : other: 4 2 (1972Hu10).
		5204.9	1.3	2693.9	(3/2,5/2)	
		6135.5	68	1763.14	5/2 ⁺	I_γ : other: 77.0 77 (1972Hu10).
		7898.2	4	0	3/2 ⁺	I_γ : other: 3.0 15 (1972Hu10).
7923.3		3084.1	1.4	4839.10		
		3153.3	2	4769.86		
		3298.7	1.9	4624.4		
		3745.2	3	4177.9	3/2	I_γ : other: 2 1 (1972Hu10).
		3749.6	5	4173.45		I_γ : other: 6.0 6 (1972Hu10).
		3863.9	1.7	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		3955.8	1.5	3967.3		I_γ : other: 1.0 5 (1972Hu10).
		4004.6	1.2	3918.48	3/2 ⁺	I_γ : other: 1.0 5 (1972Hu10).
		4920.2	0.9	3002.75	5/2	
		5229.0	11	2693.9	(3/2,5/2)	I_γ : other: 12.0 12 (1972Hu10).
		5277.2	0.9	2645.70	(5/2,7/2)	
		6159.6	1.5	1763.14	5/2 ⁺	I_γ : other: 1.0 5 (1972Hu10).
		6703.2	27	1219.39	1/2 ⁺	I_γ : other: 28.0 28 (1972Hu10).
		7922.3	41	0	3/2 ⁺	I_γ : other: 48.0 48 (1972Hu10).
7970.3		2324	3	5646		I_γ : other: 3.0 15 (1972Hu10).
		2384	3	5586		
		2754.4	3	5215.8		I_γ : other: 3.0 15 (1972Hu10).
		3089.2	10	4880.96		I_γ : other: 9.0 9 (1972Hu10).
		3200.3	5	4769.86		I_γ : other: 6.0 6 (1972Hu10).
		3622.3	8	4347.76	5/2,9/2	I_γ : other: 8.0 8 (1972Hu10).
		3792.2	6	4177.9	3/2	I_γ : other: 6.0 6 (1972Hu10).
		3856.4	6	4113.7	7/2	I_γ : other: 2 1 (1972Hu10).
		4807.0	10	3162.99	7/2 ⁻	I_γ : other: 13.0 13 (1972Hu10).
		4967.2	5	3002.75	5/2	I_γ : other: 5.0 25 (1972Hu10).
		5324.2	20	2645.70	(5/2,7/2)	I_γ : other: 22.0 22 (1972Hu10).
		6206.6	20	1763.14	5/2 ⁺	I_γ : other: 21.0 21 (1972Hu10).
		6750.2	1	1219.39	1/2 ⁺	
7987.8	3/2	1881.6	0.5	6106.2		I_γ : other: 1.0 5 (1972Hu10).
		2333	1.7	5655		I_γ : other: 1.0 5 (1972Hu10).
		3809.7	<0.8	4177.9	3/2	

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)**

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^{<i>b</i>}	δ^b	Comments
7987.8	3/2	3814.1	<0.8	4173.45				
		3928.4	1	4059.2				
		6767.7	60	1219.39	1/2 ⁺	D+Q	+0.21 4	I_γ : other: 62.0 62 (1972Hu10). Mult., δ : or -3.1 4 (1976Me12). $A_2=-0.07$ 2, $A_4=-0.05$ 3 (1976Me12).
		7986.8	36	0	3/2 ⁺	D+Q	-0.36 2	I_γ : other: 36.0 36 (1972Hu10). Mult., δ : or -11 3 (1976Me12). $A_2=-0.10$ 2, $A_4=-0.06$ 2 (1976Me12).
7995.6	5/2	2190	0.6	5806				
		2395.8	1.1	5599.7				
		2592.0	3	5403.5				I_γ : other: 3.0 15 (1972Hu10).
		2832.1	1.5	5163.34	7/2			
		3817.5	4	4177.9	3/2			I_γ : other: 3.0 15 (1972Hu10).
		3936.2	1.1	4059.2				
		4832.3	7	3162.99	7/2 ⁻			I_γ : other: 6.0 6 (1972Hu10).
		4992.5	2	3002.75	5/2			
		5301.3	0.2	2693.9	(3/2,5/2)			
		5349.5	0.6	2645.70	(5/2,7/2)			
		6231.9	0.9	1763.14	5/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		7994.6	78	0	3/2 ⁺			I_γ : other: 87.0 87 (1972Hu10).
8001.0	(7/2,9/2 ⁺)	2355	1.8	5646				I_γ : other: 3.0 15 (1972Hu10).
		2785.1	0.8	5215.8				
		3119.9	0.7	4880.96				
		3231.0	8	4769.86				I_γ : other: 8.0 8 (1972Hu10).
		3653.0	1.9	4347.76	5/2,9/2			
		3827.3	10	4173.45				I_γ : other: 8.0 8 (1972Hu10).
		4057.9	3	3942.9				I_γ : other: 2 1 (1972Hu10).
		4837.7	0.8	3162.99	7/2 ⁻			
		5354.9	1.6	2645.70	(5/2,7/2)			I_γ : other: 2 1 (1972Hu10).
		6237.3	70	1763.14	5/2 ⁺			I_γ : other: 76.0 76 (1972Hu10).
		8000.0	1.4	0	3/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		8005.0	5/2 ⁺	2350	1.8	5655		
3123.9	0.6			4880.96				
3165.8	3			4839.10				
3826.9 ^c	1.3 ^c			4177.9	3/2			I_γ : total I_γ for γ decays to 4173 and 4178 levels.
3831.3 ^c	1.3 ^c			4173.45				I_γ : total I_γ for γ decays to 4173 and 4178 levels.
4841.7	15			3162.99	7/2 ⁻			I_γ : other: 17.0 17 (1972Hu10). $A_2=-0.084$ 14, $A_4=-0.0041$ 15 (1981Bi05).
5001.9	2			3002.75	5/2			I_γ : other: 3.0 15 (1972Hu10).
5310.7	1.7			2693.9	(3/2,5/2)			
5358.9	4			2645.70	(5/2,7/2)			
6241.3	15			1763.14	5/2 ⁺			I_γ : other: 20.0 20 (1972Hu10). $A_2=-0.11$ 2, $A_4=-0.09$ 3 (1976Me12).
6784.9	0.6			1219.39	1/2 ⁺			
8004.0	55			0	3/2 ⁺			I_γ : other: 60.0 60 (1972Hu10). $A_2=-0.03$ 1, $A_4=-0.02$ 2 (1976Me12).
8035.7		2230	6.0 6	5806				I_γ : from 1972Hu10.
		3857.6	15.0 15	4177.9	3/2			I_γ : from 1972Hu10.
		3976.3	11.0 11	4059.2				I_γ : from 1972Hu10.
		5341.4	5.0 25	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.
		6272.0	5.0 25	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
		6815.6	56 6	1219.39	1/2 ⁺			I_γ : from 1972Hu10.
		8034.7	2.0 10	0	3/2 ⁺			I_γ : from 1972Hu10.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8038.5	3/2	3183.9	0.6	4854.4				
		3199.2	1.6	4839.10				
		3413.9	3	4624.4				
		3860.4	1.6	4177.9	3/2			
		3979.1	3	4059.2				I_γ : other: 4 2 (1972Hu10).
		4071.0	0.8	3967.3				
		4119.8	1.4	3918.48	3/2 ⁺			
		5035.4	4	3002.75	5/2			I_γ : other: 3.0 15 (1972Hu10).
		5344.2	57	2693.9	(3/2,5/2)	D+Q	-0.20 2	I_γ : other: 61.0 61 (1972Hu10). Mult., δ : or +19 +10-5 (1976Me12). $A_2=+0.08$ 2, $A_4=-0.03$ 2 (1976Me12).
		6274.8	15	1763.14	5/2 ⁺			I_γ : other: 17.0 17 (1972Hu10). $A_2=+0.24$ 2, $A_4=+0.01$ 3 (1976Me12).
		6818.4	4	1219.39	1/2 ⁺			I_γ : other: 6.0 6 (1972Hu10).
		8037.5	8	0	3/2 ⁺	D+Q	-0.13 4	I_γ : other: 9.0 9 (1972Hu10). Mult., δ : or +8.1 +33-18 (1976Me12). $A_2=+0.20$ 3, $A_4=-0.08$ 3 (1976Me12).
8075.9		2430	11	5646				I_γ : other: 10.0 10 (1972Hu10).
		2490	11	5586				I_γ : other: 11.0 11 (1972Hu10).
		2860.0	8	5215.8				I_γ : other: 7.0 7 (1972Hu10).
		2912.4		5163.34	7/2			I_γ : 1.0 5 from 1972Hu10.
		3194.8		4880.96				I_γ : 2 1 from 1972Hu10.
		3305.9	4	4769.86				I_γ : other: 4 2 (1972Hu10).
		3897.8	8	4177.9	3/2			I_γ : other: 7.0 7 (1972Hu10).
		3962.0	2	4113.7	7/2			I_γ : other: 2 1 (1972Hu10).
		5072.8	2	3002.75	5/2			I_γ : other: 2 1 (1972Hu10).
		6312.2	56	1763.14	5/2 ⁺			I_γ : other: 53.0 53 (1972Hu10).
		8074.9		0	3/2 ⁺			I_γ : 1.0 5 from 1972Hu10.
8096.1	(5/2,7/2 ⁺)	2290	2	5806				I_γ : other: 1.0 5 (1972Hu10).
		2450	5	5646				I_γ : other: 5.0 25 (1972Hu10).
		2496.3	5	5599.7				I_γ : other: 6.0 6 (1972Hu10).
		2880.2	1.3	5215.8				I_γ : other: 1.0 5 (1972Hu10).
		2932.6	2	5163.34	7/2			I_γ : other: 2 1 (1972Hu10).
		3256.8	3	4839.10				I_γ : other: 3.0 15 (1972Hu10).
		3326.1	2	4769.86				I_γ : other: 1.0 5 (1972Hu10).
		3918.0	1.9	4177.9	3/2			
		4036.7	2	4059.2				I_γ : other: 2 1 (1972Hu10).
		4177.4	1.8	3918.48	3/2 ⁺			I_γ : other: 3.0 15 (1972Hu10).
		4932.7	5	3162.99	7/2 ⁻			I_γ : other: 5.0 25 (1972Hu10).
		5401.8	5	2693.9	(3/2,5/2)			I_γ : other: 5.0 25 (1972Hu10).
		5449.9	17	2645.70	(5/2,7/2)			I_γ : other: 17.0 17 (1972Hu10).
		6332.4	8	1763.14	5/2 ⁺			I_γ : other: 9.0 9 (1972Hu10).
8106.4	3/2	8095.1	39	0	3/2 ⁺			I_γ : other: 40.0 40 (1972Hu10).
		2506.6	0.5	5599.7				I_γ : other: 0.5 3 (1972Hu10).
		3251.8	0.6	4854.4				I_γ : other: 0.5 3 (1972Hu10).
		3267.1	0.5	4839.10				I_γ : other: 0.5 3 (1972Hu10).
		3928.3	1.6	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		4047.0	1.3	4059.2				I_γ : other: 0.5 3 (1972Hu10).
		4138.8	3	3967.3				I_γ : other: 3.0 15 (1972Hu10).
		5103.3	9	3002.75	5/2			I_γ : other: 8.0 8 (1972Hu10).
		5412.1	9	2693.9	(3/2,5/2)	D+Q	+4.9 10	I_γ : other: 8.0 8 (1972Hu10). Mult., δ : -0.05 4 (1976Me12). $A_2=+0.32$ 4, $A_4=-0.08$ 5 (1976Me12).
		6342.6	3	1763.14	5/2 ⁺			I_γ : other: 3.0 15 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)**

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^{<i>b</i>}	δ^b	Comments
8106.4	3/2	6886.3	35	1219.39	1/2 ⁺			I_γ : other: 35.0 35 (1972Hu10). $A_2=-0.46$ 2, $A_4=-0.05$ 3 (1976Me12).
		8105.4	37	0	3/2 ⁺	D+Q	+1.2 <i>I</i>	I_γ : other: 40.0 40 (1972Hu10). Mult., δ : or -0.46 4 (1976Me12). $A_2=+0.83$ 3, $A_4=+0.01$ 2 (1976Me12). I_γ : other: 6.0 6 (1972Hu10).
8113.3	(1/2 ⁺ , 3/2, 5/2 ⁺)	1621	6	6492				
		2007.0	6	6106.2				
		2307	1.1	5806				
		2458	0.9	5655				
		2513.5		5599.7				I_γ : 3.0 15 (1972Hu10).
		2897.4	1.8	5215.8				
		3939.6	7	4173.45				I_γ : other: 5.0 25 (1972Hu10).
		4053.9	1.3	4059.2				
		4145.7	5	3967.3				I_γ : other: 5.0 25 (1972Hu10).
		5110.2	1.9	3002.75	5/2			
		5419.0	13	2693.9	(3/2, 5/2)			I_γ : other: 17.0 17 (1972Hu10).
		6349.5	5	1763.14	5/2 ⁺			I_γ : other: 6.0 6 (1972Hu10).
		6893.2	32	1219.39	1/2 ⁺			I_γ : other: 32.0 32 (1972Hu10).
		8112.3	25	0	3/2 ⁺			I_γ : other: 26.0 26 (1972Hu10).
8147.2	3/2 ⁻	2464	1.0	5683				I_γ : other: 1.0 5 (1972Hu10).
		3137.0	3	5010.1				
		3969.1	8	4177.9	3/2			
		4087.7	3	4059.2				
		4179.6	2	3967.3				
		4228.5	3	3918.48	3/2 ⁺			I_γ : other: 3.0 15 (1972Hu10).
		5452.8	3	2693.9	(3/2, 5/2)			I_γ : other: 4 2 (1972Hu10).
		6927.1	15	1219.39	1/2 ⁺			I_γ : other: 20.0 20 (1972Hu10).
		8146.2	62	0	3/2 ⁺			I_γ : other: 75.0 75 (1972Hu10).
		8156.8	(5/2 ⁺ , 7/2 ⁻)	2511	4	5646		
2571	8			5586				I_γ : other: 7.0 7 (1972Hu10).
2940.9	6			5215.8				I_γ : other: 8.0 8 (1972Hu10).
3386.8	1.0			4769.86				
3978.7	5			4177.9	3/2			I_γ : other: 8.0 8 (1972Hu10).
4042.9	7			4113.7	7/2			I_γ : other: 7.0 7 (1972Hu10).
4097.3	1			4059.2				
4213.6	2			3942.9				
4993.4	5			3162.99	7/2 ⁻			I_γ : other: 9.0 9 (1972Hu10).
5153.6	6			3002.75	5/2			I_γ : other: 4 2 (1972Hu10).
5510.6	51			2645.70	(5/2, 7/2)			I_γ : other: 50.0 50 (1972Hu10).
6393.0	1			1763.14	5/2 ⁺			
8155.8	3			0	3/2 ⁺			I_γ : other: 4 2 (1972Hu10).
8179.4				3554.8	9.0 9	4624.4		
		5016.0	8.0 8	3162.99	7/2 ⁻			I_γ : from 1972Hu10.
		5176.2	4.0 20	3002.75	5/2			I_γ : from 1972Hu10.
		5485.0	3.0 15	2693.9	(3/2, 5/2)			I_γ : from 1972Hu10.
		6415.6	68 7	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
		8178.4	8.0 8	0	3/2 ⁺			I_γ : from 1972Hu10.
8208.2	5/2 ⁺	2553	0.7	5655				I_γ : other: 1 (2001Vo24).
		3327.1	0.8	4880.96				I_γ : other: 1 (2001Vo24).
		3368.9	0.4	4839.10				I_γ : other: 1 (2001Vo24).
		4148.7	1.5	4059.2				I_γ : others: 2 (2001Vo24); 2 <i>I</i> (1972Hu10).
		4289.4		3918.48	3/2 ⁺			I_γ : 1.0 5 (1972Hu10).
		5205.0	1.1	3002.75	5/2			I_γ : others: 1 (2001Vo24); 1.0 5 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8208.2	5/2 ⁺	5513.8 6444.4	0.5 14	2693.9 1763.14	(3/2,5/2) 5/2 ⁺	D+Q	+0.33 4	I _γ : other: 1 (2001Vo24); 1.0 5 (1972Hu10). I _γ : other: 14 (2001Vo24); 16.0 16 (1972Hu10). Mult.,δ: from 2001Vo24. A ₂ =+0.51 13, A ₄ =-0.21 11, A ₆ =+0.0004 (1996Ka21). A ₂ =+0.51 13, A ₄ =-0.21 11 (2001Vo24). I _γ : other: 3 (2001Vo24); 3.0 15 (1972Hu10). I _γ : other: 78 (2001Vo24); 76.0 76 (1972Hu10). Mult.,δ: from 2001Vo24. Other: δ(Q/D)=0.52 42 from 1996Ka21. A ₂ =-0.55 8, A ₄ =+0.004 78, A ₆ =+0.006 72 (1996Ka21). A ₂ =-0.55 8, A ₄ =0.00 8 (2001Vo24).
		6988.1 8207.2	3 78	1219.39 0	1/2 ⁺ 3/2 ⁺	D+Q	-0.36 4	I _γ : other: 3 (2001Vo24); 3.0 15 (1972Hu10). I _γ : other: 78 (2001Vo24); 76.0 76 (1972Hu10). Mult.,δ: from 2001Vo24. Other: δ(Q/D)=0.52 42 from 1996Ka21. A ₂ =-0.55 8, A ₄ =+0.004 78, A ₆ =+0.006 72 (1996Ka21). A ₂ =-0.55 8, A ₄ =0.00 8 (2001Vo24).
8216.3	5/2	2561 3446.3 4038.2 4042.6 5052.9	0.8 0.8 <3 <3 41	5655 4769.86 4177.9 4173.45 3162.99	 3/2 7/2 ⁻	D+Q	+0.11 2	I _γ : other: 1 (2001Vo24). I _γ : other: 1 (2001Vo24). I _γ : other: 3 (2001Vo24). I _γ : other: 41 (2001Vo24); 44.0 44 (1972Hu10). Mult.,δ: from γ(θ) in 2001Vo24. A ₂ =-0.32 7, A ₄ =-0.11 7, A ₆ =+0.05 6 (1996Ka21). A ₂ =-0.32 7, A ₄ =-0.11 7 (2001Vo24). I _γ : other: 5 (2001Vo24); 5.0 25 (1972Hu10). I _γ : other: 3 (2001Vo24); 4 2 (1972Hu10). I _γ : other: 1 (2001Vo24). I _γ : other: 45 (2001Vo24); 47.0 47 (1972Hu10). Mult.,δ: from γ(θ) in 2001Vo24. Other: δ(Q/D)=0.59 56 from 1996Ka21. A ₂ =-0.560 8, A ₄ =+0.017 71, A ₆ =+0.066 75 (1996Ka21). A ₂ =-0.56 1, A ₄ =+0.02 7 (2001Vo24).
		5213.1 5521.9 6452.5 8215.3	5 3 1.4 45	3002.75 2693.9 1763.14 0	5/2 (3/2,5/2) 5/2 ⁺ 3/2 ⁺	D+Q	+0.06 3	
8242.4		4068.7 5239.2 7022.3	7 2 91	4173.45 3002.75 1219.39	 5/2 1/2 ⁺			
8251?		4137 5248 7031	4& 6& 82&	4113.7 3002.75 1219.39	7/2 5/2 1/2 ⁺			
		8250	8&	0	3/2 ⁺			
8268.9		3498.9 4090.7 4350.1 5105.5 5265.7 5574.5 5622.7 6505.1 8267.9	2 12 3 7 13 6 4 8 45	4769.86 4177.9 3918.48 3162.99 3002.75 2693.9 2645.70 1763.14 0	 3/2 3/2 ⁺ 7/2 ⁻ 5/2 (3/2,5/2) (5/2,7/2) 5/2 ⁺ 3/2 ⁺			I _γ : from 1976Sp08.
8277.2		2622 3396.1 3437.9 3507.2 3652.6 3929.2 4103.5 4217.7	1.5 5 11 1.2 3 1.8 2 1.5	5655 4880.96 4839.10 4769.86 4624.4 4347.76 4173.45 4059.2	 5/2,9/2 			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	Comments
8277.2		4358.4	5	3918.48	3/2 ⁺		
		5274.0	10	3002.75	5/2		
		5582.8	3	2693.9	(3/2,5/2)		
		5631.0	3	2645.70	(5/2,7/2)		
		6513.4	23	1763.14	5/2 ⁺		
		8276.2	29	0	3/2 ⁺		
8282.4	(3/2 ⁻ ,5/2)	2696	8	5586			
		3066.5	8	5215.8			
		3118.9	3	5163.34	7/2		
		4104.2	14	4177.9	3/2		
		4222.9	6	4059.2			
		5119.0	26	3162.99	7/2 ⁻		
		6518.6	7	1763.14	5/2 ⁺		
		8281.4	28	0	3/2 ⁺		
8287.82	1/2 ⁻	2633	2	5655			
		3277.6	3	5010.1			
		4109.7	9	4177.9	3/2		
		4228.4	2	4059.2			
		4320.2	3	3967.3			
		4369.05	1	3918.48	3/2 ⁺		
		5593.4	23	2693.9	(3/2,5/2)		$A_2=-0.01$ 4, $A_4=-0.02$ 4 (1976Me12).
		7067.66	44	1219.39	1/2 ⁺		δ : $\delta(\text{Q/D})=+0.21$ 2 or -3.1 2 for $J_i=3/2$ (1976Me12).
							$A_2=-0.08$ 2, $A_4=-0.01$ 3 (1976Me12).
		8286.77	13	0	3/2 ⁺		δ : $\delta(\text{Q/D})=-0.35$ 4 or -11 3 for $J_i=3/2$ (1976Me12).
							$A_2=-0.11$ 3, $A_4=-0.01$ 4 (1976Me12).
8298.2	3/2	3287.9	3	5010.1			
		3417.1	4	4880.96			
		3443.6	5	4854.4			
		4124.5	6	4173.45			
		4238.7	10	4059.2			
		5134.8	1	3162.99	7/2 ⁻		
		5295.0	1	3002.75	5/2		
		5603.8	10	2693.9	(3/2,5/2)		$A_2=+0.09$ 6, $A_4=-0.07$ 7 (1976Me12).
		5652.0	6	2645.70	(5/2,7/2)		
		6534.4	8	1763.14	5/2 ⁺		$A_2=+0.43$ 11, $A_4=-0.51$ 12 (1976Me12).
		7078.0	24	1219.39	1/2 ⁺	D+Q	Mult., δ : -1.7 1 or -0.02 2 (1976Me12).
							$A_2=-0.53$ 3, $A_4=+0.11$ 3 (1976Me12).
		8297.1	13	0	3/2 ⁺	D+Q	Mult., δ : -0.15 4 or $+9.5$ +48-24 (1976Me12).
							$A_2=+0.18$ 3, $A_4=-0.10$ 4 (1976Me12).
8318.1		2512	0.7	5806			
		2672	0.7	5646			
		2718.3	1.5	5599.7			
		3548.1	0.3	4769.86			
		3693.5	1.8	4624.4			
		4139.9	<3	4177.9	3/2		
		4144.4	<3	4173.45			
		4258.6	1.2	4059.2			
		4350.5	3	3967.3			
		5154.7	0.5	3162.99	7/2 ⁻		
		5314.9	3	3002.75	5/2		
		5623.7	1.6	2693.9	(3/2,5/2)		
		5671.9	0.9	2645.70	(5/2,7/2)		
		6554.3	1.8	1763.14	5/2 ⁺		
		7097.9	2	1219.39	1/2 ⁺		
		8317.0	78	0	3/2 ⁺		
8323.1		4209.1	1 &	4113.7	7/2		

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8323.1		5319.9	3&	3002.75	5/2			
		5676.9	4&	2645.70	(5/2,7/2)			
		6559.3	<2&	1763.14	5/2 ⁺			
		7102.9	8&	1219.39	1/2 ⁺			
		8322.0	82&	0	3/2 ⁺			
8381.8	5/2 ⁺	3611.7		4769.86				I_γ : 1 (2001Vo24).
		3757.2		4624.4				I_γ : 1 (2001Vo24).
		4267.8	8.0 8	4113.7	7/2			I_γ : from 1972Hu10. Other: 7 (2001Vo24).
		4463.0	24.0 24	3918.48	3/2 ⁺	D+Q	+0.15 3	I_γ : from 1972Hu10. Other: 24 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24.
								$A_2=+0.25$ 3, $A_4=-0.04$ 2 (2001Vo24).
		5378.6	25.0 25	3002.75	5/2	D(+Q)	+0.02 3	I_γ : from 1972Hu10. Other: 25 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24. Other:
								$\delta=-0.11$ 1 from 1967Ko20, 1967Ko23.
								$A_2=+0.60$ 7, $A_4=-0.02$ 7 (1967Ko20,1967Ko23).
		5687.4	1.0 5	2693.9	(3/2,5/2)			$A_2=+0.51$ 23, $A_4=-0.05$ 24 (2001Vo24).
								I_γ : from 1972Hu10. Other: 1 (2001Vo24).
		5735.6	5.0 25	2645.70	(5/2,7/2)			$A_2=+0.22$ 9, $A_4=-0.02$ 11 (1967Ko20,1967Ko23).
								I_γ : from 1972Hu10. Others: 5 (2001Vo24).
								Mult., δ : D+Q with $\delta=+0.72$ +42-20 and $J_f=3/2$
								for level 2645 (1967Ko20).
		6618.0	34.0 34	1763.14	5/2 ⁺	D+Q	+0.06 3	I_γ : from 1972Hu10. Other: 34 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24. Other:
								$\delta=-0.12$ 10 from $\gamma(\theta)$ in 1967Ko23.
								$A_2=+0.79$ 15, $A_4=-0.08$ 15
								(1967Ko20,1967Ko23).
								$A_2=+0.02$ 13, $A_4=-0.08$ 13 (2001Vo24).
		8380.7	3.0 15	0	3/2 ⁺	D+Q	1.65	I_γ : from 1972Hu10. Other: 2 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 1967Ko23.
								$A_2=-0.83$ 22, $A_4=+0.72$ 33
								(1967Ko20,1967Ko23).
8402.9		3632.8	4.8 24	4769.86				
		4224.7	9 5	4177.9	3/2			
		4343.4	23 6	4059.2				
		5708.5	10 5	2693.9	(3/2,5/2)			
		5756.7	7 4	2645.70	(5/2,7/2)			
		6639.1	15 4	1763.14	5/2 ⁺			
		8401.8	31 8	0	3/2 ⁺			
8405.7		3524.6	15 4	4880.96				
		4232.0	1.6 8	4173.45				
		4291.7	1.4 7	4113.7	7/2			
		5402.5	15 4	3002.75	5/2			
		6641.9	43 11	1763.14	5/2 ⁺			
		8404.6	24 6	0	3/2 ⁺			
8416.7		3406.4	5&	5010.1				
		4243.0	11&	4173.45				
		4473.5	3&	3942.9				
		5413.5	7&	3002.75	5/2			
		5770.5	8&	2645.70	(5/2,7/2)			
		6652.9	22&	1763.14	5/2 ⁺			
		7196.5	9&	1219.39	1/2 ⁺			
		8415.6	35&	0	3/2 ⁺			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
8464.3		2809	8 4	5655		
		3248.3	5.8 29	5215.8		
		3454.0	8 4	5010.1		
		3609.7	3.3 17	4854.4		
		3839.7	4.3 22	4624.4		
		4286.1	5.2 26	4177.9	3/2	
		4290.6	5.5 28	4173.45		
		4496.7	5.2 26	3967.3		
		4545.5	2.6 13	3918.48	3/2 ⁺	
		5461.1	17 4	3002.75	5/2	
		5769.9	16 4	2693.9	(3/2,5/2)	
		7244.1	9 4	1219.39	1/2 ⁺	
		8463.2	10 3	0	3/2 ⁺	
8484.4	3/2 ⁺	2829	5.6 28	5655		I_γ : other: 6 2001Vo24 .
		3603.2	7.0 35	4880.96		I_γ : other: 7 2001Vo24 .
		3859.8	1.0 5	4624.4		I_γ : other: 1 2001Vo24 .
		4516.8	1.2 6	3967.3		I_γ : other: 1 2001Vo24 .
		4565.6	5.2 26	3918.48	3/2 ⁺	I_γ : other: 5 2001Vo24 .
		5481.2	7 4	3002.75	5/2	I_γ : other: 7 2001Vo24 .
		5790.0	20 5	2693.9	(3/2,5/2)	I_γ : other: 20 2001Vo24 .
		5838.2	2.8 14	2645.70	(5/2,7/2)	I_γ : other: 3 2001Vo24 .
		6720.6	46 12	1763.14	5/2 ⁺	I_γ : other: 46 2001Vo24 .
		8483.3	4 2	0	3/2 ⁺	I_γ : other: 4 2001Vo24 .
8486.1		2728	1.1 6	5758		
		3861.5	1.8 9	4624.4		
		4307.9 ^c	4.8 ^c 24	4177.9	3/2	Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4312.4 ^c	4.8 ^c 24	4173.45		Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4426.6	1.9 10	4059.2		
		4518.5	7 4	3967.3		
		5482.9	8 4	3002.75	5/2	
		5791.7	19 5	2693.9	(3/2,5/2)	
		6722.3	3.6 18	1763.14	5/2 ⁺	
		7265.9	6.6 33	1219.39	1/2 ⁺	
		8485.0	46 12	0	3/2 ⁺	
8514.3		3659.7	4.4 22	4854.4		
		4336.1	2 1	4177.9	3/2	
		4454.8	27 7	4059.2		
		4546.7	23 6	3967.3		
		4595.5	6.6 33	3918.48	3/2 ⁺	
		7294.1	10 3	1219.39	1/2 ⁺	
		8513.2	27 7	0	3/2 ⁺	
8571.9		2848.3	2.2 11	5723.52		
		2917	1.3 7	5655		
		2972.1	0.7 4	5599.7		
		3408.4	1.8 9	5163.34	7/2	
		3690.7	0.9 5	4880.96		
		4398.2	0.7 4	4173.45		
		4457.9	1.1 6	4113.7	7/2	
		4653.1	1.1 6	3918.48	3/2 ⁺	
		5408.5	1.6 8	3162.99	7/2 ⁻	
		5568.7	4.5 23	3002.75	5/2	
		5877.5	0.6 3	2693.9	(3/2,5/2)	
		5925.7	8 4	2645.70	(5/2,7/2)	
		7351.7	5.4 27	1219.39	1/2 ⁺	
		8570.8	70 7	0	3/2 ⁺	
8580.5		2088	4.4 22	6492		

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ [†]	I_γ [#]	E_f	J_f^π	Mult. ^b	δ ^b	Comments
8580.5		3741.2	1.3 7	4839.10				
		4402.3	4.4 22	4177.9	3/2			
		4521.0	2 1	4059.2				
		4661.7	2.3 12	3918.48	3/2 ⁺			
		5886.1	5.6 28	2693.9	(3/2,5/2)			
		7360.3	54 6	1219.39	1/2 ⁺			
		8579.4	26 7	0	3/2 ⁺			
8590.0		2832	4.8 24	5758				
		2935	3.7 19	5655				
		3819.9	16 4	4769.86				
		4416.3	13 3	4173.45				
		4476.0	3.6 18	4113.7	7/2			
		5426.6	8 4	3162.99	7/2 ⁻			
		5586.8	22 6	3002.75	5/2			
		5943.8	5.5 28	2645.70	(5/2,7/2)			
		6826.2	14 4	1763.14	5/2 ⁺			
		8588.9	10 3	0	3/2 ⁺			
8612.0		6848.1	50 5	1763.14	5/2 ⁺			
		7391.8	17 4	1219.39	1/2 ⁺			
		8610.9	33 8	0	3/2 ⁺			
8613.7		3732.5	1.6 8	4880.96				
		4554.2	5.4 27	4059.2				
		4670.5 ^d	4&	3942.9				I_γ : from 1967Da12 only; not observed by others.
		4694.9	7 4	3918.48	3/2 ⁺			
		5610.5	2.5 13	3002.75	5/2			
		5919.3	4.3 22	2693.9	(3/2,5/2)			
		5967.5	23 6	2645.70	(5/2,7/2)			
		6849.8 ^d	9&	1763.14	5/2 ⁺			I_γ : from 1967Da12 only; not observed by others.
		7393.5 ^d	2&	1219.39	1/2 ⁺			I_γ : from 1967Da12 only; not observed by others.
8618.1	(3/2 ⁺ ,5/2 ⁺)	8612.6	56 6	0	3/2 ⁺			
		2894.5	2.1 11	5723.52				
		2963	1.7 9	5655				
		3763.5	1.4 7	4854.4				
		3993.5	1.5 8	4624.4				
		4650.5	3.0 15	3967.3				
		4699.3	2.3 12	3918.48	3/2 ⁺			
		5614.9	16 4	3002.75	5/2			
		5923.7	14 4	2693.9	(3/2,5/2)			$A_2=+0.147$ 33, $A_4=+0.114$ 52 (1970Ko33).
		5971.9	1.7 9	2645.70	(5/2,7/2)			
		6854.2	23 6	1763.14	5/2 ⁺			$A_2=+0.662$ 23, $A_4=+0.319$ 39 (1970Ko33 reported for 7855 γ , 7855 is likely a misprint).
		8617.0	33 8	0	3/2 ⁺			$A_2=-0.293$ 29, $A_4=-0.028$ 47 (1970Ko33).
8630.1	7/2	2906.5	0.3 2	5723.52				
		3466.6	8 4	5163.34	7/2	D+Q	+0.60 10	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09. $A_2=+0.60$ 3, $A_4=-0.08$ 4 (1976Sp09).
		3860.0	1.4 7	4769.86				
		4282.1	12 3	4347.76	5/2,9/2	D+Q	-0.184 14	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09. $A_2=+0.11$ 2, $A_4=+0.02$ 2 (1976Sp09).
		4456.4	3.6 18	4173.45				

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8630.1	7/2	4516.1	10 5	4113.7	7/2	D+Q	-0.06 2	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.43$ 2, $A_4=-0.01$ 2 (1976Sp09).
		4686.9	10 5	3942.9				
		5466.7	3.5 18	3162.99	7/2 ⁻			
		5626.9	1.4 7	3002.75	5/2			
		5983.9	2.6 13	2645.70	(5/2,7/2)			
		6866.2	47 12	1763.14	5/2 ⁺	D+Q	-0.011 3	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.10$ 33, $A_4=-0.65$ 46 (1970Ko33). $A_2=-0.387$ 7, $A_4=+0.012$ 8 (1976Sp09). $A_2=+0.398$ 46, $A_4=+0.019$ 78 (1970Ko33).
		8629.0	0.5 3	0	3/2 ⁺			
8640	5/2	4697	23 [@]	3942.9				E_γ : other: 4689 7 (1970Ko33).
		5477	20 [@]	3162.99	7/2 ⁻			E_γ : other: 5473 7 (1970Ko33).
		5637	6 [@]	3002.75	5/2			E_γ : other: 5630 7 (1970Ko33).
		5946	13 [@]	2693.9	(3/2,5/2)			E_γ : other: 5941 7 (1970Ko33).
		6876	38 [@]	1763.14	5/2 ⁺			E_γ : other: 6874 7 (1970Ko33). $A_2=-0.52$ 12, $A_4=-0.29$ 19 (1970Ko33).
8641.4	3/2,5/2	2986	2.9 15	5655				
		3055	2.5 13	5586				
		3237.7	4.7 24	5403.5				
		3802.1	2.6 13	4839.10				
		3871.3	24 6	4769.86				
		4016.8	1.9 10	4624.4				
		4463.2 ^c	14 ^c 4	4177.9	3/2			Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4467.6 ^c	14 ^c 4	4173.45				Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4581.9	9 5	4059.2				
		4722.6	5.9 30	3918.48	3/2 ⁺			
		5947.0	5.2 26	2693.9	(3/2,5/2)			
		8640.3	28 7	0	3/2 ⁺			$A_2=-0.12$ 2, $A_4=-0.01$ 2 (1976Sp09).
8686.2		3100	2.1 11	5586				
		3282.5	3.1 16	5403.5				
		4508.0	2 1	4177.9	3/2			
		4626.7	3.5 18	4059.2				
		5522.7	3.7 19	3162.99	7/2 ⁻			
		5991.8	2.8 14	2693.9	(3/2,5/2)			
		6922.3	4.8 24	1763.14	5/2 ⁺			
		7466.0	1.9 10	1219.39	1/2 ⁺			
		8685.0	76 8	0	3/2 ⁺			
8696.9		2891	2.9 15	5806				
		3097.1	4.8 24	5599.7				
		7476.7	12 3	1219.39	1/2 ⁺			
		8695.7	80 8	0	3/2 ⁺			
8750.9		2945	2.2 11	5806				
		3347.2	4.9 25	5403.5				
		3534.9	2.4 12	5215.8				
		4572.7	2.1 11	4177.9	3/2			
		4577.1	2.3 12	4173.45				
		6056.4	3.0 15	2693.9	(3/2,5/2)			
		6987.0	30 8	1763.14	5/2 ⁺			
		7530.6 ^d	2	1219.39	1/2 ⁺			I_γ : from 1967Da12 only; not observed by others.
		8749.7	53 5	0	3/2 ⁺			
8779.8		3179.9	4.3 22	5599.7				

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
8779.8		3563.8	6.6 33	5215.8		
		4601.6 ^c	15 ^c 4	4177.9	3/2	Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4606.0 ^c	15 ^c 4	4173.45		Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4861.0	4.2 21	3918.48	3/2 ⁺	
		5776.5	27 7	3002.75	5/2	
		6085.3	12 3	2693.9	(3/2,5/2)	
		6133.5	13 3	2645.70	(5/2,7/2)	
		7015.9	6.7 34	1763.14	5/2 ⁺	
		8778.6	11 3	0	3/2 ⁺	
8787.2		3063.5	6.8 34	5723.52		
		3623.7	19 5	5163.34	7/2	
		4017.1	6.7 34	4769.86		
		4609.0	9 5	4177.9	3/2	
		5623.7	5.8 29	3162.99	7/2 ⁻	
		5783.9	20 5	3002.75	5/2	
		7023.3	25 6	1763.14	5/2 ⁺	
		8786.0	8 4	0	3/2 ⁺	
8798.4		4738.9	8 4	4059.2		
		5795.1	16 4	3002.75	5/2	
		7034.5	7.0 35	1763.14	5/2 ⁺	
		7578.1	7.0 35	1219.39	1/2 ⁺	
		8797.2	62 6	0	3/2 ⁺	
8820.9		5817.6	9 5	3002.75	5/2	
		6126.4	18 5	2693.9	(3/2,5/2)	
		7057.0	21 5	1763.14	5/2 ⁺	
		8819.7	52 5	0	3/2 ⁺	
8824.2		5820.9	37 9	3002.75	5/2	
		6129.7	6.0 3	2693.9	(3/2,5/2)	
		7060.3	16 4	1763.14	5/2 ⁺	
		7603.9	22 6	1219.39	1/2 ⁺	
		8823.0	19 5	0	3/2 ⁺	
8829.3		3174	10 5	5655		
		3990.0	1.3 7	4839.10		
		4059.2	1.5 8	4769.86		
		4910.5	4.5 23	3918.48	3/2 ⁺	
		5826.0	22 6	3002.75	5/2	
		6134.8	13 3	2693.9	(3/2,5/2)	
		6183.0	33 8	2645.70	(5/2,7/2)	
		7065.4	9 5	1763.14	5/2 ⁺	
		7609.0	2.6 13	1219.39	1/2 ⁺	
		8828.1	3.8 19	0	3/2 ⁺	
8833.1		3617.1	6.9 35	5215.8		
		4063.0	2.6 13	4769.86		
		4659.3	14 4	4173.45		
		4719.1	4.7 24	4113.7	7/2	
		4773.6	11 3	4059.2		
		4914.3	11 3	3918.48	3/2 ⁺	
		5669.6	3.7 19	3162.99	7/2 ⁻	
		5829.8	6.7 34	3002.75	5/2	
		6138.6	4.5 23	2693.9	(3/2,5/2)	
		6186.8	23 6	2645.70	(5/2,7/2)	
		7069.2	9 5	1763.14	5/2 ⁺	
		8831.9	2.9 15	0	3/2 ⁺	
8838.1	7/2 ⁻	3674.6 [‡]		5163.34	7/2	
		3956.9 [‡]		4880.96		

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
8838.1	$7/2^-$	4068.0	8 4	4769.86		
		4490.0 [‡]		4347.76	$5/2, 9/2$	
		4778.6 [‡]		4059.2		
		5674.6	29 7	3162.99	$7/2^-$	
		7074.2	63 6	1763.14	$5/2^+$	$A_2 = -0.20$ 2, $A_4 = -0.00$ 2 (1981Bi05).
8856.1		3974.9	3.4 17	4880.96		
		4231.4	5.1 26	4624.4		
		4677.9	3.5 18	4177.9	$3/2$	
		4937.3	11 3	3918.48	$3/2^+$	
		5692.6	51 5	3162.99	$7/2^-$	
		6161.6	2.8 14	2693.9	$(3/2, 5/2)$	
		6209.8		2645.70	$(5/2, 7/2)$	
		7092.2	7 4	1763.14	$5/2^+$	
		8854.9	16 4	0	$3/2^+$	
8868.6		4690.4	2.6 13	4177.9	$3/2$	
		5865.3	16 4	3002.75	$5/2$	
		6174.1	12 3	2693.9	$(3/2, 5/2)$	
		7104.7	4.4 22	1763.14	$5/2^+$	
		7648.3	16 4	1219.39	$1/2^+$	
		8867.4	49 12	0	$3/2^+$	
8886.1		3231	6 3	5655		
		3286.2	6.6 33	5599.7		
		3670.1	2.2 11	5215.8		
		3722.6	9 4	5163.34	$7/2$	
		4116.0	4.5 23	4769.86		
		4707.9	8 4	4177.9	$3/2$	
		4967.2	4.1 21	3918.48	$3/2^+$	
		5882.8	8 4	3002.75	$5/2$	
		6239.8	26 7	2645.70	$(5/2, 7/2)$	
		7122.2	12 3	1763.14	$5/2^+$	
		8884.9	14 4	0	$3/2^+$	
8893.3	$(5/2, 7/2)^+$	3293.4 ^d		5599.7		I_γ : 1 from 2001Vo24 only; not observed by others.
		3307	1.6 8	5586		
		4779.3	9 4	4113.7	$7/2$	I_γ : other: 9 2001Vo24 .
		4950.0	3.9 20	3942.9		I_γ : other: 4 2001Vo24 .
		5729.8	29 7	3162.99	$7/2^-$	I_γ : other: 29 2001Vo24 .
		6198.8	37 9	2693.9	$(3/2, 5/2)$	I_γ : other: 37 2001Vo24 .
		7129.4	19 5	1763.14	$5/2^+$	I_γ : other: 19 2001Vo24 .
		8892.1	1.0 5	0	$3/2^+$	I_γ : other: 1 2001Vo24 .
8904.8		2724	4.3 22	6181		
		3250	3.4 17	5655		
		3894.5	3.9 20	5010.1		
		4050.2	6.2 31	4854.4		
		4065.5	4.5 23	4839.10		
		4726.6	18 5	4177.9	$3/2$	
		4845.2	5.7 29	4059.2		
		6210.3	15 4	2693.9	$(3/2, 5/2)$	
		7684.5	10 3	1219.39	$1/2^+$	
		8903.6	29 7	0	$3/2^+$	
8906.8	$5/2^+$	3252		5655		I_γ : 4 from 2001Vo24 only; not observed by others.
		3261	3.7 19	5646		I_γ : other: 2 2001Vo24 .
		3306.9	2.2 11	5599.7		
		3321	4	5586		I_γ : other: 4 2001Vo24 .
		3387	2.5 13	5520		
		3690.8	4.4 22	5215.8		

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8906.8	5/2 ⁺	4136.7	5.6 28	4769.86				I_γ : other: 6 2001Vo24 .
		4963.5	13 3	3942.9				I_γ : other: 15 2001Vo24 .
		7142.9	69 7	1763.14	5/2 ⁺			I_γ : other: 69 2001Vo24 .
8919.8		3274	5.6 28	5646				
		4571.7	8 4	4347.76	5/2,9/2			
		5756.3	57 6	3162.99	7/2 ⁻			
		6273.5	26 7	2645.70	(5/2,7/2)			
		7155.9	4 2	1763.14	5/2 ⁺			
8953.0	3/2	7732.7		1219.39	1/2 ⁺	D+Q		Mult.: from $\gamma(\theta)$ in 1976Sp08 ; $\delta=-0.34$ 2 or $\delta=-5.0$ 4 (1976Sp09).
		8951.8		0	3/2 ⁺	D+Q		$A_2=+0.12$ 3, $A_4=-0.02$ 4 (1976Sp09).
								Mult.: from $\gamma(\theta)$ in 1976Sp08 ; $\delta=-0.39$ 5 or $\delta=-8.2$ 7 (1976Sp09).
								$A_2=-0.19$ 2, $A_4=+0.02$ 2 (1976Sp09).
8984.2		3260.5	0.8 4	5723.52				
		4144.8	1.2 6	4839.10				
		4214.1	0.5 3	4769.86				
		4359.5	0.5 3	4624.4				
		4806.0	1.5 8	4177.9	3/2			
		4810.4	1.5 8	4173.45				
		5065.3	9 5	3918.48	3/2 ⁺			
		5820.7	6 3	3162.99	7/2 ⁻			
		6289.7	1.8 9	2693.9	(3/2,5/2)			
		6337.9	1.2 6	2645.70	(5/2,7/2)			
		7220.3	16 4	1763.14	5/2 ⁺			
		8983.0	60 6	0	3/2 ⁺			
9081.4	5/2 ⁺	3357.7	1	5723.52				I_γ : from 2001Vo24 .
		4200.2	1	4880.96				I_γ : from 2001Vo24 .
		4242.0 [‡]		4839.10				
		4311.3 [‡]		4769.86				
		4456.7 [‡]		4624.4				
		4903.1	2	4177.9	3/2			I_γ : from 2001Vo24 .
		4907.6	2	4173.45				I_γ : from 2001Vo24 .
		5162.5	9	3918.48	3/2 ⁺	D(+Q)	-0.002 9	I_γ : from 2001Vo24 .
								Mult., δ : from $\gamma(\theta)$ in 1976Sp09 .
								$A_2=-0.43$ 3, $A_4=+0.05$ 3 (1976Sp09).
		5917.9	6	3162.99	7/2 ⁻			I_γ : from 2001Vo24 .
		6386.9	2	2693.9	(3/2,5/2)			I_γ : from 2001Vo24 .
		6435.1	1	2645.70	(5/2,7/2)			I_γ : from 2001Vo24 .
		7317.4	16	1763.14	5/2 ⁺	D+Q	+0.11 2	I_γ : from 2001Vo24 .
								Mult., δ : from $\gamma(\theta)$ in 1976Sp09 .
								$A_2=+0.555$ 14, $A_4=+0.03$ 2 (1976Sp09).
		9080.1	60	0	3/2 ⁺	D+Q	+0.089 9	I_γ : from 2001Vo24 .
								Mult., δ : from $\gamma(\theta)$ in 1976Sp09 .
								$A_2=-0.243$ 11, $A_4=+0.05$ 2 (1976Sp09).
9157.1	5/2	3941.1	7.4	5215.8				
		4386.9	6.9	4769.86				
		4978.8 ^c	13 ^c 3	4177.9	3/2			Final level is uncertain, either 4173 or 4178 (1976Sp08).
		4983.3 ^c	13 ^c 3	4173.45				Final level is uncertain, either 4173 or 4178 (1976Sp08).
		5993.6	9.6	3162.99	7/2 ⁻	D+Q	+0.10 2	Mult., δ : from $\gamma(\theta)$ in 2001Vo24 .
								$A_2=-0.15$ 8, $A_4=-0.31$ 9, $A_6=-0.05$ 8 (1996Ka21).
		6153.8	8.4	3002.75	5/2			

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
9157.1	6462.6	1.3	2693.9	(3/2,5/2)			
	6510.8	2.9	2645.70	(5/2,7/2)			
	7393.1	7.5	1763.14	5/2 ⁺			
	9155.8	43	0	3/2 ⁺	D+Q	+0.09 1	Mult., δ : from $\gamma(\theta)$ in 2001Vo24 . Other: $\delta(\text{Q/D})=0.47$ 50 (1996Ka21). $A_2=-0.29$ 6, $A_4=-0.38$ 8, $A_6=-0.10$ 6 (1996Ka21).

[†] From level-energy differences, unless otherwise noted.

[‡] γ branches observed in **1981Bi05**. Branching ratios are not given.

[#] For levels below 8.27 MeV ($E_p < 1.95$ MeV), I_γ is mainly from **1976Me12** and for levels above 8.27 MeV ($E_p > 1.95$ MeV), I_γ is mainly from **1976Sp08**, unless otherwise noted.

@ From **1970Ko33**.

& From **1967Da12**.

^a From **1963Ha32**.

^b From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ data as given under comments.

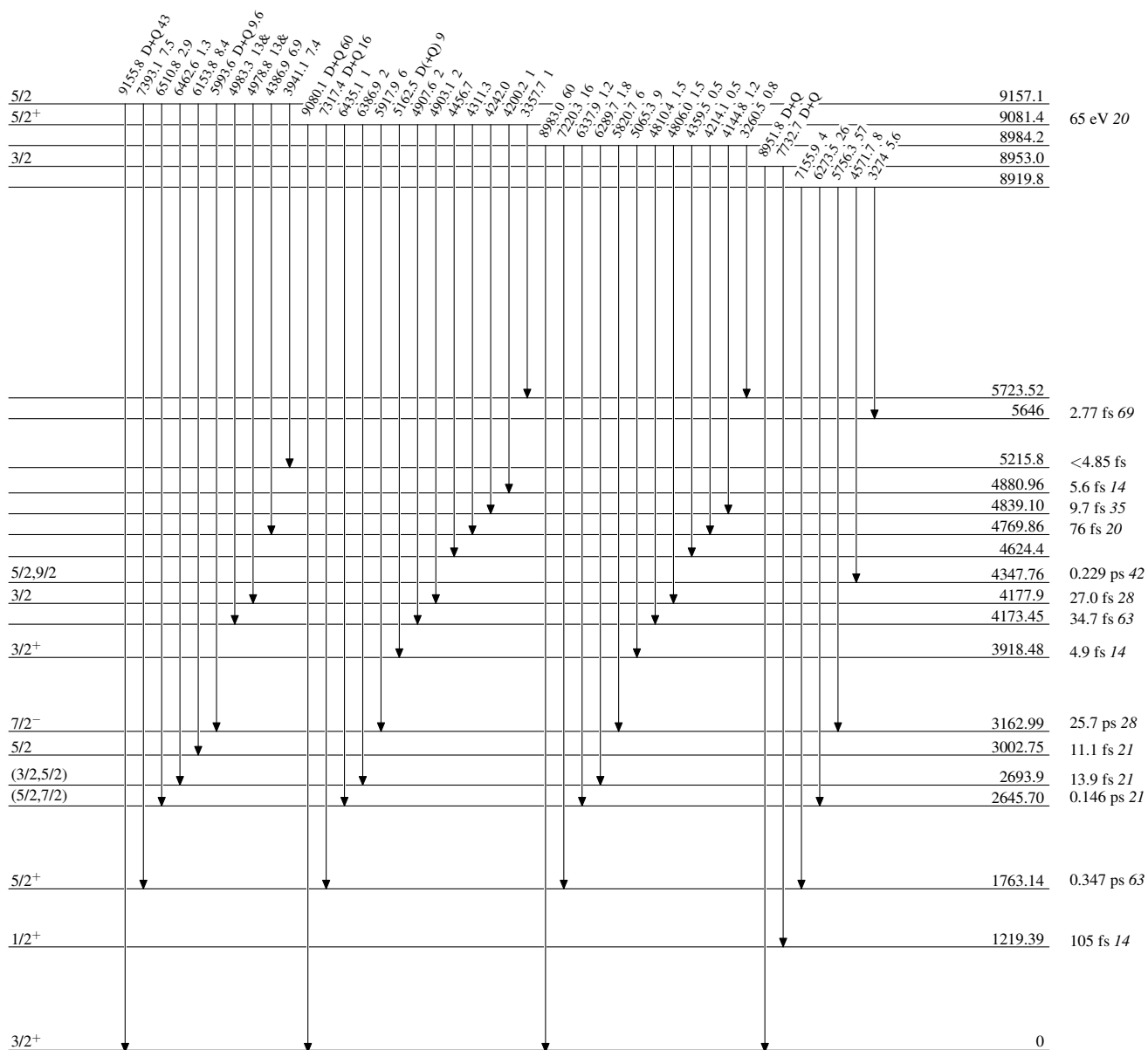
^c Multiply placed with undivided intensity.

^d Placement of transition in the level scheme is uncertain.

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

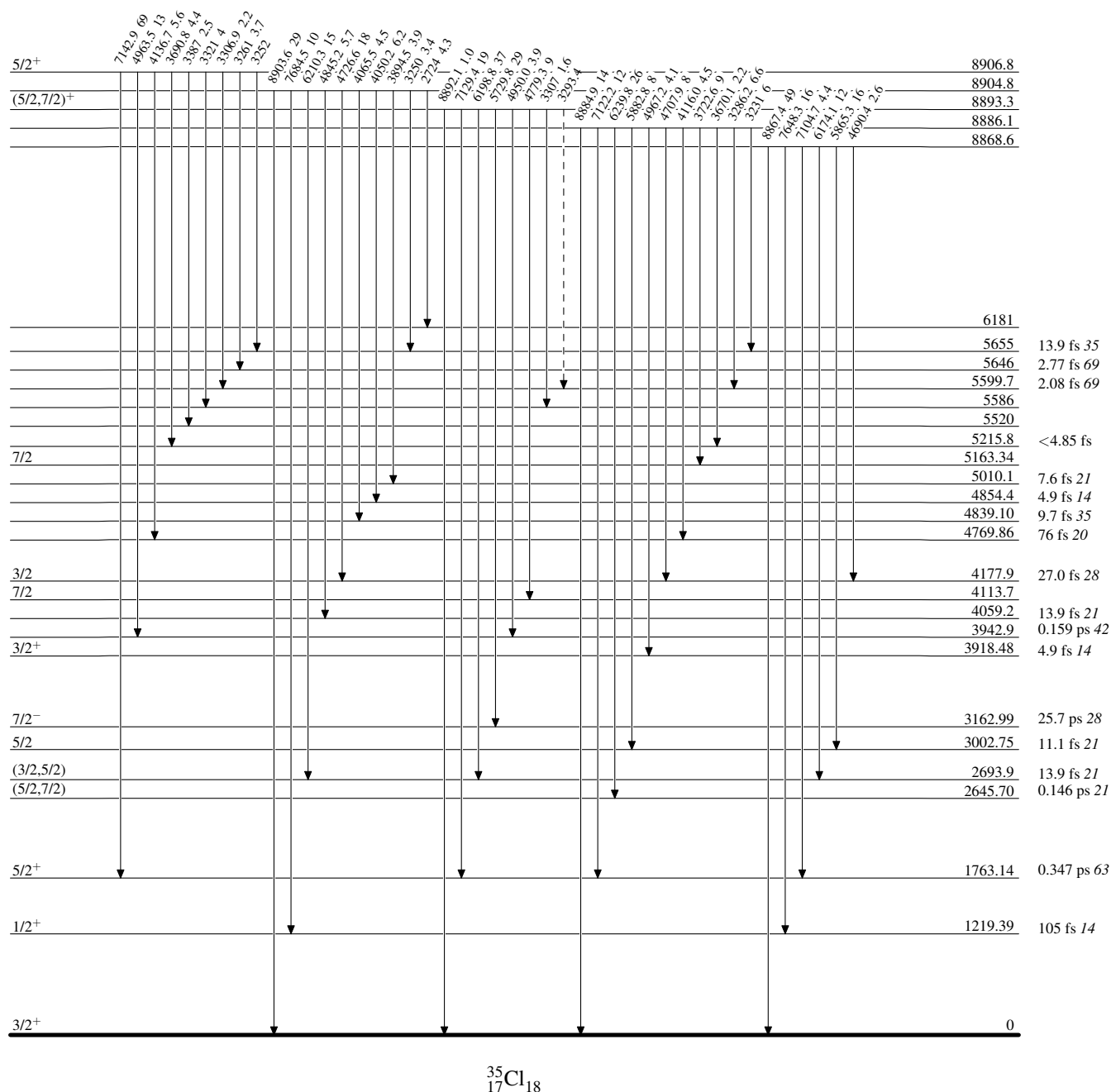
 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

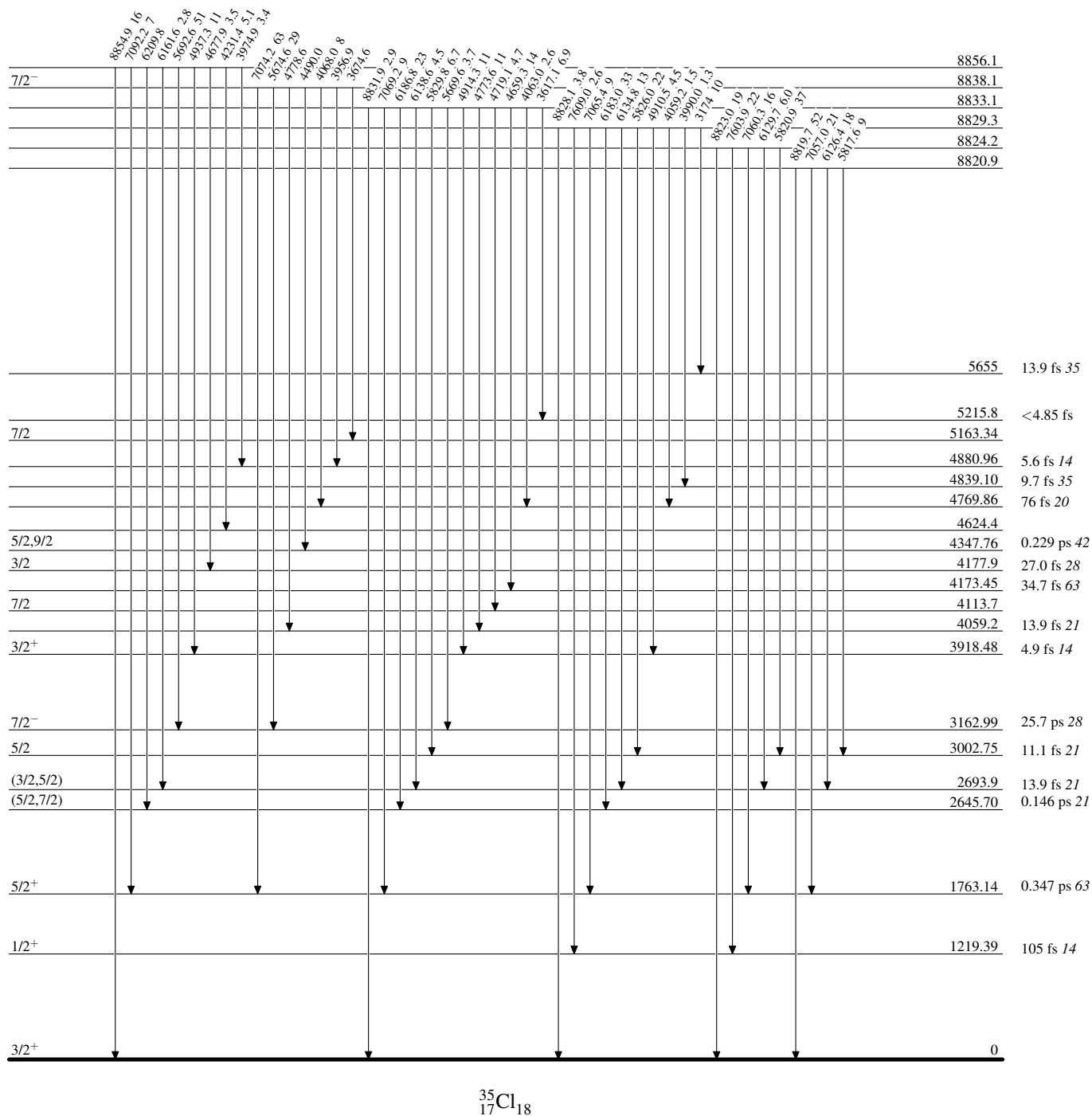
Intensities: % photon branching from each level
& Multiplicity placed: undivided intensity given

-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

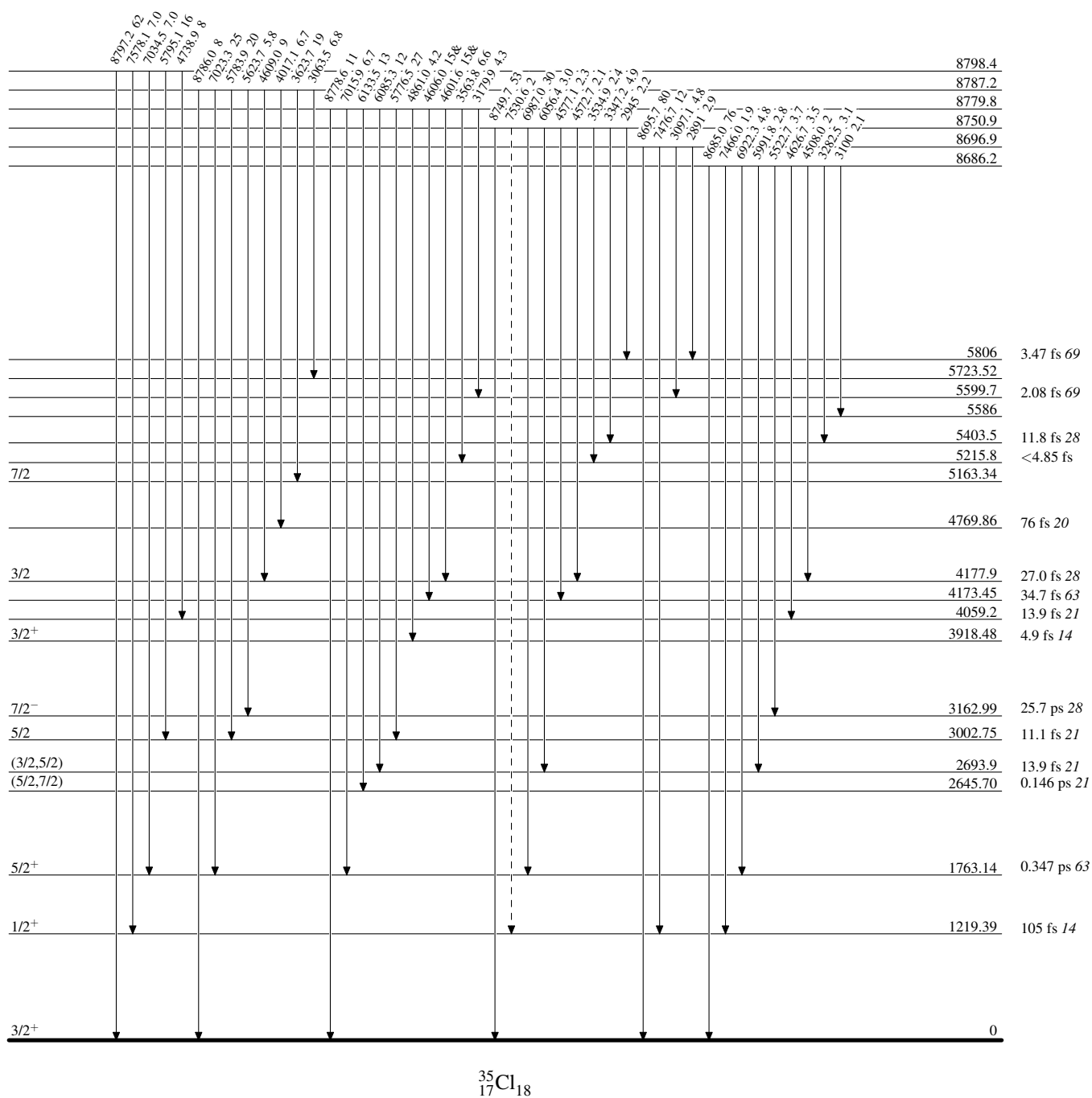
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

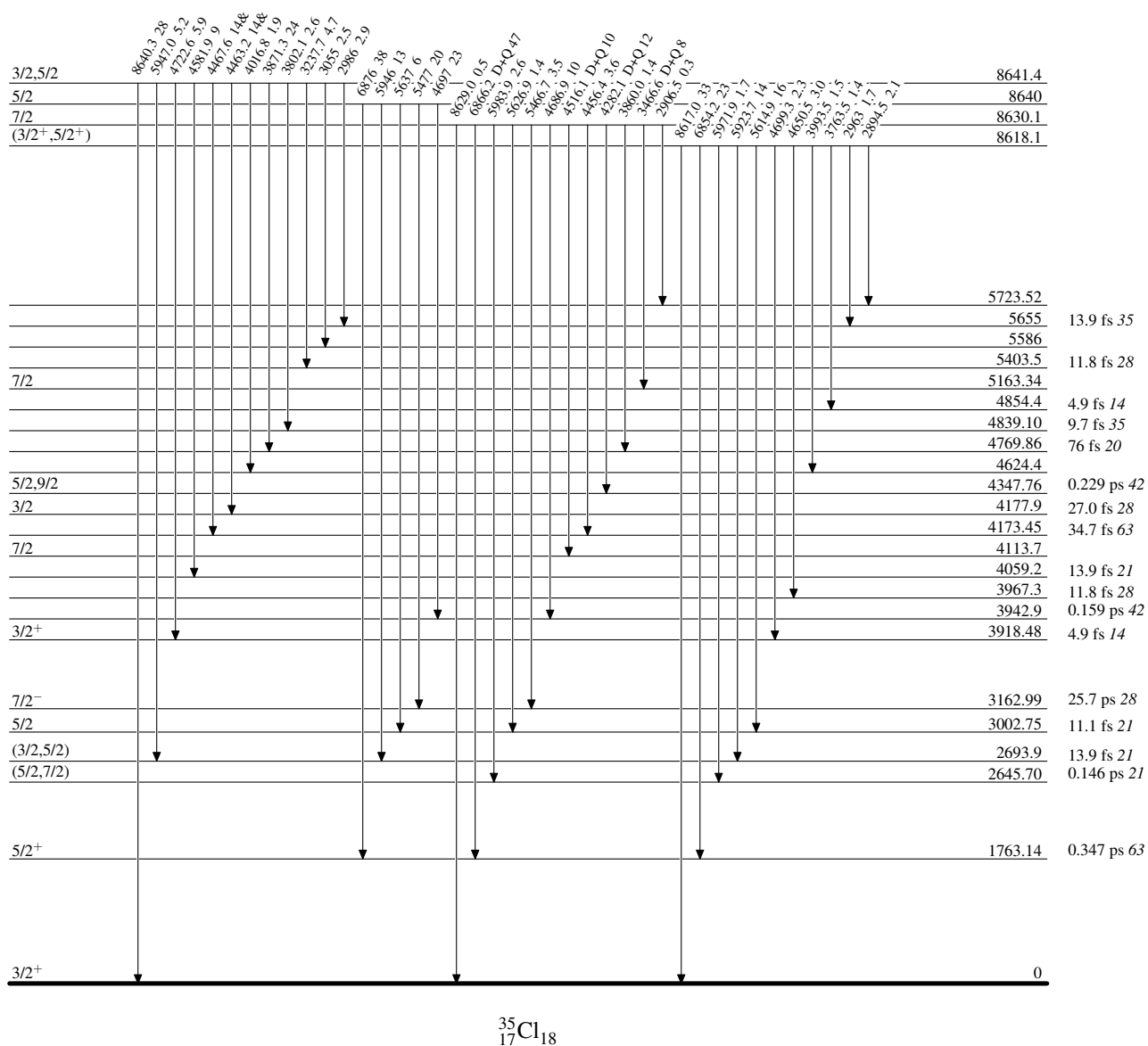
Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

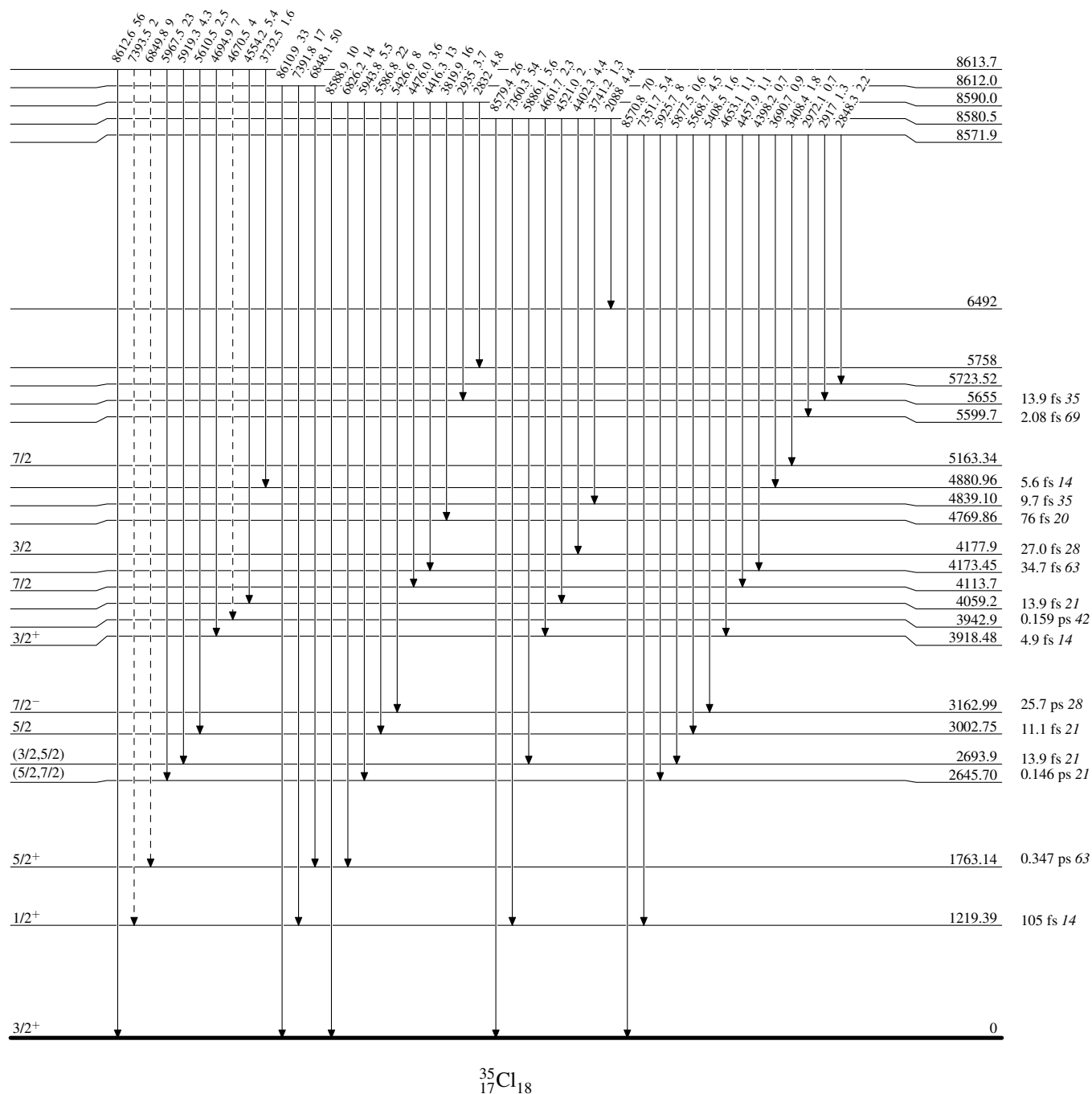


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

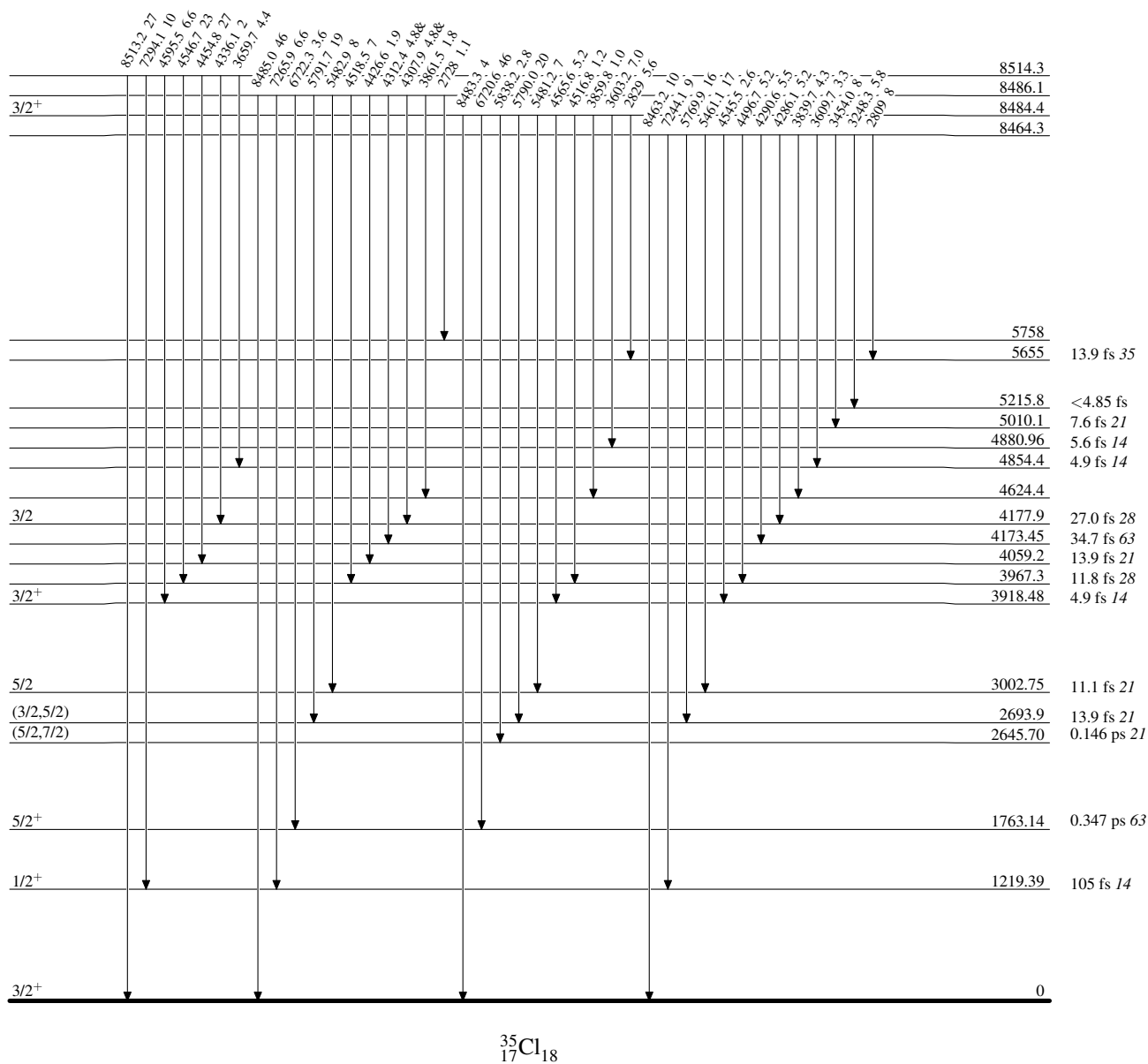
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

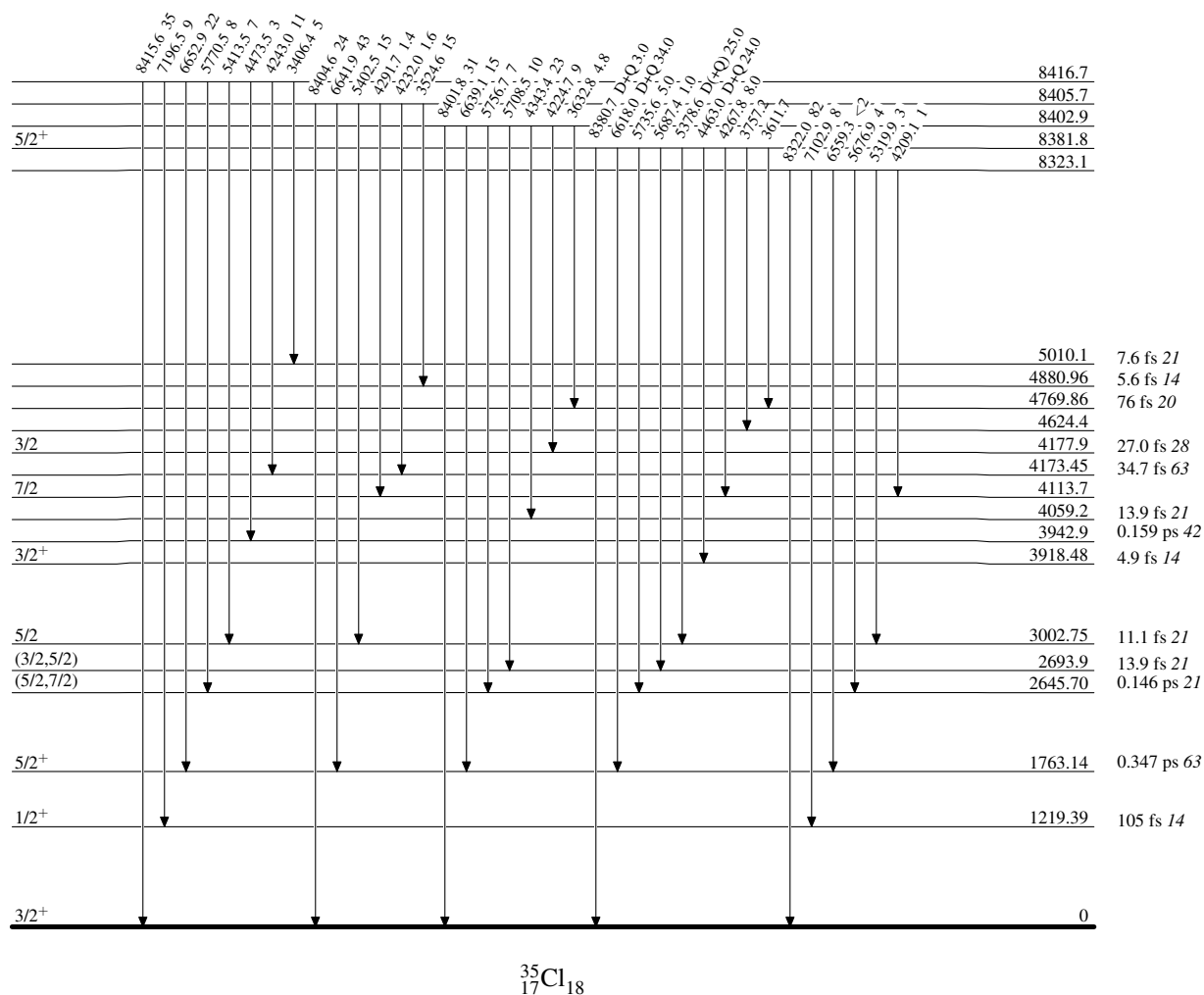
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

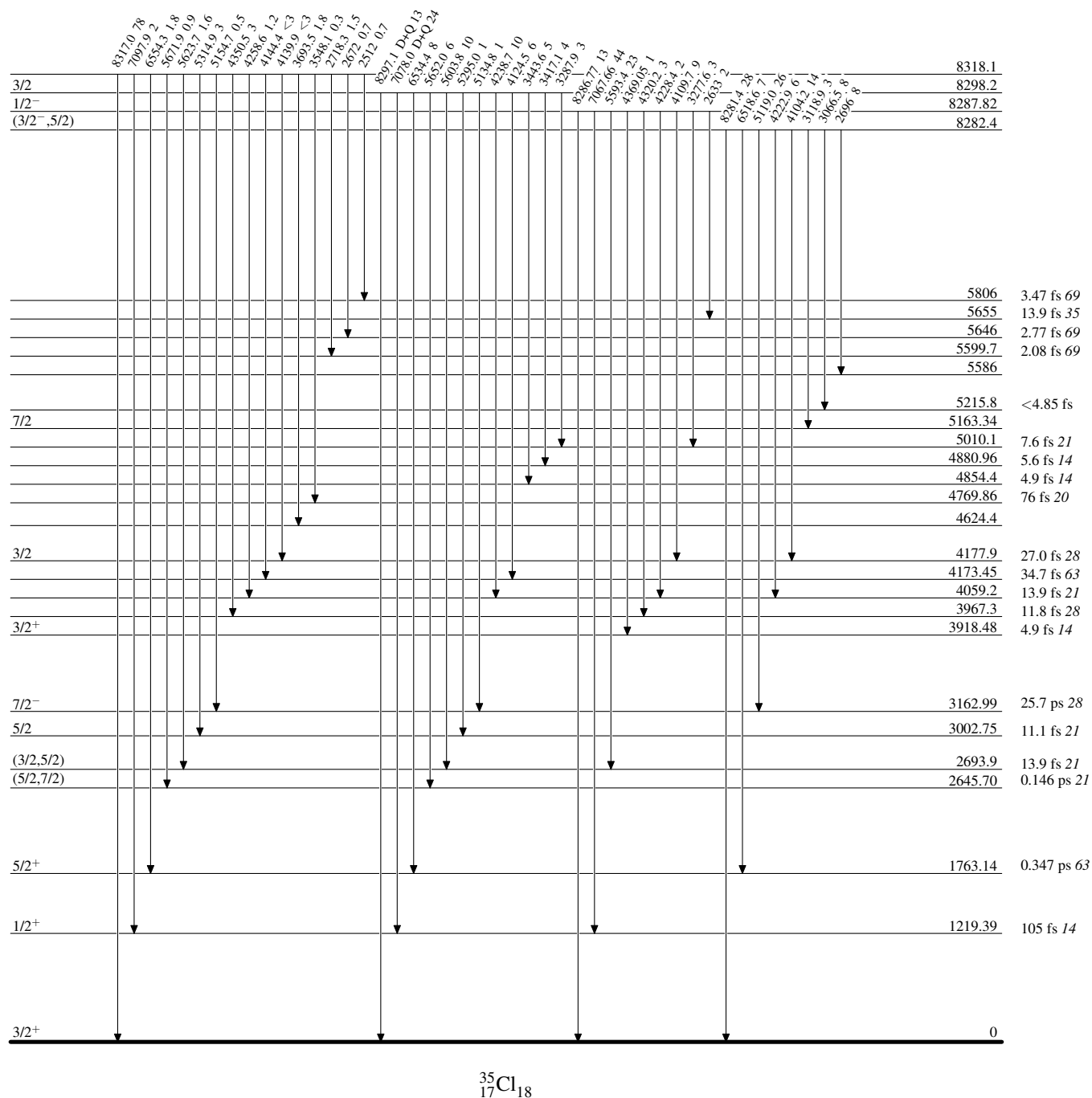
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

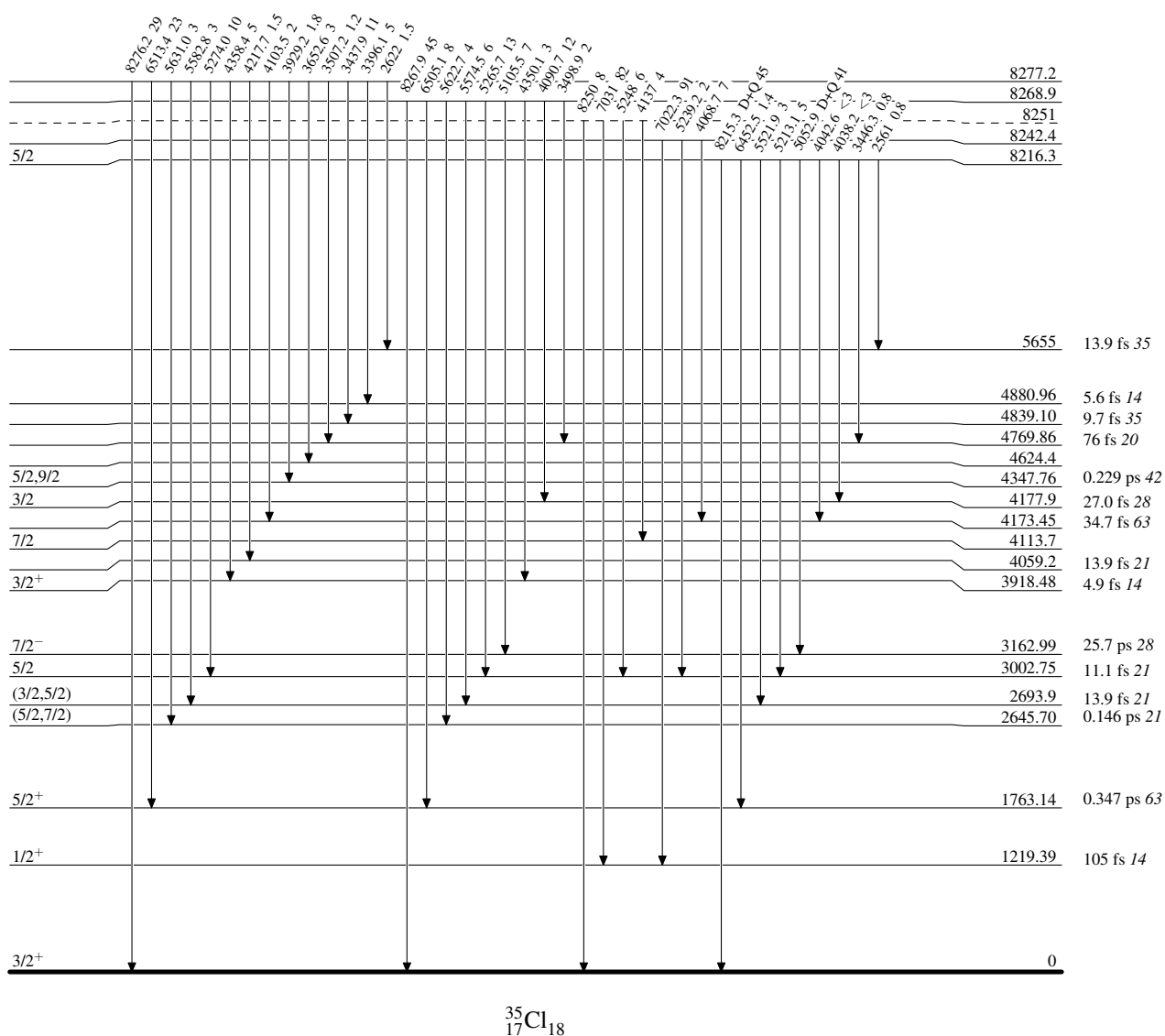
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

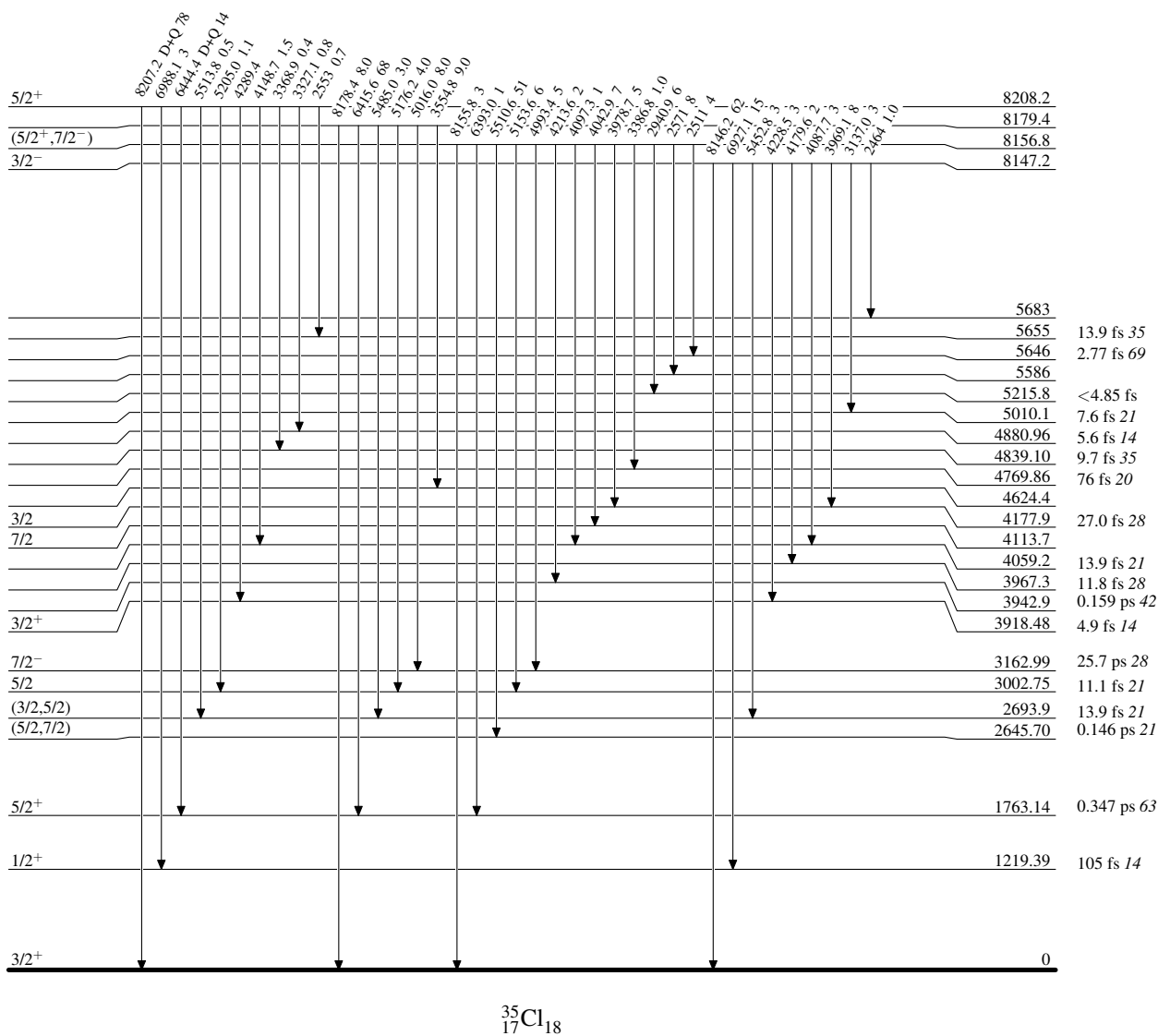
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

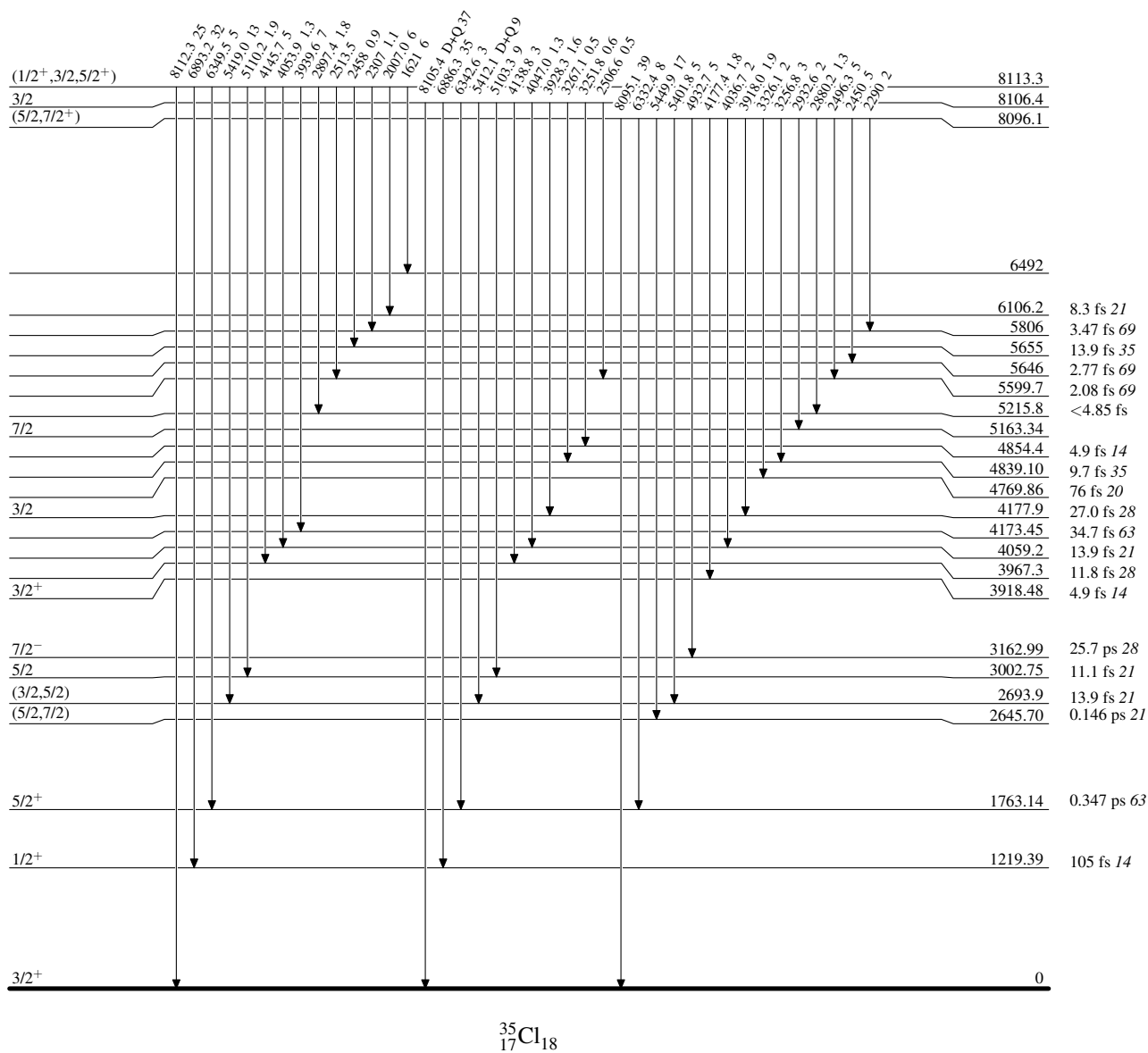
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

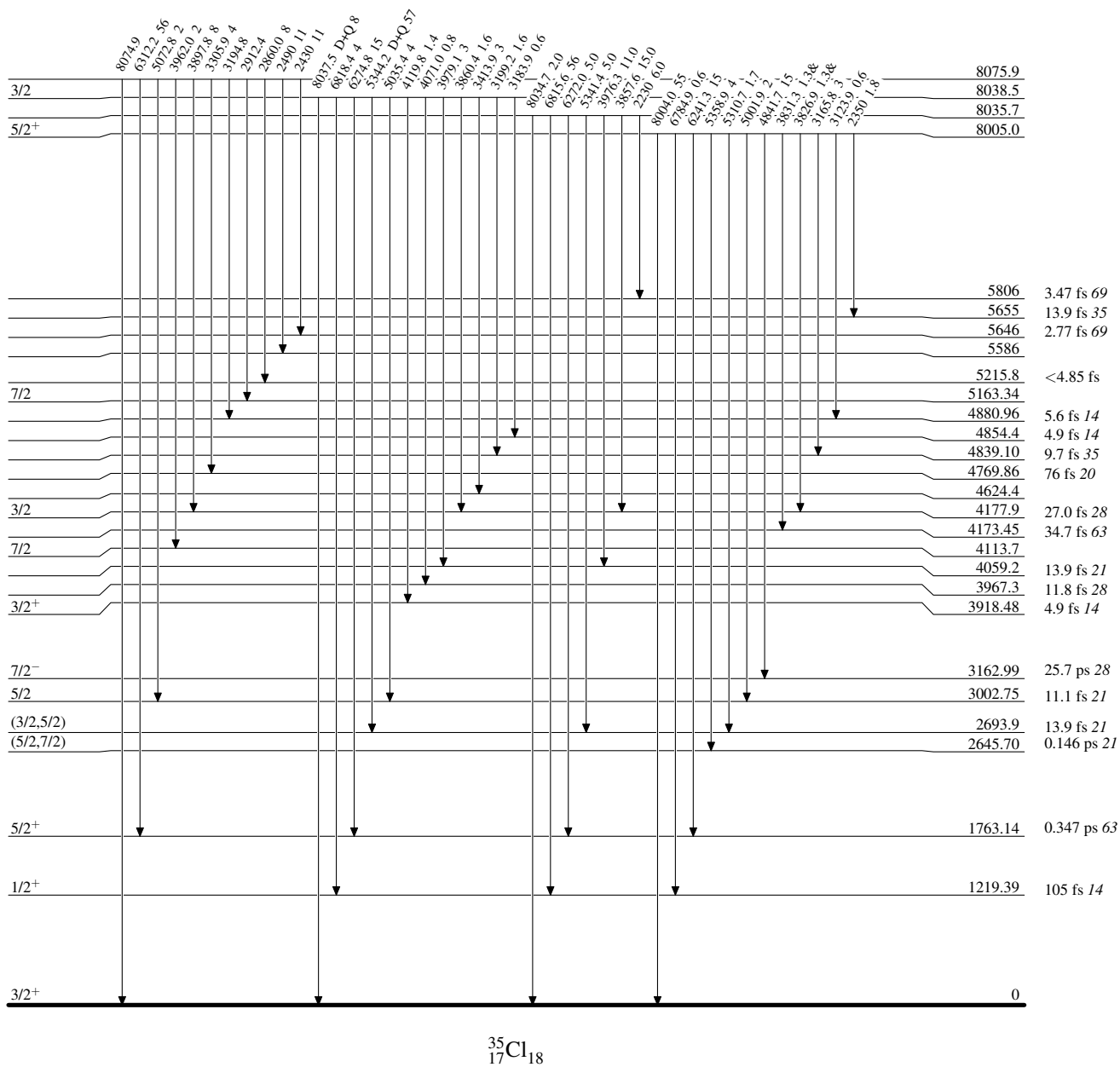
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

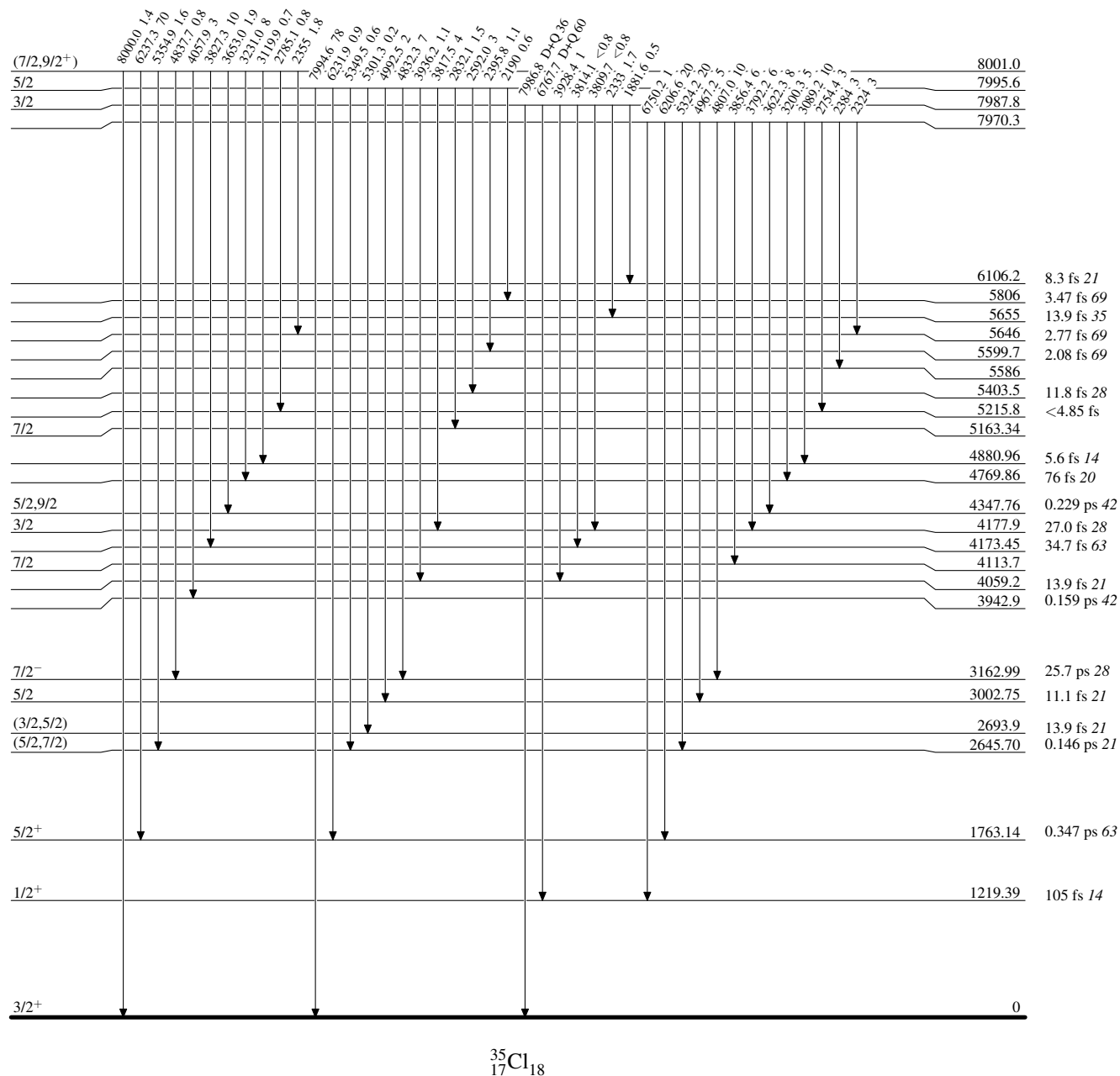
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

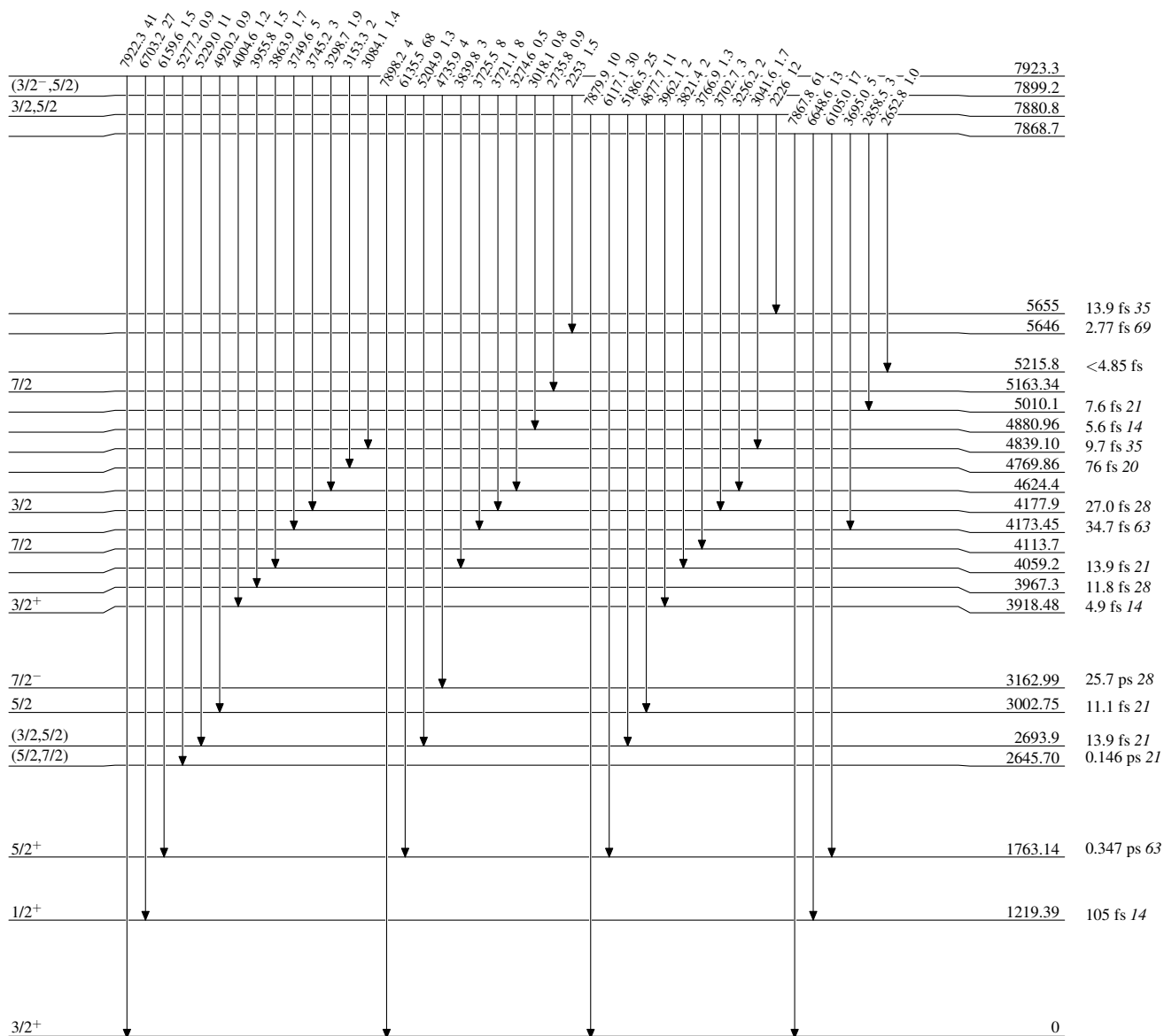
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

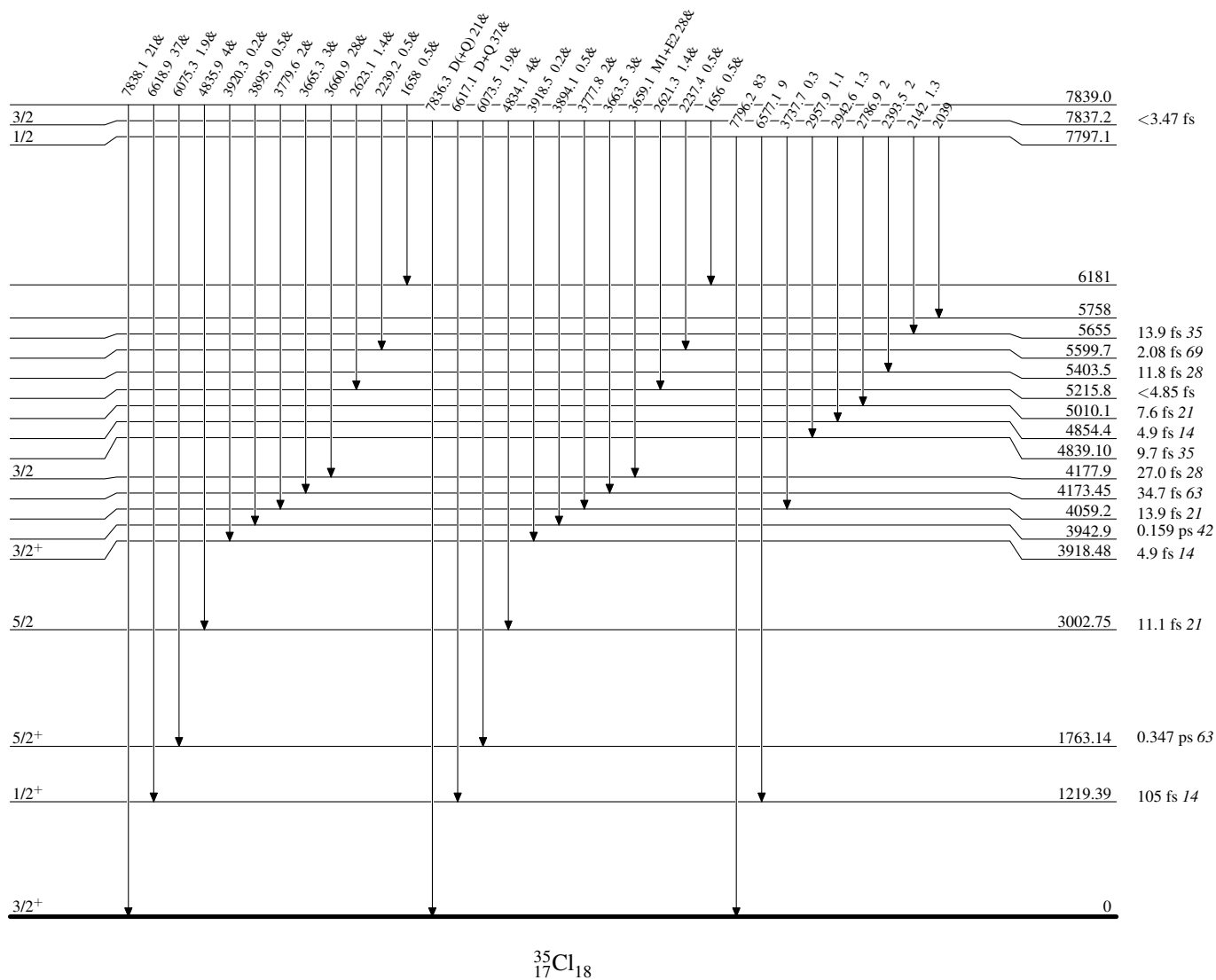
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

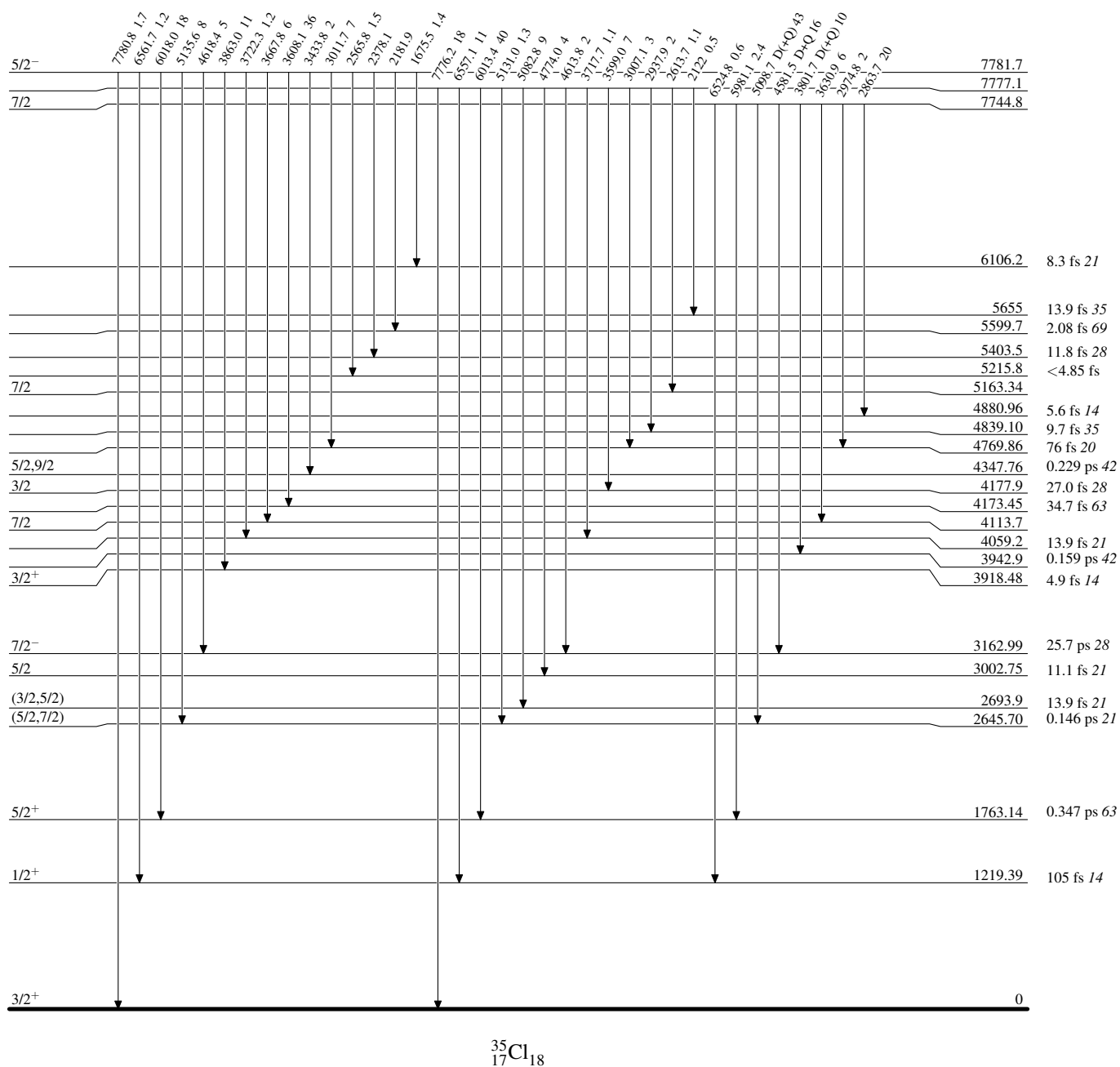
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

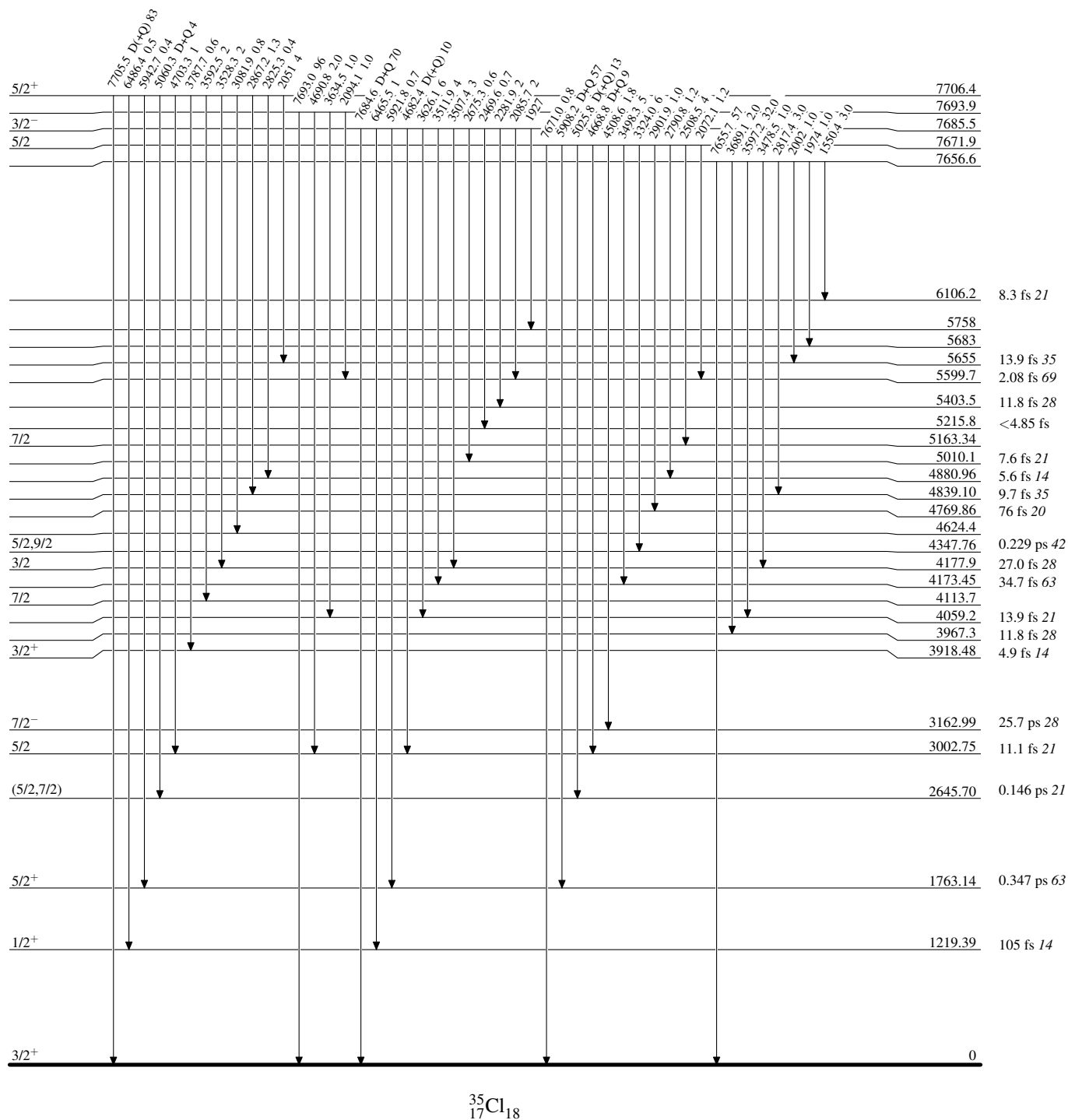
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

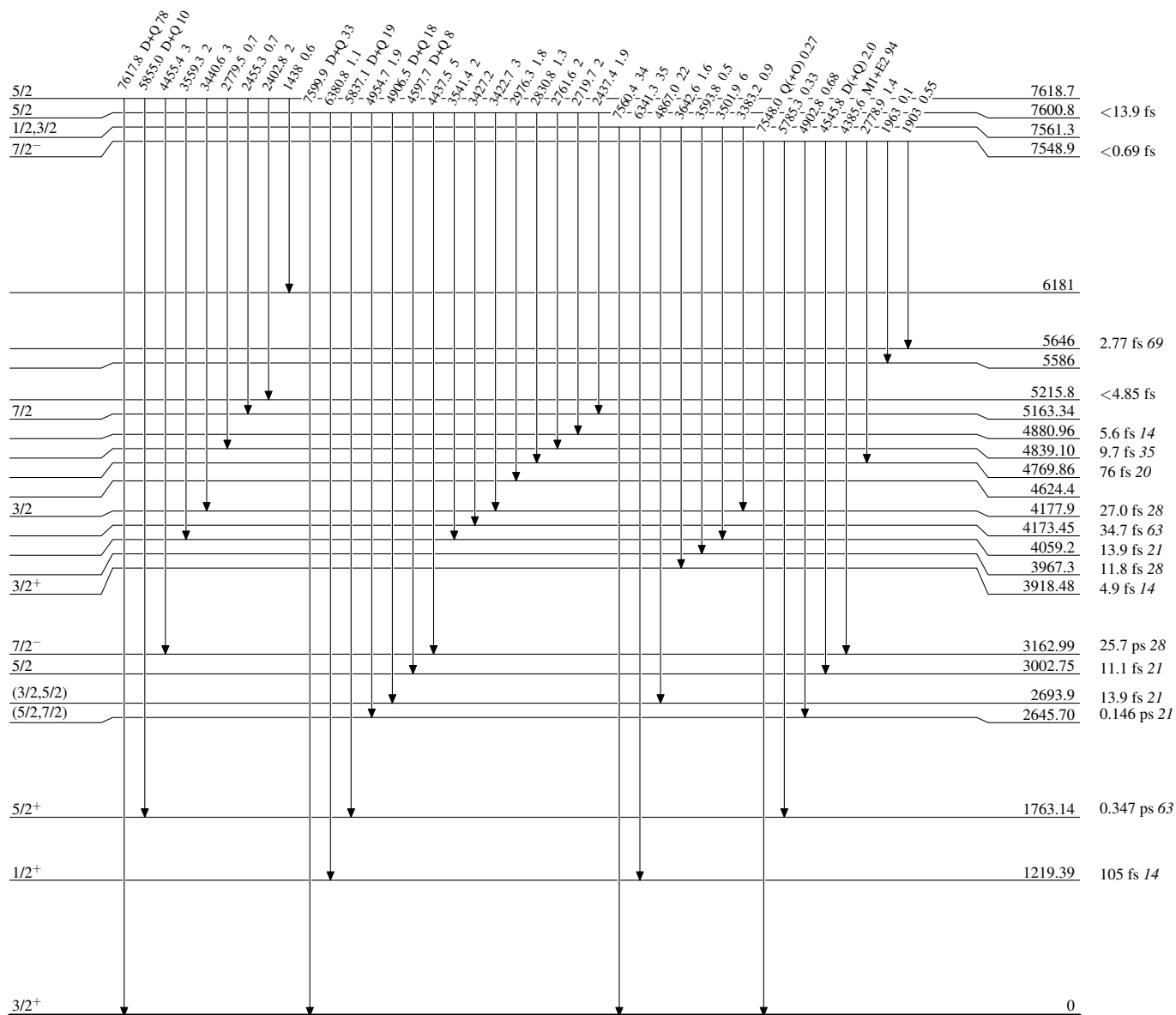
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

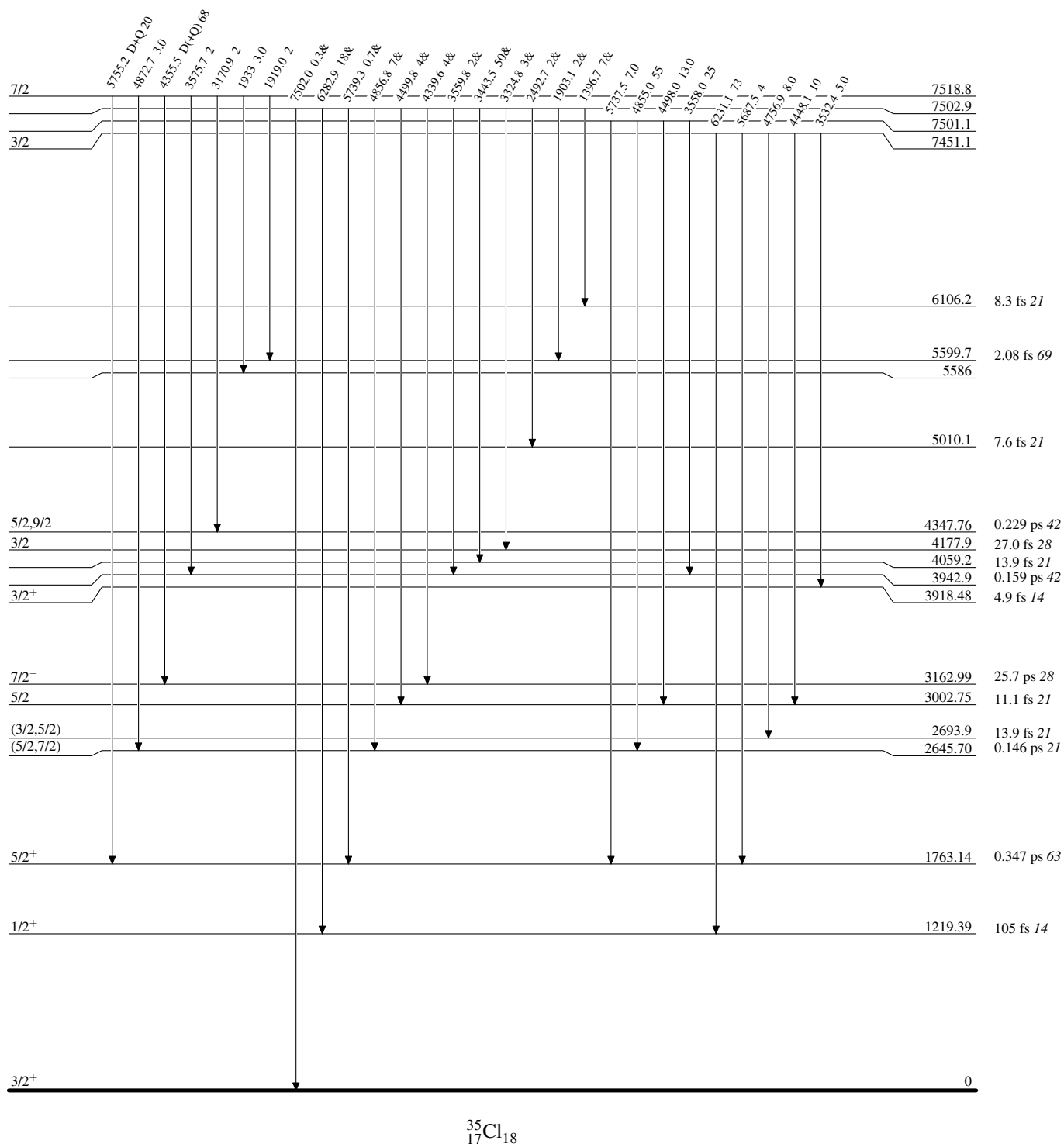
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

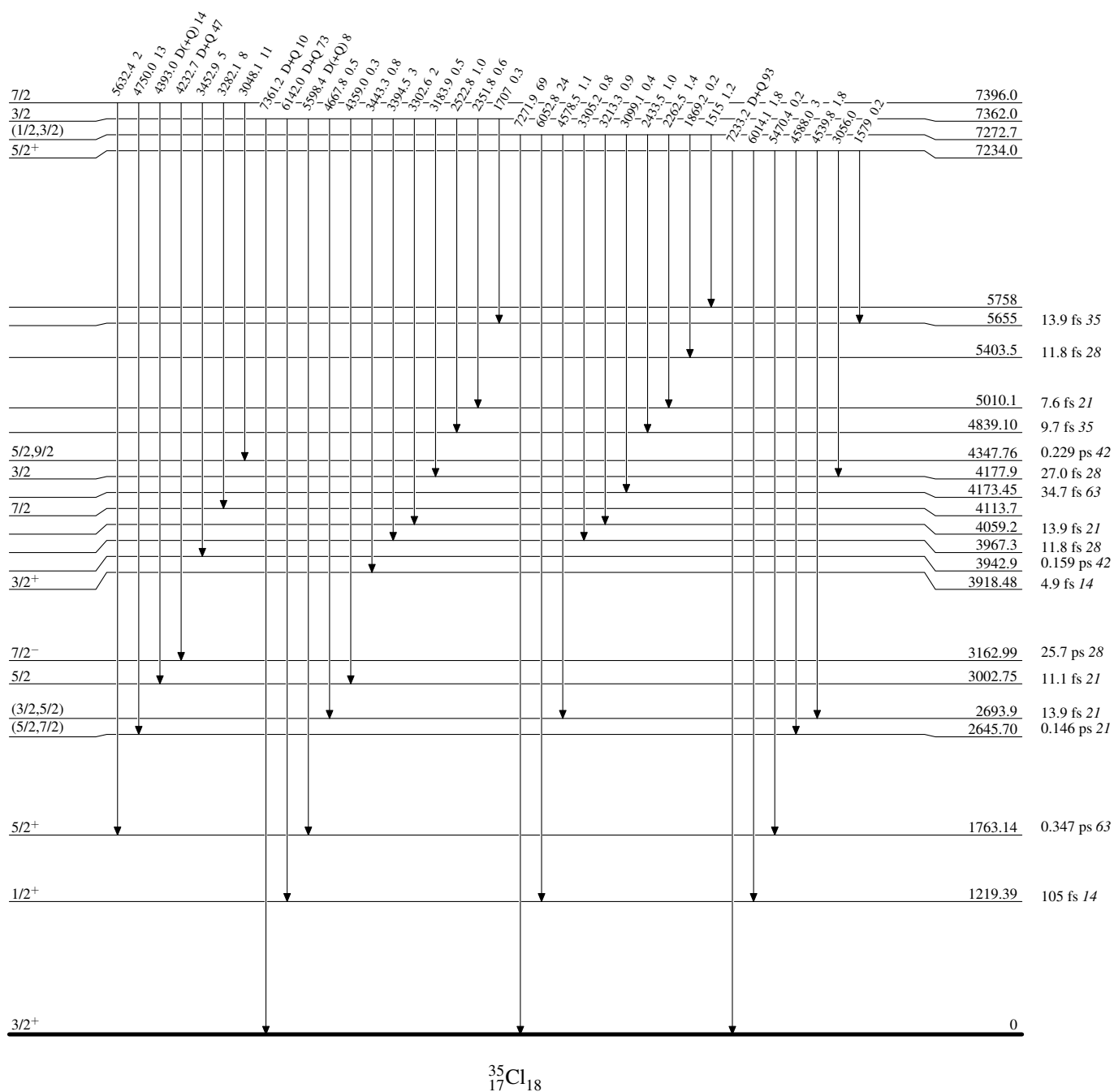
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

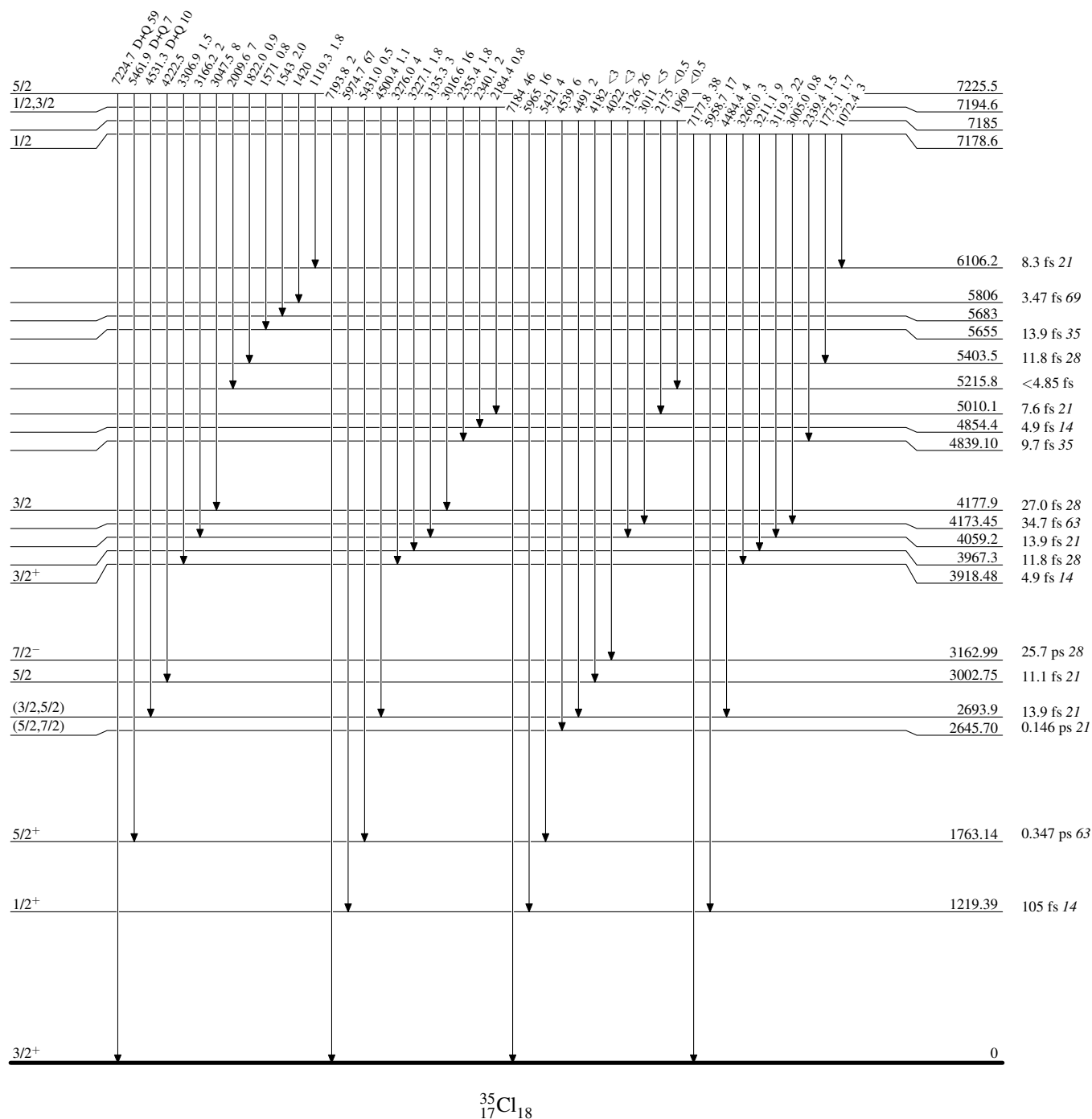
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

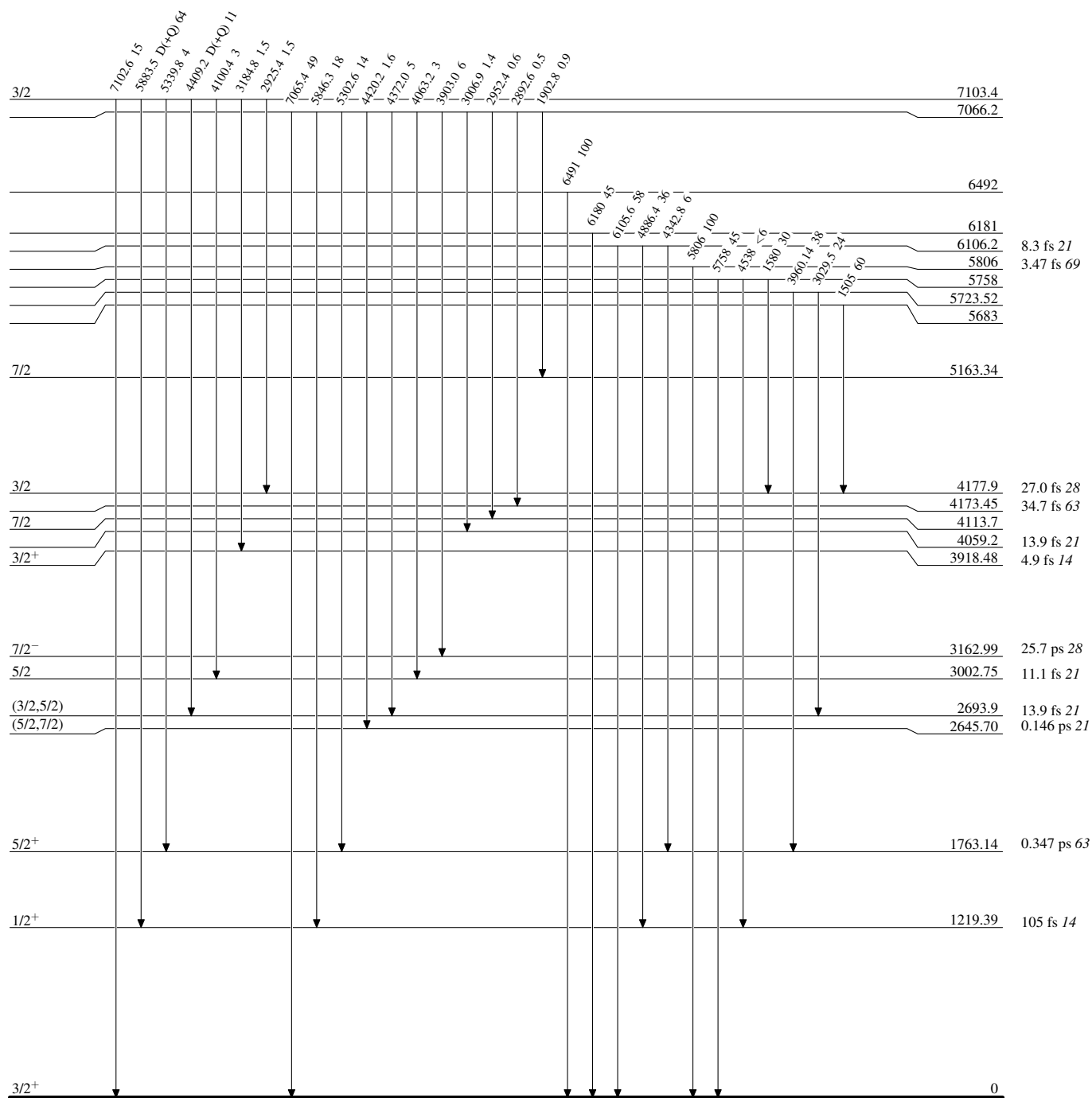
Intensities: % photon branching from each level
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

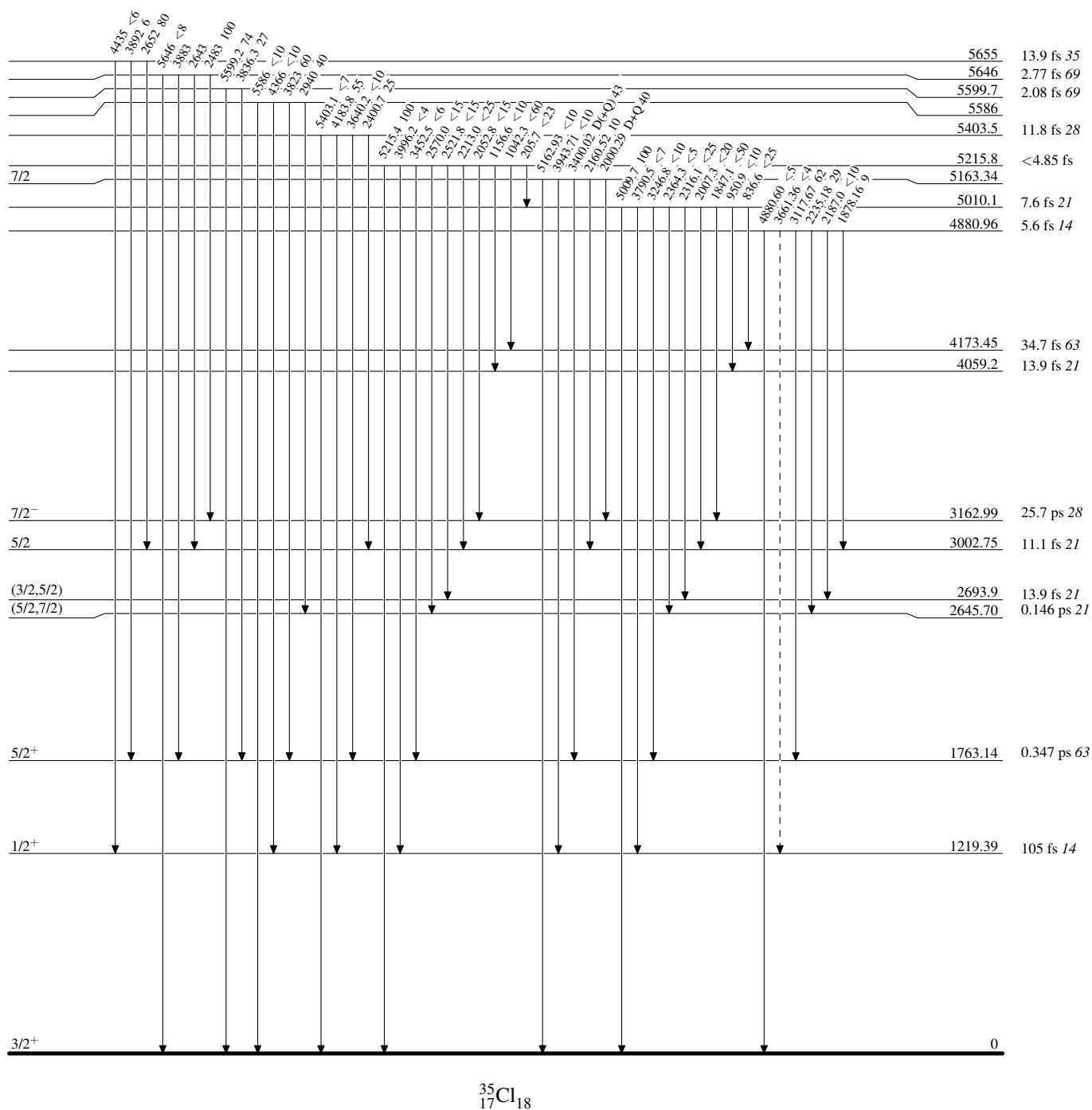


$^{34}\text{S}(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

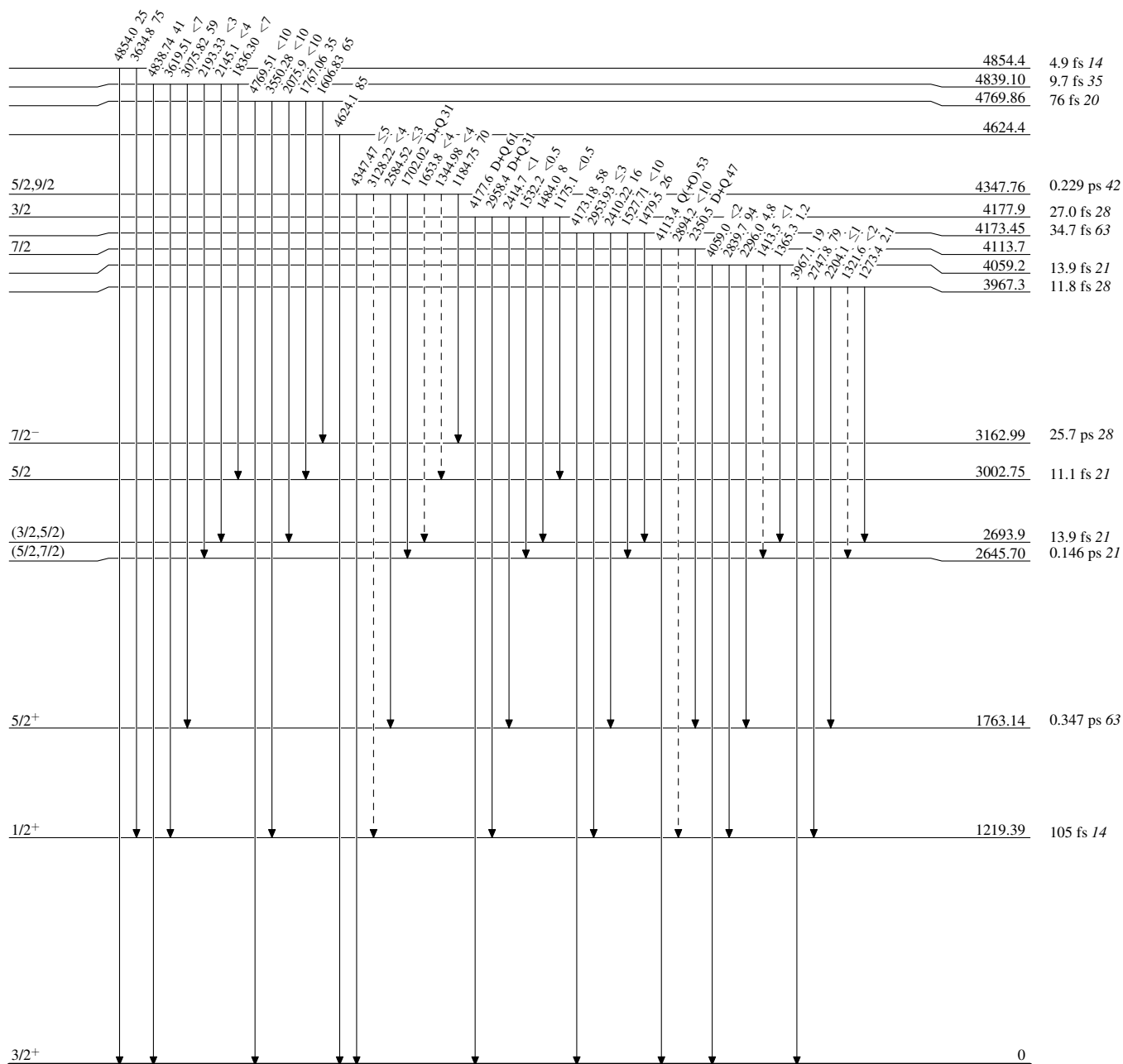
-----► γ Decay (Uncertain)


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)


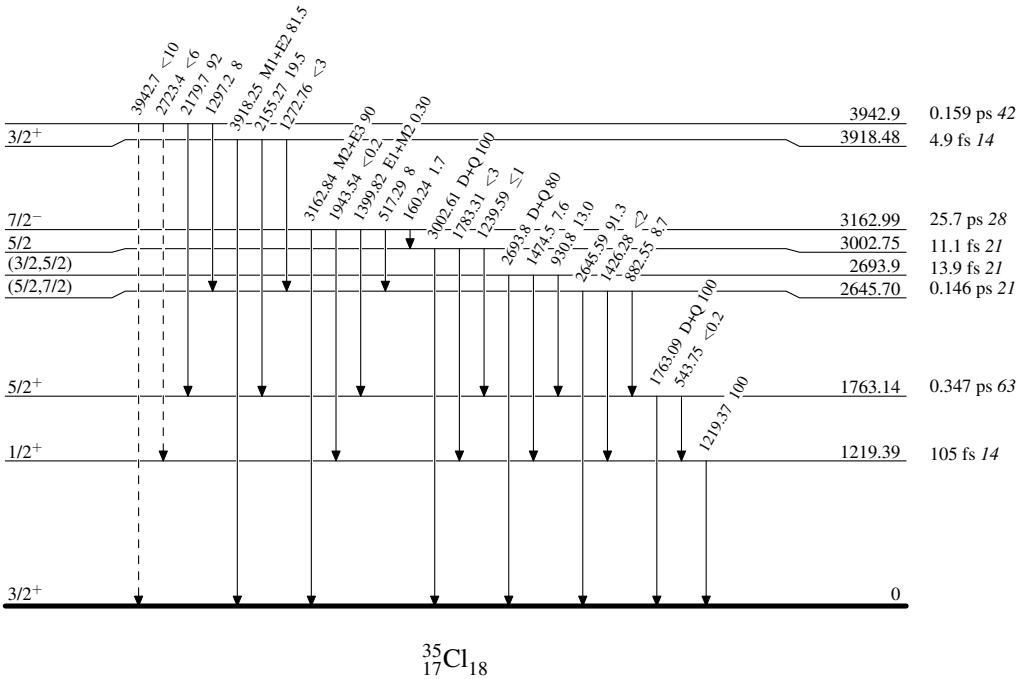
³⁴S(p,γ) 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)



³⁵Cl₁₈